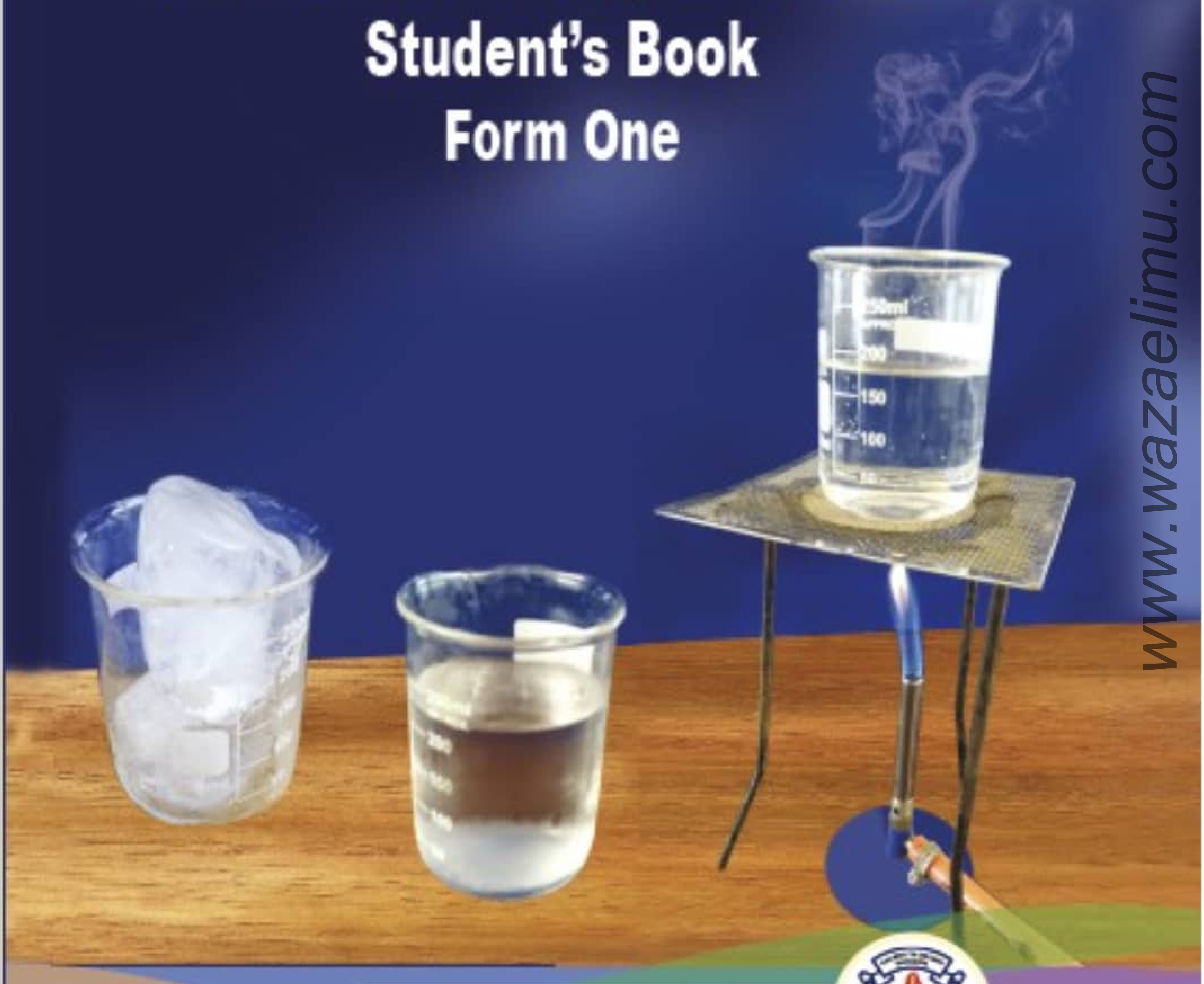
Chemistry

for Secondary Schools

Student's Book

Form One



Tanzania Institute of Education

Chemistry

for Secondary Schools

Student's Book
Form One

THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF EDUCATION,
SCIENCE AND TECHNOLOGY

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Dr Lyabwene M. Mtahabwa
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Tanzania Institute of Education

Mikocheni Area

132 Ali Hassan Mwinyi Road

P. O. Box 35094
14112 DAR ES SALAAM

Mobile numbers: +255 735 041 168/ +255 735 041 170
E-mail: director.general@tie.go.tz
Website: www.tie.go.tz

Table of Contents

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Editors:	Prof. John A. M. Mahugija (UDSM), Prof. Stelyus L. Mkoma (SUA), Dr Grace A. Kinunda (UDSM), Dr James E. Mgaya (DUCE), Dr Matobola Mihale (OUT), Dr Godbless Shao & Dr Stephano Mlozi (MUCE)
Designer:	Ms Victoria R. Mwinyi
Photographer:	Mr Chrisant A. Ignas (TIE)
Illustrator:	Mr Fikiri Msimbe (TIE)
Coordinator:	Mr Joseph Chamadali (TIE)

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Dr Aneth A. Komba
Director General
Tanzania Institute of Education

Preface

This textbook, *Chemistry for Secondary Schools*, is written specifically for Form One Students in the United Republic of Tanzania. The textbook is prepared in accordance with the 2023 Chemistry Syllabus for Ordinary Secondary Education, Form I–IV, issued by the Ministry of Education, Science and Technology (MoEST). It is a revised edition of Chemistry for Secondary Schools Student's Book for Form One that was published in 2021 in accordance with the 2005 syllabus issued by the then Ministry of Education and Vocational Training (MoEVT).

The book consists of five chapters, namely: Introduction to chemistry; Laboratory techniques and safety; Fire, firefighting and flames; Matter; and Elements, compounds and mixtures. In addition to the contents, the chapters contain illustrations, activities, tasks, projects and exercises. You are encouraged to do all the activities, tasks, projects and attempt all questions in the exercises. You are also required to prepare a portfolio for keeping records of activities performed in different lessons. Doing so will enable you to develop the intended competencies.

Additional learning resources are available at TIE e-Library *https://ol.tie.go.tz* or *ol.tie.go.tz*

Tanzania Institute of Education

Chapter One

Introduction to Chemistry

Introduction

Chemistry is fundamental to our world. It has a variety of roles and impacts on several aspects of life, including food, medicine, manufactured products, advancement of technology, and the environment. In this chapter, you will learn about the concept of chemistry, the relationships of chemistry with other related disciplines, and applications of chemistry in the development of a modern society. The competencies developed will enable you to apply chemistry skills, theories and principles to solve daily life challenges.

Think

The contributions of chemistry in various aspects of life.

Concept of chemistry

Task 1.1

1. Use reliable online sources to search for images or pictures of various materials produced by chemical processes and identify those commonly found in your environment.
2. Examine the properties such as colour, size and physical state of the materials.
3. Watch animated videos illustrating the composition of chalk, soft drinks or water, and how water undergo changes from liquid to solid state.
4. Explain the meaning of chemistry based on the task performed.

Chemistry is a branch of science that deals with the composition, structure and properties of matter, and the changes that matter undergoes. Matter is anything that has mass and occupies space. It includes the materials or substances of nature which make up our environment.

Chemistry is divided into different areas such as analytical chemistry, inorganic chemistry, organic chemistry, physical chemistry, chemical engineering and biochemistry. Analytical chemistry deals with the studies of instruments and methods used to separate, identify, and quantify chemical species in matter. Inorganic chemistry deals with properties and behaviour of *inorganic compounds*. Organic chemistry deals with properties and behaviour of *organic compounds*. Physical chemistry is concerned with the relations between the physical properties of substances and their chemical composition and transformations. Chemical engineering deals with operations, designs and methods of improving production. Biochemistry deals with the study of chemical processes that occur in living things.

The people who study chemistry are called chemists. Since chemistry is a science that involves experiments and practical work, chemists must acquire scientific skills to successfully obtain facts and verify them. These skills include:

1. Making thorough observations;
2. Recording accurately what has been observed;
3. Organising the observed and recorded information;
4. Repeating tests to ensure observations are accurate;
5. Drawing conclusion from observations; and
6. Predicting possible outcomes of similar experiments.

Therefore, chemistry students should apply these skills to be able to think logically and critically.

Relationships of chemistry with other related disciplines

Task 1.2

Conduct a library and internet search on the relationships between chemistry and other related disciplines and prepare a report.

Chemistry is not an isolated discipline, because it is related to other disciplines such as, biology, physics, agriculture, geology, environmental sciences and mathematics. Studying the relationships of chemistry with other related disciplines is important as it helps to understand the world around us and give insight to physical, chemical and biological processes. It also provides useful integrated skills and acts as a foundation for understanding scientific skills through researching different phenomena. Furthermore, the relationships of chemistry with other related disciplines help to understand different professions resulting from learning chemistry. A person studying chemistry with other disciplines may become a medical doctor, pharmacist, nurse, chemical analyst, laboratory technician, laboratory scientist, researcher, chemical engineer, teacher and environmentalist. This means chemists find employment opportunities in hospitals, pharmaceutical industries, research centres, testing laboratories, schools, energy sectors, chemical industries, food processing industries, biotechnology firms and other manufacturing industries. The following are some disciplines which are related to chemistry:

Biology: Chemistry and biology are closely related disciplines. These disciplines converge in biochemistry and biotechnology fields. Biochemistry explores the chemical processes that occur within living organisms for understanding biological systems. On the other hand, biotechnology is the use of living organisms to produce economically useful products. These products are produced by integrating the knowledge and skills from chemistry, biology and chemical engineering. Therefore, the relationships of these disciplines bring development that resolves daily life challenges related to agriculture, medicine, nutrition and environmental science.

Physics: Chemistry has a strong interrelation with physics and forms the foundation for understanding the physical world and the behaviours of matter. Chemistry and physics converge in physical chemistry. This field drives innovations and technological advancements in various areas such as medical instruments, materials science and products from manufacturing industries.

Agriculture: Chemistry is closely related to agriculture through agro-chemistry. Agro-chemistry examines the chemical processes related to plants, animals and soil health, crop protection and fertilisers. This field helps in ensuring food availability and security as well as good agricultural practices which enhance the quality of agricultural products.

Geology: Chemistry and geology disciplines integrate through geochemistry field. This field applies chemistry theories and principles in exploring the chemical composition and processes of Earth and other planets. Geochemists use their expertise in investigating the abundance and distribution of chemical elements in rocks and minerals which is the foundation of mining industry.

Environmental Sciences: Environmental chemistry and atmospheric chemistry are among the fields which provide relationships between chemistry and environmental science. These fields apply chemistry theories and principles to study the interactions of chemicals in the environment, including pollution and their impacts to the natural world.

Mathematics: Mathematics is applied in almost all areas of chemistry including physical chemistry, analytical chemistry, inorganic chemistry, organic chemistry and biochemistry. It serves as a powerful tool for various aspects of chemistry such as analysis of chemical processes that involve interactions and changes of matter, and the interpretation of experimental data through statistical analysis which may be used to draw conclusions in chemistry experiments. It is also applied in measuring mass, volume, temperature and concentration of chemical substances, calculating percentage and ratios that express concentrations, composition and yields of chemical reactions as well as computational chemistry which uses mathematics to simulate chemical processes, predict structures and calculate properties of chemical systems.
The relationships of these disciplines make meaningful contributions for advancement of the chemistry related fields such as medicines, pharmacy, agriculture and environmental chemistry. Therefore, mathematics allows individuals to conduct precise experiments, make predictions and develop deep understanding on the principles governing the behaviours of matter.

Applications of chemistry in the development of a modern society

Project

Visit a nearby manufacturing industry, health facility, mining site or agricultural farm to identify materials produced through chemical processes, study the practical applications of chemistry in the development of modern society, and prepare a report.

Chemistry plays a significant role in the development of a modern society. It is applied in a wide range of industrial processes and daily life activities. It has influenced most aspects of life including healthcare, energy production and environmental sustainability. The quality of life for most people worldwide has improved due to practical applications of chemistry. Many of the products that are obtained by the applications of chemistry are useful for development of modern societies. These products are processed in different areas such as agriculture, medicine, nutrition and cooking, and manufacturing industries.

Agriculture

Agriculture is the practice of growing crops and keeping animals. Farmers use many products made through chemical processes to obtain better agricultural yields. These products include fertilisers, pesticides, animal vaccines and processed animal feeds. Fertilisers are used to improve the quality and quantity of crops. Some of the fertilisers are shown in Figure 1.1.

Figure 1.1: *Fertilisers*

Pesticides are used to destroy or control pests harmful to crops, animals, or their products. Pesticides are sprayed or sprinkled on crops to destroy pests. Figure 1.2 shows a farmer spraying pesticides on the crops.

Figure 1.2: *Spraying pesticides on the crops*

Pesticides are also used in animal dips (Figure 1.3) to kill pests such as ticks that attack animals.

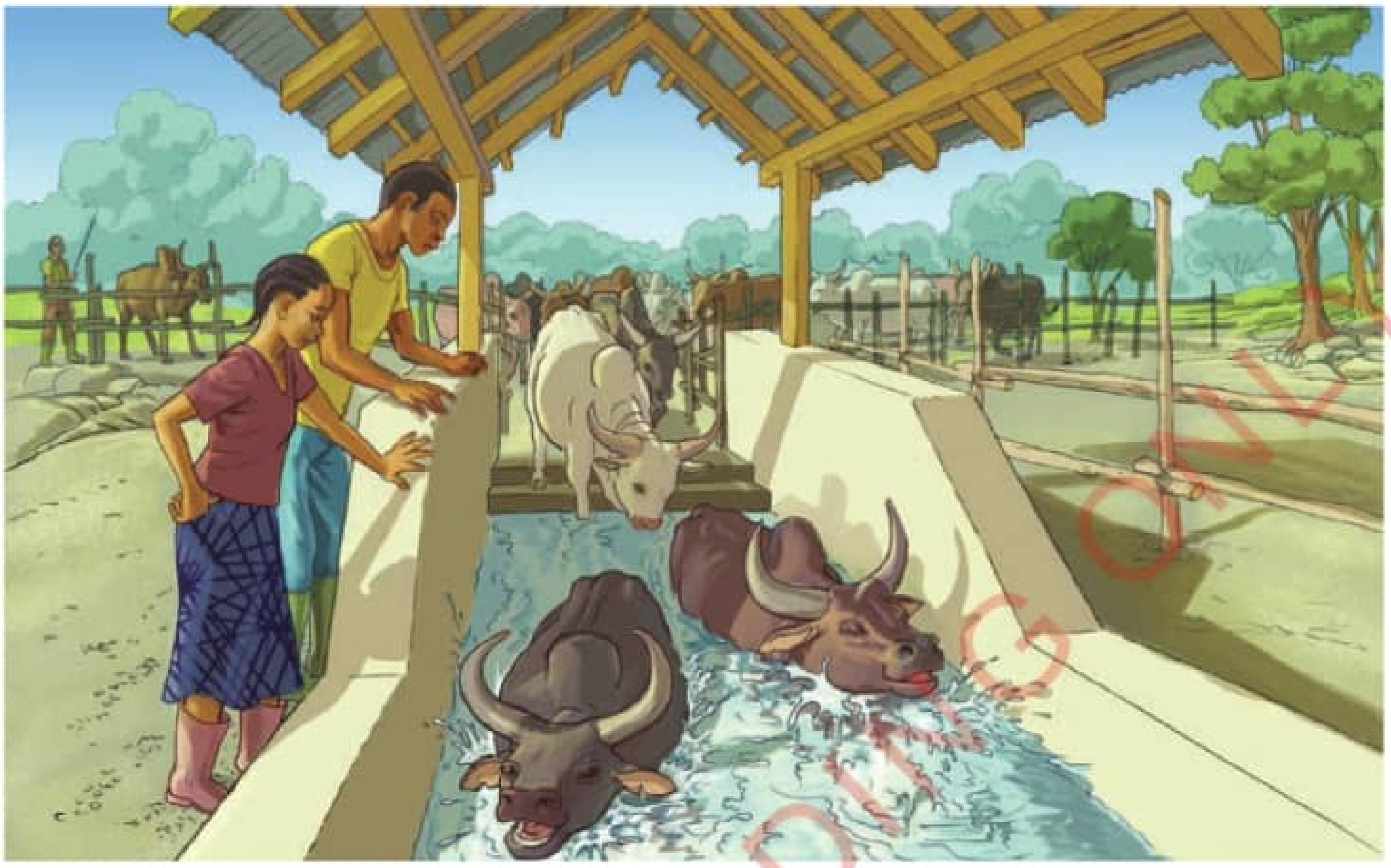


Figure 1.3: *Animal dip*

Animal vaccines are used to protect animals against various infectious diseases. They are also important in preventing the transmission of diseases between domesticated animals and human being. Figure 1.4 shows a dog being vaccinated.

Figure 1.4: *Vaccinating a dog*

Medicines

The field of medicine ensures our well-being through the prevention and treatment of diseases. Chemically produced substances in the forms of vaccines and medicines are used in this field. Medicines are found in medical stores and pharmacies. Some medicines are shown in Figure 1.5.

Figure 1.5: *Medicines in a pharmacy*

Nutrition and cooking

Chemistry has an impact on the tastes, textures and nutritional value of food. Understanding chemistry helps in identification of different food components including carbohydrates, proteins, fats and oils, vitamins and minerals. Chemical reactions involved in cooking help to have foods that are healthy and palatable. The reactions taking place during cooking bring impact on the taste and texture of the food. For example, chemical processes are useful in the preparation of baked foods such as bread, cakes, rock buns and biscuits (Figure 1.6).

Figure 1.6: *Examples of baked foods*

Manufacturing industries

Manufacturing industries such as textile, chemical, paper, food and beverage, transport industries, communication, plastic manufacturing, and construction materials industries rely heavily on chemical processes. Most of the machine parts are made of metals, plastics and rubber. These materials are produced through chemical processes. Raw materials from different sources are also processed chemically to obtain products. Examples of such products are cement, plastic containers, fabrics, chemicals, rubber and paper. Figure 1.7 shows some of the products manufactured using various chemical sources and processes.

Figure 1.7: *Some of the products from the manufacturing industries*

Food and beverage industries

The food and beverage industries have benefited greatly from chemical processes and products. Among these processes is the preservation of foods, especially canned or bottled. Examples of chemicals commonly used as food preservatives are common salt, sodium nitrite, citric acid, sodium benzoate and alcohol. Chemically preserved food stuffs last longer than those unpreserved. Some foods which are preserved using chemicals are shown in Figure 1.8.

Figure 1.8: *Canned and bottled foods*

Transport

Fuels and lubricants used in motor vehicles, aircraft, motorcycles, ships and other means of transportation are produced by chemical processes. Their parts such as engines, batteries and tyres are made through chemical processes. Other materials such as seals, plastics, paints, coolants, carpets and seat covers found in the means of transportation are also made through chemical processes. Figure 1.9 shows a car engine and its associated parts made through the application of chemistry.

Figure 1.9: *Car engine and its associated parts*

Communication and technology

Chemistry plays a significant role in communication and technology, from the development of batteries and electronic devices to the transmission of data through various networks. Applications of chemistry principles give impact on innovation and advancement of technology. It also improves communication and brings entertainment. Newspapers and magazines are written using papers and ink that are manufactured through chemical processes. Telephones, radios, televisions, and computers rely much on wires, batteries, chemicals and other materials that are chemically manufactured. Figure 1.10 indicates some of the communication media whose components are made by chemical processes.

Figure 1.10: *Some of the communication media*

Home care products and cosmetics

Home care products are chemically made materials used to clean the home and its surroundings, making them more comfortable to live in while ensuring effective hygiene and personal care. These products include soaps, detergents, disinfectants, air fresheners and polish. Some of the home care products are shown in Figure 1.11.

Figure 1.11: *Examples of home care products*

Cosmetics are also produced chemically. They are used to improve one's physical appearance or odour. Such products include deodorants, lotions, lipsticks and face foundation creams (Figure 1.12).

Figure 1.12: *Examples of cosmetics*

Exercise 1.1

1. Using examples of chemical substances, explain how chemistry is commonly applied in your community.
2. How do the chemistry skills contribute to the understanding of environmental science field?
3. How does chemistry influence the development of a modern society?

Chapter summary

1. Chemistry is a branch of science that deals with the composition, structure and properties of matter, and the changes that matter undergoes.
2. The people who study chemistry are called chemists.
3. Chemists use various theories and principles from chemistry and other related disciplines including biology, physics, agriculture, geology, environmental science and mathematics to understand the world around us and give insight to physical, chemical and biological processes.
4. The skills acquired in chemistry and other related disciplines can enable a person

to develop different professions such as medical doctor, pharmacist, chemical analyst, chemical engineer, teacher, laboratory technician, laboratory scientist, environmentalist, nurse, biotechnologist and researcher.

5. A lot of materials are made through the application of the knowledge, skills and principles of chemistry. The materials include soaps and detergents, foods and drinks, medicines, fuels and cosmetics.

6. Chemistry is important because it is applied in many areas such as agriculture, medicines, food and beverage industries, home care and cosmetics industries, research, transport, and communications and technology. Therefore, the applications of chemistry in these areas assist in the development of modern society.

Revision exercise 1

Choose the correct answer in items 1–5.

1. If a Form One Student aspires to be a professional farmer, which among the following subjects a student should be advised to focus on in order to develop the necessary competencies?
   (a) Geography, physics and chemistry
   (b) Biology, physics and mathematics
   (c) Agriculture, chemistry and biology
   (d) Chemistry, physics and biology

2. Imagine that, you are playing a chemistry-related trivia game with your friends. If you are to choose one term which does not relate to any recognised branches of chemistry among the following, which term could you choose?
   (a) Organic chemistry
   (b) Arithmetic chemistry
   (c) Inorganic chemistry
   (d) Physical chemistry

3. What are some of the main purposes of vaccines as chemical substances?
   (a) Cure animals and human beings from diseases.
   (b) Protect animals and human beings from diseases.
   (c) Introduce vitamins and proteins in animals.
   (d) Make animals grow faster than normal.

4. Which of the following terms refer to a person with specialised knowledge and expertise in the field of chemistry?

(a) Alchemist
(b) Chemistry
(c) Chemist
(d) Pseudo-chemist

5. A person at home can apply chemistry in:
   (a) playing games.
   (b) watching television.
   (c) cooking food in a kitchen.
   (d) dusting the floor in a house.
6. Explain how chemistry principles are used in making water clean and safe.
7. What is the role of chemistry in overcoming health problems?
8. Use relevant examples to support the statement, "agriculture and chemical industry can work together rather than separate".
9. In what ways have modern societies benefited from chemistry through communication and advancement of technology?
10. How does the application of chemistry contribute to a clean environment?

Chapter Two

Laboratory techniques and safety

Introduction

Most of the chemistry experiments in schools take place in a special room called laboratory. Figure 2.1 shows an example of a school laboratory. In this chapter, you will learn about laboratory safety measures, First Aid and basic chemistry laboratory apparatus. The competencies developed will enable you to protect yourself in the laboratory and apply the best ways of handling apparatus and chemicals as well as home delicate appliances. Moreover, you will develop knowledge and skills in providing First Aid to sick or persons injured in accidents that may occur in the laboratory, in public places and at home.

Think

The appropriate use of the laboratory enhances safety to users and equipment.

Figure 2.1: *Example of a school laboratory*

Task 2.1

Watch a video which illustrates the laboratory safety measures. Note all features illustrated in the video and compare them with the features of your laboratory.

Laboratory safety measures

Safety in the laboratory is of great importance to its users. This is because the experiments performed in a chemistry laboratory often involve the use of delicate or complicated equipment and harmful chemicals. In this regard, activities carried out in the laboratory always require measures to prevent risks. It is important to minimize these risks to protect people and the environment. It is therefore reasonable to put in place safety measures when constructing the laboratory and when doing day-to-day laboratory activities to minimise or eliminate injuries and exposure to harmful substances. The following are some of the safety measures to be taken in the laboratory:

1. The laboratory should be equipped with working fire extinguishers fitted in accessible positions with clear instructions on how to use them in case of fire accidents.
2. Cupboards, storage cabinets and drawers should have locks. This is to ensure that one does not accidentally get into contact with harmful substances or interfere with the equipment/apparatus.
3. Laboratory floors should not be polished to avoid slipperiness.
4. All apparatus should be regularly checked to ensure that they are safe for use.
5. All laboratory users should wear appropriate protective gears to minimise exposure to hazards.
6. Emergency exits should be present, easy to access and use. Doors should open outward.
7. There should be a manual or instructions on how to treat spills of different chemical substances present in the laboratory.
8. Gas cylinders should be labelled, well stored and supported. They should always be in good working conditions.
9. Each laboratory should contain a First Aid Kit with all the necessary items.
10. Refrigerators and freezers used in the laboratory should be labelled "*For chemical use only*" to avoid contamination with other substances. They should be clean and free of any spills.
11. The fume chambers should be labelled. They should be kept in good working conditions to minimise unexpected gas leaks or emissions.

12. Containers for chemicals should be checked regularly to ensure that they do not leak. They should have secured stoppers or covers when the chemicals are not in use.
13. Stored chemicals should be regularly inspected to ensure no use of expired chemicals.
14. Chemicals that can easily react with each other should never be stored together.
15. All chemical containers should be well-labelled to prevent accidental use of wrong substances.
16. Equipment for monitoring accidents such as smoke detectors for fire accidents should be installed to give alerts of any possible dangers.

Handling chemicals safely

Working in the chemistry laboratory requires great care because many chemicals are poisonous and can cause death. The hazards of the chemicals in the school laboratory should be identified by checking the *safety signs* labelled on their containers. All chemicals which are not correctly labelled should be removed from the laboratory and destroyed using proper methods.

Safety signs

Some safety signs have words and clear messages while many rely on visual symbols to warn for the potential hazards or to provide instructions for safe usage of the chemicals. On chemical containers, symbols are used specifically for labelling chemicals that are hazardous (Figure 2.2). It is important to take note of these symbols to ensure safety in the laboratory. For example, a container of nitric acid has the symbol for corrosive substance. That symbol indicates that the chemical corrodes substances such as metals, clothes, books, tables and causes burns to the skin. Table 2.1 shows interpretations of some safety symbols.

Figure 2.2: *Some safety symbols on chemical containers*

Table 2.1: *Interpretations of some safety signs*

S/N	Sign	Meaning
1.	 Corrosive	Corrodes surfaces as well as the human skin
2.	 Toxic	Poisonous and can cause death
3.	 Oxidant	Readily gives off oxygen or other oxidising materials. May intensify fire and cause explosions
4.	 Flammable/inflammable	Catches fire easily
5.	 Explosive	Blasts easily
6.	 Caution	Hazardous, can cause damage to organs from a short-term exposure

S/N	Sign	Meaning
7.	 Health hazard	Hazardous, can cause serious long-term health effects such as cancer, breathing difficulties and damaging of organs

Task 2.2

Watch simulations or videos illustrating how concentrated acids affect different materials, petrol catches fire and detonators explode. Relate the safety signs on the containers containing the substances with your observations.

Activity 2.1

Aim: To demonstrate the effect of concentrated acids on different materials.

Requirements: **Concentrated sulfuric acid, 3 white tiles, dropper, pieces of paper, cloth and pieces of metals**

Caution: **Sulfuric acid is highly corrosive. Handle it with care.**

Procedure

1. **Place each item on a separate white tile.**
2. **Using a dropper, pour few drops of concentrated sulfuric acid onto each item on a white tile. Be very careful not to spill the acid on yourself, others or on the worktable.**
3. **Observe and record the results.**

Question

What happened when few drops of concentrated sulfuric acid came into contact with each item?

Project

Visit various places like industries, kitchen, health facilities and other public places to study the use of safety signs. Write a report on your study visit that includes the places they are found, names of the signs observed, their meaning and where else they can be found.

Exercise 2.1

1. Imagine you have been given a chance to advise the school building committee on how should the laboratory building look like. What could be your advice?
2. What dangers are likely to occur if a laboratory user fails to follow the laboratory safety measures?
3. How can safety signs be used to prevent accidents and injuries in the laboratory?

Laboratory rules

A laboratory rule is a statement that explains what to do or not to do in the laboratory. Experiments should be conducted in the laboratory by following proper procedures. It is important to follow laboratory rules in order to have successful practical activities in the laboratory and to avoid some hazards. In this section, you will learn about laboratory rules to be followed before, during and after practical activities.

Task 2.3

Make a chart of at least ten important laboratory rules. You can use pictures, signs and artwork in your charts. The best chart will be pinned up on the laboratory noticeboard. Why should laboratory rules be obeyed by everyone?

Laboratory rules before practical activities

The following are the laboratory rules to be followed before the practical activities:

1. Do not enter in the laboratory without the permission or presence of the teacher or laboratory technician.
2. Dress appropriately for the laboratory activities. Wear a laboratory coat and safety goggles. Do not wear loose or floppy clothing. Tie back long hair. Roll up long sleeves. Do not wear shorts, sandals or walk barefooted.

3. Keep the windows open for proper ventilation.
4. Identify and note locations of all exits.

Laboratory rules during practical activities

The following are the laboratory rules to be followed during the practical activities:

1. Read the instructions carefully before you start any activity.
2. If you do not understand something, ask your teacher or the laboratory technician before proceeding.
3. Read the labels on chemical containers to be sure you have the right substance. Do not interchange labels.
4. Do not eat, drink, smoke, play or run in the laboratory.
5. Do not taste or smell a chemical or anything unless you have been instructed on how to do it.
6. Use a fume chamber when carrying out experiments which produce harmful gases and vapours.
7. Perform the intended experiments only. Do not set up your own experiments or interfere with someone's experiment.
8. Do not spill liquids on the floor.
9. Report any breakages or accidents to the teacher or laboratory technician immediately.
10. When heating substances in the test tube, direct the mouth of the test tube away from yourself and others. Do not point burners or hot substances towards yourself or other people.
11. Use a lighter or wooden splint to light burners. Do not use papers. Always strike the match before turning on the gas tap.
12. In case of a gas leakage, turn off all the gas taps and open the windows. Leave the room immediately.
13. Do not touch any electrical equipment with wet hands.
14. Do not use dirty, cracked or broken apparatus.
15. Turn off any gas or water taps that are not in use.
16. Do not remove any chemical or equipment from the laboratory without permission.
17. Replace covers and stoppers on the containers after using the chemicals.
18. Keep inflammable substances away from open flames.

19. Wash off any chemical spills on your skin or clothes with plenty of clean water.

Laboratory rules after practical activities

The following are laboratory rules to be followed after practical activities:

1. Appropriately dispose-off any wastes. Use the litter bins and not the sink to dispose any solid waste. Do not return unused chemicals to their original containers.
2. Do not pour concentrated chemicals into the sinks.
3. Clean up the equipment and store them safely.
4. Turn off gas and water taps.
5. Clean the working surfaces, benches and sinks.
6. Wash your hands with soap and running water.

Exercise 2.2

1. Why should chemicals in the laboratory be labelled and always well closed?
2. You have been assigned by your chemistry teacher to present the benefits of having laboratory rules. Use the following laboratory rules to prepare your presentation:
   (a) Do not eat, drink, smoke, play or run in the laboratory.
   (b) Do not taste or smell a chemical or anything unless you have been instructed on how to do it.
   (c) Use a fume chamber when carrying out experiments which produce harmful gases and vapours.
   (d) Perform the intended experiments only.
3. Give reasons to support each of the following statements:
   (a) A laboratory should be equipped with working fire extinguishers.
   (b) A laboratory should have large windows.
   (c) Concentrated chemicals should not be poured into the sinks.

First Aid

First Aid is an immediate help given to a person who is sick or injured before a professional medical assistance is available. It is important to learn the basics of First Aid because accidents can occur in the laboratory. The First Aid helps to:

1. reduce the likelihood of death.
2. shorten the recovery time.
3. prevent permanent disability.
4. reduce pain and suffering.
5. prevent the victim's condition from getting worse.

Causes of accidents in the laboratory

There are various causes of accidents in the laboratory. These include:

1. Failure to follow the right procedure for the given experiments that can lead to explosion and damage of equipment.
2. Spilled liquids left on the floor may cause slipping and falling.
3. Chemical spills and exposure that can lead to burns and damage to body parts such as eyes and skin.
4. Wrong use of equipment that can result in breakage, and in turn lead to cuts.
5. Accidental swallowing of harmful chemicals.
6. Improper disposal of chemical wastes that may result in explosions, burns and even fires.
7. Poor ventilation in the laboratory may cause poisoning by inhaling harmful gases and fainting due to lack of oxygen.
8. Gas leakage from taps or cylinders may lead to fires or even explosions. The cylinders may leak after a faulty refill or due to a faulty regulator.
9. Use of wrong reagents due to wrong labelling or use of expired reagents may cause burns, poisoning or damage to equipment.
10. Electric shock can occur if electrical appliances are not plugged in properly or are touched with wet hands.
11. Lack of adequate information on procedure and hazards related to certain practical activities may result in burns or explosions.
12. Eating or drinking in the laboratory may cause poisoning through contamination of food or drinks, or inhalation of harmful vapours while eating or drinking. Food particles or spilled drinks can damage apparatus or contaminate the chemicals.

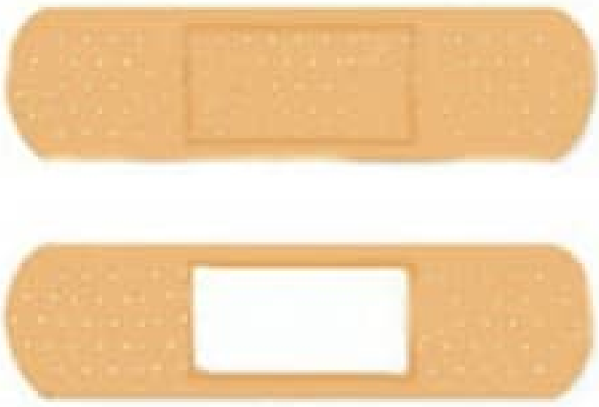
First Aid Kit

A First Aid Kit is a box that contains items used to give help to a sick or injured person. An example of the First Aid Kit is shown in Figure 2.3.

Figure 2.3: First Aid Kit

Every laboratory should have a First Aid Kit. The kit can be bought or schools can assemble their own kits. The main items in the First Aid Kit and their uses are listed in Table 2.2.

Table 2.2: Components of First Aid Kit and their uses

S/N	Item	Photograph	Uses
1.	**First Aid guide**		**Contains guidelines on how to use the items in the First Aid Kit**
2.	**Plaster or adhesive bandage**		**Covering small cuts or wounds to protect them from dirt and germs**

S/N	Item	Photograph	Uses
3.	Sterile gauze		Covering wounds to protect them from dirt and germs
4.	Antiseptic		Cleaning wounds to kill germs
5.	Soap		Washing hands, wounds and equipment
6.	Razor blade or pair of scissors	 	Cutting dressing materials
7.	Safety pin		Securing bandages
8.	Bandage		Keeping dressings in place and immobilising injured limbs

S/N	Item	Photograph	Uses
9.	**Cotton wool**		**Cleaning and drying wounds**
10.	**Clinical thermometer**		**Measuring the body temperature**
11.	**Disposable sterile gloves**		**Covering the hands to avoid infecting wounds and to prevent direct contact with a** victim's body fluids
12.	**Petroleum jelly**		**Smoothing dry, cracked and sore skin**
13.	**Liniment**		**Reducing muscle pains**
14.	**Torch**		**Source of light**

S/N	Item	Photograph	Uses
15.	Whistle		Blown to call for help
16.	Painkiller		Relieving pain
17.	Gentian violet		Treating fungal infections of the skin and mouth. Also used for the treatment of serious heat burns
18.	Iodine tincture		Preventing infections in fresh cuts, burns and scrapes

First Aid procedure

Physical injuries and sudden illness may occur in the laboratory due to accidents caused by various sources, such as gas leaks, fire, liquid and chemical spills, or incorrect use of chemicals or apparatus. These can result into poisoning, fainting, falls, explosion, cuts, burns, electric shock and damage to equipment. Such emergencies require urgent attention and First Aid must be administered. However, it is important to consider your own safety, even when you provide First Aid to another person. The following are some of the situations/conditions that may require First Aid and the procedure to be followed in giving the help.

Suffocation

Suffocation is a condition in which the lungs are not getting enough oxygen causing difficulty in breathing. Foam can also appear in the mouth and nostrils. The victim may eventually lose consciousness. Some chemical substances such as sulfur dioxide gas and nitric acid vapour can cause suffocation when inhaled, leading to breathing difficulties.

Procedure

1. Remove the cause if possible, or remove the victim from the cause of suffocation.
2. Ensure the victim's airways are open for air to reach the lungs. Do this by placing the victim on his or her back. With one hand on the victim's forehead and the other on the chin, tilt the head backwards in order to open the airways as shown in Figure 2.4.

Figure 2.4: *Tilting the victims head*

3. Blow air into the victim's airways using a clean material such as a piece of cloth or paper (Figure 2.5) and compress the chest with both hands. Do not use mouth to mouth *resuscitation* as it can cause infections.
4. Keep the victim warm using a light blanket.
5. Take the victim to a nearby health care facility.

Figure 2.5: *Blowing air into the victim's airways*

Burns

Burns are injuries resulting from the body coming into contact with heat or harmful chemicals. Burns usually cause blisters on the skin, and if severe, the skin becomes charred and peels off. This causes a lot of pains and may result into infections. Burns caused by hot liquids or vapour are called *scalds*.

Procedure

1. Lay the victim down and if possible, protect the burnt area from coming into contact with the ground.
2. Gently pour cold water on the burn for about 10 minutes to cool it and reduce pains. If the burn is severe, immediately call for medical help.
3. Check the breathing and pulse, and be prepared to blow air into the victim's airways if necessary.
4. Gently remove any jewellery, shoes or burnt clothing from the injured person. Loosen any tight clothing. Do not remove any clothing that is sticking to the skin.
5. Cover the burn with a sterile gauze and wrap it loosely to avoid pressure on the skin. Do not use fluffy cotton. Bandaging the wound reduces pain and prevents infections.
6. Give the victim a pain reliever.
7. Take the victim to a nearby health care facility.

Caution

(a) Do not use ice as it further damages the skin.

(b) Do not apply ointment or butter to the burn since this prevents proper healing.

(c) Do not break any blisters as that can cause infections.

(d) Burns to the face and in the mouth or throat are serious as they cause rapid inflammation of the air passage and may cause suffocation. In such cases, seek medical help immediately.

Choking

Choking is the blockage of the upper part of the airways by food, drink or other objects. This prevents proper breathing. Signs of choking include difficulty in speaking and breathing.

Procedure

1. Encourage the victim to cough up the object.
2. If the object remains stuck, give firm but gentle taps between the shoulder blades.
3. If the object is still stuck, perform the *Heimlich manoeuvre*. This procedure involves the following:

   (a) Make the person lean forward slightly and stand behind him or her as shown in Figure 2.6.

Figure 2.6: *Standings behind the leaning forward victim*

   (b) Make a fist with one hand and put your arms around the person. Grasp your fist with the other hand near the top of the victim's stomach as shown in Figure 2.7.

Figure 2.7: *Grasped fist placed near the top of the victim's stomach*

(c) Press your fist into the victim's abdomen and make quick upward thrusts to dislodge the object (Figure 2.8).

Figure 2.8: *Thrusting to dislodge the object*

(d) Repeat the thrusts until the object comes out.

(e) Take the victim to a nearby health care facility.

Bruises

A bruise is a skin injury that causes a change in colour of the skin. It is usually caused by a blow (hard hit) or impact that damages the blood vessels that are below the skin. Blood from the damaged vessels collects near the surface of the skin.

Procedure

1. Apply a cold compress such as a cloth dipped in cold water or ice wrapped in a cloth to the injury. Keep on compressing the injury for 20 to 30 minutes. This helps to reduce the swelling and speeds up recovery.
2. If the bruise is on the leg or foot and it covers a large area, keep the leg elevated as much as possible for the first 24 hours.
3. After 48 hours, apply a warm clean cloth to the bruise for 10 minutes, three times a day. This will help to increase blood flow to the affected area and thus speeding up the healing.

Shock

Shock is a condition in which the body system is unable to take enough blood to the vital organs. The vital organs include the heart, lungs and brain. Shock is common with many injuries, regardless of how severe they are. A victim of shock usually has a fast pulse rate and pale skin, lips and fingernails. The skin becomes cool and moist, while the limbs may tremble and become weak. As shock develops, the victim may experience nausea and even vomiting. Eventually, the victim may become restless, anxious, aggressive and finally unconscious.

Procedure

1. Control any cause of shock that can be put right, such as external bleeding.
2. Lay the victim down, then put him or her in a shock position as shown in Figure 2.9. If the victim vomits, turn him or her over to the side.

Figure 2.9: *Victim placed in a shock position*

3. Loosen tight clothing, laces and belts.
4. Maintain the victim's body temperature using warm covering, but do not overheat.
5. Be prepared to blow air into the victim's airways using a clean material as described under suffocation.
6. Take the victim to a nearby health care facility.

Electric shock

An electric shock occurs when a person comes into contact with electric current and it may result into injury or even death. Injuries may be burns or physical injuries that result from being shocked by the electric current.

Procedure

1. Break the contact between the electrical source and the victim by switching off the power source.
2. If it is not possible to switch off the current, move the person from his or her position by a dry non-conducting material made of cardboard, plastic or wooden plank or wood as shown in Figure 2.10.

Figure 2.10: *Rescuing electric shock victim*

3. If there is an immediate danger such as severe burns, muscle pain, difficulty breathing and confusion call for a professional help immediately.
4. If the victim is unconscious, check the pulse rate and whether the victim is breathing. If he or she has breathing problems, try to revive his or her breathing rate to normal. (Refer to the procedure in suffocation)

5. Administer First Aid for burns, shock or other injuries the victim may have sustained.

6. Take the victim to a nearby health care facility.

Caution: **Do not either touch a person who is still in contact with electric current or go near the area if high voltage electricity is suspected. Instead, call for help immediately.**

Fainting

Fainting is a sudden loss of consciousness caused by lack of sufficient blood and oxygen to the brain. The victim feels weak, sweats and then falls down.

Procedure

1. Loosen or remove any tight clothing from the victim.
2. Lay down the victim on his or her back.
3. Raise the legs of the victim to increase the flow of blood to the brain.
4. Ensure that the victim is in an open place with plenty of air.
5. Take the victim to a nearby health care facility immediately, if he or she does not recover in a few minutes.

Bleeding

Task 2.4

Watch the videos illustrating a person giving First Aid to a wounded victim.

1. What items did the person use to administer the aid?
2. How can the lesson learnt in the video be applied in your daily interactions with friends?

Bleeding is the loss of blood and usually occurs from a visible wound. It can also be from an internal organ. Bleeding may be light or severe. Excessive loss of blood can cause death.

Light bleeding

Procedure

1. Wear protective gloves for prevention of any infections.
2. Place the victim in a comfortable resting position.

3. Elevate the injured part.

4. Gently, but thoroughly clean the wound using clean water and antiseptic or common salt. Cover the wound using a sterile gauze. Gently clean the surrounding skin and dry it using sterile dressing (Figure 2.11).

Figure 2.11: *Cleaning a light bleeding wound*

5. Dress the wound and bandage it.

6. If the bleeding continues, take the victim to a nearby health care facility.

Severe bleeding

Procedure

1. Wear protective gloves for prevention of any infections.

2. Use your fingers to apply direct pressure onto the bleeding point for five to fifteen minutes. If the wound area is large, press the sides of the wound together gently but firmly. This should be done only if there is no fracture.

3. Raise the injured part and support it in a comfortable position that does not cause pain. This should not be done if a fracture is suspected or if it causes pain.

4. Carefully clean the wound, but do not remove any object stuck in the wound as this would lead to more bleeding.

5. Place a sterile gauze on the wound and press it down firmly. Cover it with a pad of soft material and hold it in position using a firm bandage as shown in Figure 2.12.

6. Take the victim to a nearby health care facility.

Figure 2.12: *Dressing a severe bleeding wound*

Nose bleeding

Procedure

1. Loosen clothing around the neck and chest.
2. Let the victim sit with the head tipped slightly forward.
3. Let the victim pinch in the nose and ask him or her to breathe through the mouth for few minutes.
4. Place a wet piece of cloth at the back of the victim's neck.
5. When bleeding stops, gently clean the nostrils.
6. If the bleeding continues, take the victim to a nearby health care facility.

Poisoning

A poison is any substance that can harm the body if swallowed, inhaled or absorbed into the body through the skin. Poisons include laboratory chemicals, drugs and medicines. Some of the poisons are indicated in Figure 2.13.

Figure 2.13: *Poisonous substances*

Some of the symptoms of poisoning are nausea, vomiting, abdominal cramps or pain, difficulty breathing, diarrhoea and abnormal skin colour.

Procedure

1. Call for medical assistance immediately.
2. Meanwhile, find out what caused the poisoning.
3. If the poison is in the eye:
   (a) Wash the eye with a lot of clean water.
   (b) Ask the victim to blink as much as possible.
   (c) Do not rub the eye.
4. If the poison is on the skin:
   (a) Remove any clothing from the affected part.
   (b) Wash the affected area thoroughly with a lot of water. Do not apply any ointment.
5. If the poison has been swallowed:
   (a) Induce vomiting if the poison is non-corrosive. Non-corrosive substances include medicines. This can be done by putting your fingers in the victim's throat.
   (b) Do not induce vomiting if the poison is corrosive. Corrosive substances

include kerosene, bleach, detergents, acids, bases, disinfectants, pesticides and certain toiletries.

6. If the poison has been inhaled, move the person to where there is plenty of fresh air. Make sure you protect yourself from inhaling the poison.
7. Take the victim to a nearby health care facility.

Vomiting

Vomiting is a forceful discharge of the contents of the stomach through the mouth. It can result from food poisoning, drinking contaminated water, inhaling poisonous fumes, or over-eating.

Procedure

1. Give the victim a lot of clear fluids including an oral rehydration drink.
2. Get medical assistance to a nearby health care facility if:
   (a) there is persistent vomiting.
   (b) the victim vomits blood.
   (c) the victim has high fever.
   (d) the victim is very dehydrated. This will be observed when the mouth and skin become very dry.

Exercise 2.3

1. Explain the importance of shock positioning a person suspected to be in shock.
2. Why is it necessary to take precautions while working in the laboratory?
3. If you were told to highlight the possible causes of accidents in the laboratory, what could be your highlights?
4. What are the different types of burns and how would you provide appropriate First Aid to each?
5. Why should a laboratory contain a First Aid Kit with all necessary items?
6. What are the key considerations when treating a wound to prevent infections?
7. On your field trip, you found two people surrounding a person lying down suspected of fainting. Describe how you would assess and provide the right First Aid to the victim?

Basic chemistry laboratory apparatus

Chemistry laboratory apparatus are special tools or equipment used in the laboratory. They serve various purposes such as measuring, testing, heating, filtering, grinding, holding, storing, scooping and safety.

Apparatus for measuring

The following are some of the apparatus used for measuring volume, temperature, mass, weight and time:

Pipette

A pipette (Figure 2.14) is a narrow glass tube into which small amount of a liquid is sucked for transfer to other containers. It is used to measure specific volume of liquids.

Figure 2.14: *Pipette*

Measuring cylinder

A measuring cylinder (Figure 2.15) is a glass or plastic apparatus that is graduated to measure volume of liquids.

Figure 2.15: *Measuring cylinder*

Beaker

A beaker is a cylindrical shaped apparatus with a flat bottom and a slight taper at the top. It can be made of glass or plastic materials. A glass beaker can be used for both measuring and heating liquids, while a plastic beaker is not used for heating. The beaker cannot measure liquids accurately, but can be used to estimate their volumes. Beakers can also be used for mixing and stirring solutions, transferring liquid, dissolving solid substances and observing chemical reactions. An example of a glass beaker is shown in Figure 2.16.

Figure 2.16: *Beaker*

Burette

A burette (Figure 2.17) is used to accurately measure and dispense known volume of liquids. It is commonly used in titration.

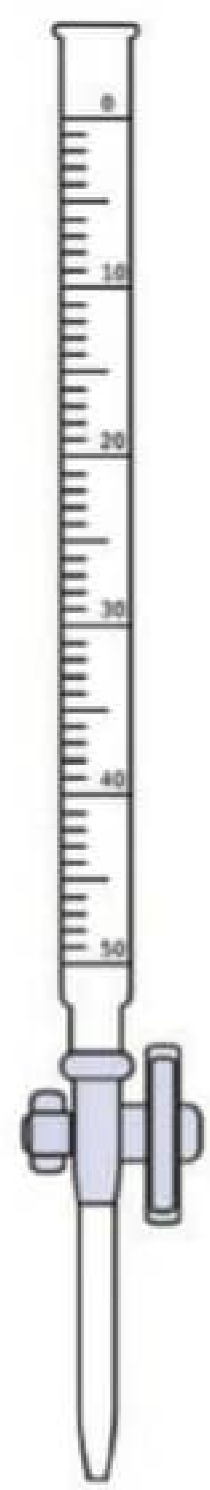


Figure 2.17: *Burette*

Measuring syringe

A measuring syringe (Figure 2.18) is used for sucking in and measuring volumes of liquids or gases.

Figure 2.18: *Measuring syringe*

Thermometer

A thermometer is an instrument used to measure and indicate the temperature of substances. The thermometer is able to measure temperature values that range from negative to positive values. Different thermometers have different temperature ranges. **An example of a thermometer is shown in Figure 2.19.**

Figure 2.19: *Thermometer*

Triple beam balance

A triple beam balance (Figure 2.20) is used to measure mass of substances. It has three beams that carry the weights. On one side there is a pan on which the object to be weighed is placed. Each of the three weights can be moved along the beams to indicate different units of mass of the object being measured.

Figure 2.20: *Triple beam balance*

Electronic analytical balance

An electronic analytical balance is used to measure the mass of substances. It gives more accurate readings than the triple beam balance. An example of the electronic balance is shown in Figure 2.21.

Figure 2.21: *Electronic analytical balance*

Spring balance

Spring balance (Figure 2.22) is used to measure the weight of an object. The object to be measured is hung on a fixed spring at one end with a hook. The reading on the scale is influenced by the force of gravity on the object making them suitable only for rough estimate of weight.

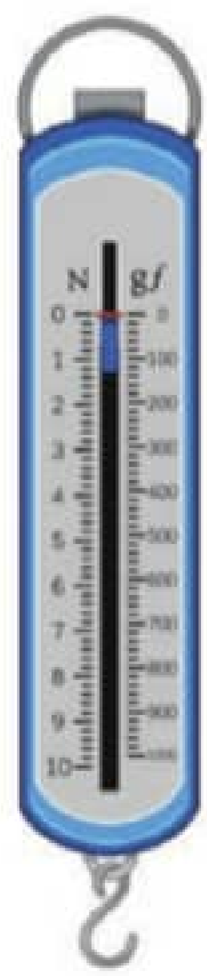


Figure 2.22: *Spring balance*

Stopwatch

This is a special watch that is used to accurately measure time during laboratory experiments. Examples of stopwatches are shown in Figure 2.23.

Figure 2.23: *Examples of stopwatches*

Apparatus for testing

The following are some of the chemistry laboratory apparatus used for various laboratory testing:

Test tube

Test tubes are used for holding chemicals or heating substances for short periods of time. They are also used when testing some simple chemical reactions. Figure 2.24 shows some test tubes.

Figure 2.24: *Test tubes*

Dropper

A dropper (Figure 2.25) is used to add liquids drop by drop during an experiment.

Figure 2.25: *Dropper*

Flasks

There are different types of flasks used in the laboratory. These include the volumetric flasks, round-bottomed flasks, flat-bottomed flasks, conical flasks and distilling flasks (Figure 2.26). They are used for measuring and holding liquids during experiments.

Volumetric flask

Round-bottomed flask

Flat-bottomed flask

Conical flask

Distilling flask

Figure 2.26: *Different types of flasks*

Watch glass

This is a very shallow circular glass container (Figure 2.27) used to hold substances that are being weighed or observed and as a cover of a beaker.

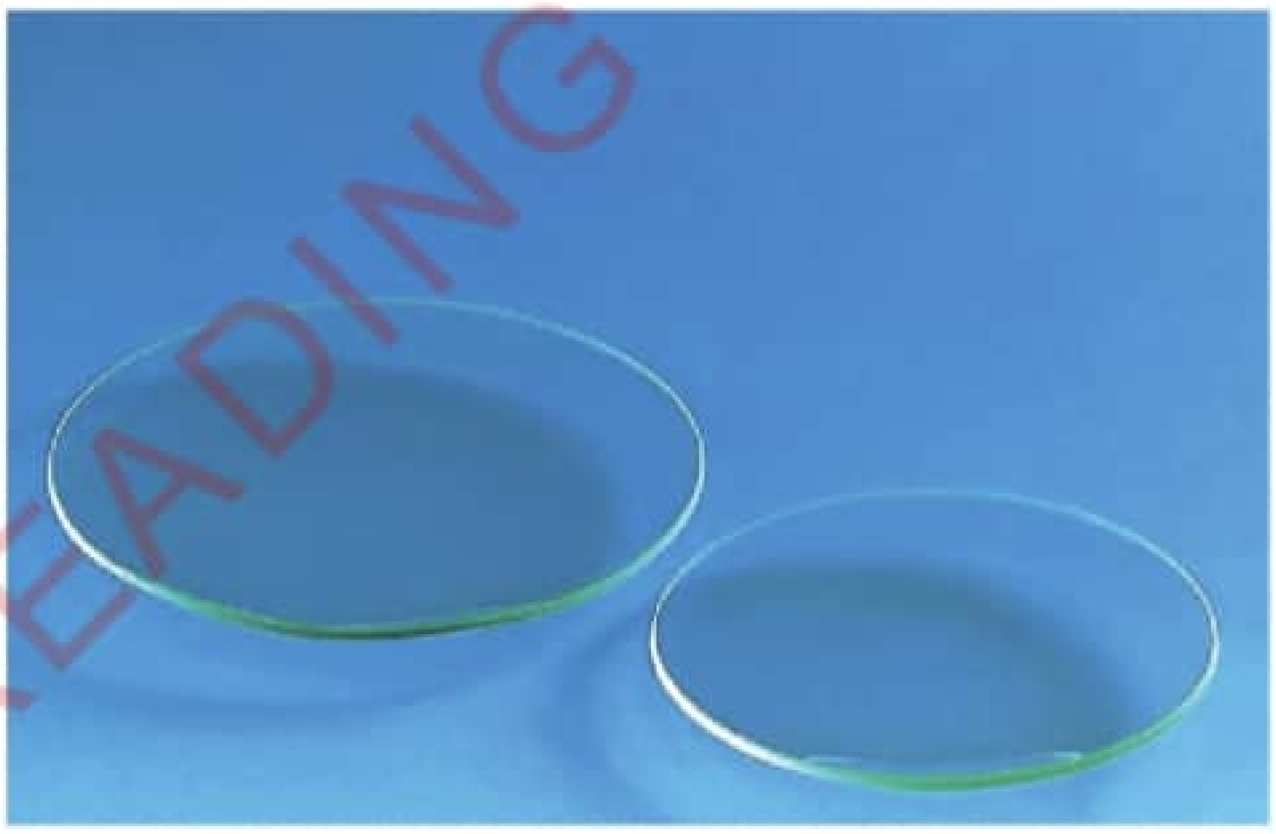


Figure 2.27: *Watch glasses*

Gas jar

This is a glass container (Figure 2.28) used for collecting and testing the properties of gases during experiments.

Figure 2.28: *Gas jar*

Thistle funnel

This is a glass funnel with a wide top and a long stem (Figure 2.29). It is used to add small volumes of liquid reagents to an exact point during experiments.

Figure 2.29: *Thistle funnel*

Apparatus for heating

The following apparatus are used when heating substances:

Burners and lamps

Burners and lamps are used as a source of heat for heating substances in the laboratory. Examples are the spirit lamp, Bunsen burner, gas stove and kerosene stove (Figure 2.30).

Spirit lamp

Bunsen burner

Gas stove

Kerosene stove

Figure 2.30: *Examples of heat sources*

Boiling tube

A boiling tube (Figure 2.31) is a hard and large test tube used to heat substances that need to be heated strongly. It is also used when the amount of a substance is too large to be contained in a test tube.

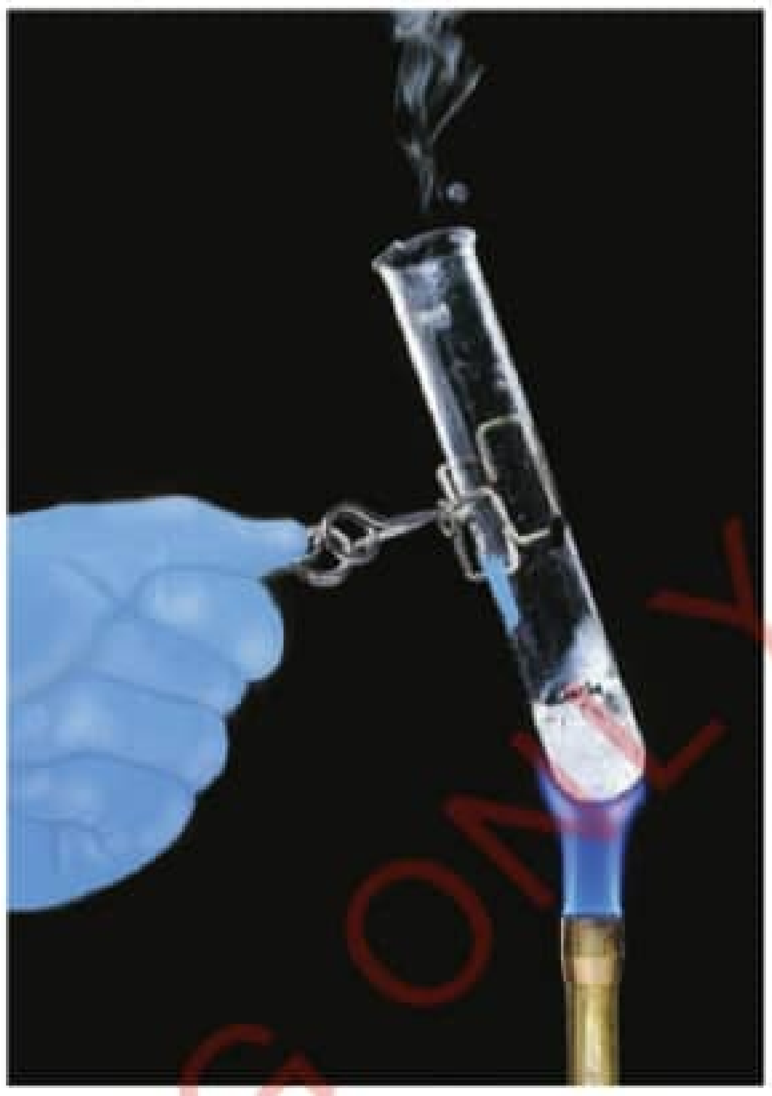


Figure 2.31: *Heating a substance in a boiling tube*

Tripod stand

A tripod stand is a support platform that has three legs (Figure 2.32). It is usually placed above the burner when heating or boiling substances.

Figure 2.32: *Tripod stand*

Wire gauze

A wire gauze (Figure 2.33) is usually placed on top of a tripod stand. A flask or beaker is placed on the wire gauze during heating. The wire gauze helps to spread out the flame and heat the substance evenly.

Figure 2.33: *Wire gauze*

Crucible

A crucible (Figure 2.34) is a container in which substances can be heated to very high temperatures. It is made of porcelain or non-reactive materials.

Figure. 2.34: *Crucible and its lid*

Evaporating dish

An evaporating dish is a shallow bowl with a curled lip commonly made of porcelain. It is used to heat and evaporate liquids and solutions. It can be heated to very high temperatures. Examples of evaporating dishes are shown in Figure 2.35.

Figure 2.35: *Evaporating dishes*

Deflagrating spoon

A deflagrating spoon (Figure 2.36) is a long-handled spoon used to heat small amount of substances inside a gas jar. Its cover prevents escaping of the gas from a gas jar during burning.

Figure 2.36: *Deflagrating spoon*

Apparatus for filtering

The following apparatus are usually used for filtering substances:

Filter funnel

A filter funnel is a device that is wide at the top and narrow at the bottom (Figure 2.37). It is used to filter substances or for transferring liquids between containers. The filter funnel is designed to hold a filter paper during filtration.

Figure 2.37: *Filter funnel*

Filter paper

A filter paper is a piece of porous paper (Figure 2.38) used for filtering liquids especially in chemical processes. It is usually folded into a cone shape and placed in a filter funnel to separate solids from liquids.

Figure 2.38: *Filter papers*

Apparatus for grinding

Mortar and pestle (Figure 2.39) are used for crushing or grinding substances. A mortar is a small hard bowl, while a pestle is a small heavy tool used for crushing things.

Figure 2.39: *Mortar and pestle*

Apparatus for holding

The following apparatus are used for supporting or holding other apparatus:

Test tube rack

A test tube rack (Figure 2.40) is an apparatus that is designed for placing test tubes so that they do not roll or break.

Figure 2.40: *Test tube rack*

Test tube holder

A test tube holder (Figure 2.41) is an apparatus that is used for holding a test tube securely while heating. It is also used to hold test tubes when dealing with substances that may be corrosive or hazardous.

Figure 2.41: *Test tube holder*

Retort stand and clamp

A retort stand is used with its clamp (Figure 2.42) to hold apparatus such as burettes during experiments. The clamp can be adjusted to an appropriate height from the base of the retort stand.

Figure 2.42: *Retort stand with a clamp*

Tongs

Tongs (Figure 2.43) consist of two long slender arms that are joined at one end and have a hinged mechanism near the other end. They are used to hold hot substances and apparatus during experiment.

Figure 2.43: *Pair of tongs*

Apparatus for storage

The following containers are used for storing substances in chemistry laboratories

Reagent bottles

These are containers specifically designed to store chemicals, reagents or solutions (Figure 2.44). Each reagent bottle is therefore labelled to show the chemical substance it contains, together with other relevant information.

Figure 2.44: *Reagent bottles with their stoppers*

Plastic wash bottle

A plastic wash bottle (Figure 2.45) is a container which is used to store distilled water used for washings or solvent commonly used for rinsing laboratory equipment.

Figure 2.45: *Plastic wash bottle*

Apparatus for scooping

Scooping is the picking up of a substance especially in powder form using a spoon. A spatula (Figure 2.46) is an apparatus used for scooping a small quantity of powder or crystalline chemicals.

Figure 2.46: *Spatula*

Apparatus for safety

Safety apparatus are used to protect individuals from potential hazards. The specific choice of the safety apparatus depends on the nature of the work and the potential risk involved. The following are examples of safety apparatus.

Safety glasses or goggles

Safety goggles (Figure 2.47) are used to protect the eyes from chemical spills, strong light and harmful vapour in the laboratory.

Figure 2.47: *Safety goggles*

Laboratory coats

Laboratory coats fall under the category of personal protective equipment designed to protect the clothing and skin from chemical spills and splashes.

Gloves

These are worn in order to protect hands from exposure to hazardous chemicals. The choice of gloves depends on specific chemicals to be handled.

Face masks

These are worn over the nose and mouth to protect against airborne particles, gases and fumes. The choice of the face mask depends on the specific hazards present.

Task 2.5

1. Watch videos or simulations showing different chemistry laboratory apparatus **and identify the apparatus and their uses.**
2. **Study the apparatus in your chemistry laboratory. Make a list of all the apparatus and classify them according to what they are made of. For example, apparatus made of glass, apparatus made of wood, apparatus made of plastic and apparatus made by clay.**
3. **Classify the apparatus according to their uses. For example, apparatus used for testing, apparatus used for storing substances and apparatus used for measuring.** Present your classification in a table as follows:

Apparatus for testing	Apparatus for storing substances	Apparatus for measuring

4. **Select any 10 laboratory apparatus found in your school chemistry laboratory and draw them.**

Activity 2.2

Aim: To measure the volume of a liquid using different apparatus.

Requirements: **Pipettes, burettes, measuring cylinders, beakers and coloured water**

Procedure

1. Pour some coloured water into a beaker.
2. Measure 20 cm^3 of the coloured water from the beaker using a pipette. Be careful not to suck the water into your mouth. Alternatively, you can use a pipette filler.

   Note: The pipette must be cleaned after each use to avoid spreading germs if sucking was done by using mouth.
3. Pour the measured coloured water into a measuring cylinder labelled A. Look at the scale on the side of the cylinder. Each number is a measurement in cubic centimetres.
4. Pour some coloured water into a burette. Read the volume of the water from the scale shown on the side of the burette. Write down your reading, for example, 10 cm^3. This is V_1 as shown in Figure 2.48(a).
5. Measure the required volume V by opening the burette tap and carefully let the water out into a measuring cylinder labelled B until the level of the water reaches final volume, V_2. In this case, V_2 is 30 cm^3 (Figure 2.48(b)). Close the tap.

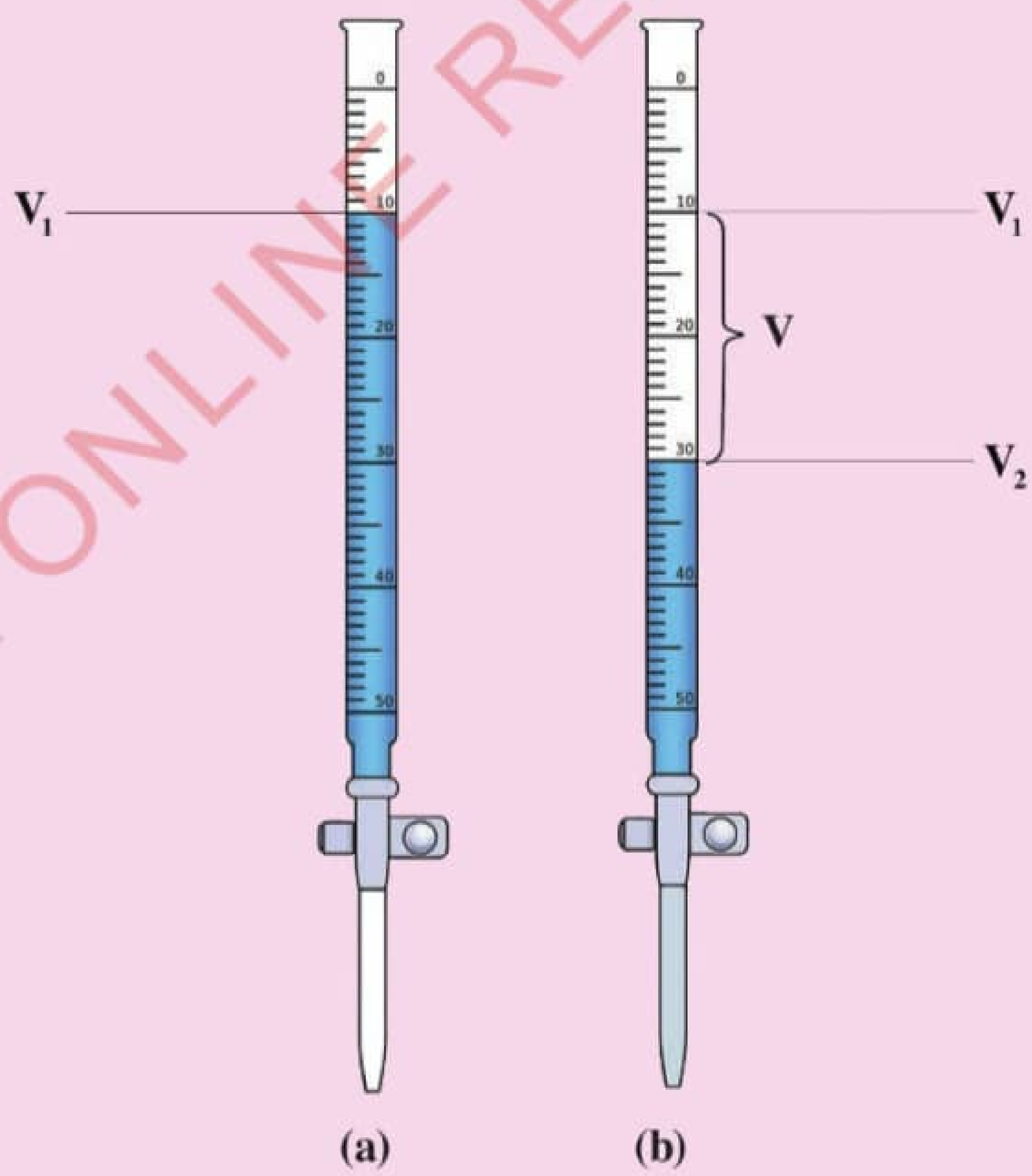


Figure 2.48: *Initial burette reading (a) and final burette reading (b)*

Question

What is the volume of the coloured water in each of the measuring cylinders?

Note: Measuring the volume of a liquid using a pipette is not difficult provided the liquid does not pass the mark near the top end of the pipette. Most pipettes used in school laboratories are made to an accuracy of one drop when they are used correctly. Measuring the volume of a liquid using a burette is not as direct as in the case of a pipette. It involves taking the initial reading of the level of the liquid in the burette. This is followed by draining the liquid to the required level (final volume). The measured volume is obtained by taking the difference between the final and initial readings.

Activity 2.3

Aim: **To measure the temperature of liquids.**

Requirements: **Thermometer, beakers, stopwatch, pair of tongs, water, source of heat and ice or fridge**

Procedure

1. **Pour some tap water into two beakers. Measure the temperature of the water by dipping a thermometer in each of the beakers for one minute.**
2. **Remove the thermometer and record the temperature.**
3. Place the first beaker in a fridge or in a bucket of cold water or ice cubes. Let it **remain there for about 20 minutes.**
4. **Remove the beaker from the fridge or bucket of cold water or ice cubes, and dip a thermometer in the water for one minute. Record the temperature.**
5. Heat the water in the second beaker for five minutes then remove the beaker from **the heat source.**
6. **Dip the thermometer in the second beaker for one minute without touching the base of the beaker. Record the temperature.**

Questions

1. **What was the reading on the thermometer when placed in:**

(a) tap water?
(b) cold water?
(c) heated water?

2. Why is it not advised to touch the base of the heated beaker with the thermometer while recording the temperature?

3. How applicable are the skills obtained from this activity in your daily life activities?

Activity 2.4

Aim: To measure the mass of different substances.

Requirements: Triple beam or an electronic balance, watch glasses, various items such as salt, sugar, flour, cooking fat, erasers and sand

Procedure

1. Place an empty watch glass on the weighing balance. Note down its mass. This will be M_1.
2. Place one item provided on the weighed watch glass and measure the mass of the watch glass with the item. This will be M_2.
3. Repeat steps 1 and 2 for all the items provided. Record your data in the following table:

S/N	Mass of the watch glass (M_1)	Mass of the watch glass with the item (M_2)
1.		
2.		

Calculate the mass of each item.

Question

How will the skills acquired in this activity be used in your day-to-day activities?

Exercise 2.4

1. A beaker is one of the apparatus used for volume measurement in the laboratory. How does it differ from other volumetric apparatus like measuring cylinders and pipettes?
2. Why is it necessary to familiarise yourself with the laboratory apparatus?
3. Using relevant examples, explain why the uses of common chemistry laboratory apparatus depend on what they are made of.

Chapter summary

1. A chemistry laboratory is a special room where chemistry experiments are carried out.
2. Safety signs have symbols or words that warn for the potential hazards or provide instructions for safe usage of the substance.
3. First Aid is the help given to a sick or injured person before a professional medical help is available.
4. A First Aid Kit is a box that contains items used to give First Aid to a sick or injured person.
5. Laboratory apparatus is special tools and equipment used in a laboratory.

Revision exercise 2

Choose the correct answer in items 1–8.

1. In rendering First Aid to an electric shocked victim, you must primarily think of:
   (a) checking whether the victim is breathing.
   (b) putting the victim in a recovery position.
   (c) administering First Aid for burns or injuries
   (d) breaking the contact with an electric source.
2. Some experiments require powdery chemicals. Which apparatus will you use to make them available out of the given granules?
   (a) Pestle and filter funnel
   (b) Round-bottomed flask and trough

(c) Mortar and pestle

(d) Bunsen burner and filter paper

3. In a discussion of the argument "access to safety equipment should never be blocked by any object." Which among the reasons can justify the statement?

   (a) It is just a simple law.

   (b) There must be spaces for people to move around in the laboratory.

   (c) The equipment is used every day.

   (d) It is important to reach safety equipment quickly in case of an accident

4. A Form One Student saw a flammable sign on a box. This suggest that the box is most likely to contain:

   (a) firewood.

   (b) papers.

   (c) radioactive materials.

   (d) spirit used in lamps.

5. Your chemistry teacher denied a student to enter the laboratory because of a floppy clothing. This is because the cloth may:

   (a) be blown by wind.

   (b) affect the measurements.

   (c) catch fire or cause one to fall.

   (d) cause poor ventilation in the body.

6. Imagine you are a class monitor/monitress and your chemistry teacher asked you to remind your fellow students on the laboratory rules before the practical session. Which rule will you insist to be adhered before a practical session?

   (a) Do not use dirty, cracked or broken apparatus.

   (b) Do not taste or smell chemicals.

   (c) Report any accident however small it may be.

   (d) Do not enter in the laboratory without permission.

7. Fainting is different from shock because fainting is a sudden loss of:

   (a) weight of the body.

   (b) consciousness.

   (c) confidence in mind.

   (d) water in the body.

8. **The chemistry teacher gave students a list of laboratory apparatus and asked them** to drop those that do not relate with chemistry. Which list could be dropped?

(a) crucible and lid, evaporating dish and filter funnel.

(b) wire gauze, test tube rack and deflagrating spoon.

(c) burette, pipette and conical flask.

(d) **microscope, drawing board and prism.**

9. **Match the following apparatus with their uses by writing the letter of the apparatus and the Roman number of its corresponding use:**

Apparatus	Uses
(a) Test tube	(i) Holding, heating and estimating the volume of liquids
(b) Conical flask	(ii) Measuring the mass of chemicals in the laboratory
(c) Measuring syringe	(iii) Holding substances that are being weighed or observed
(d) Beaker	(iv) Usually placed above the Bunsen burner with a wire gauze during heating or boiling
(e) Electronic balance	(v) Source of heat used for heating
(f) Watch glass	(vi) Holding chemicals and heating small portions of the chemicals in liquid or solid form
(g) Tripod stand	(vii) Used with a clamp to support apparatus like round-bottomed flask during experiments
(h) Retort stand	(viii) Adding reagents to an exact point during experiments
(i) Thistle funnel	(ix) Holding liquids during experiments
(j) Bunsen burner	(x) Sucking in and measuring volumes of liquids
	(xi) Titrating chemicals

10. **A student accidentally dropped a test-tube rack and broke three of the test tubes while the teacher was in the preparation room. To whom should the student report** the accident? Give reasons.

11. When your group is performing an experiment in the laboratory, and one of the students accidentally touches his/her eyes and starts shouting for help, briefly explain how you will render a First Aid to your colleague.
12. What are the main safety considerations and precautions when working with basic chemistry laboratory apparatus?
13. Most of the chemistry laboratory apparatus are made from glass. What could be the reasons behind?
14. Which reasons will you use to convince your friend that a chemistry laboratory must adhere to safety measures?
15. A Form One Student was boiling water on a Bunsen burner without using a wire gauze. If you were to advise him/her, what could be your advice on the necessity of using the wire gauze when using a Bunsen burner in a laboratory setting?
16. Your sister wants to know where your laboratory apparatus skills will be applied in real life. How could you help her using a kitchen as a laboratory setting?

Chapter Three

Fire, firefighting and flames

Introduction

In chemistry laboratories, flames from various sources of heat are used for heating purposes. The flames produced should be handled well to ensure safety. When flames are not properly handled can lead to fire accidents. In this chapter, you will learn about fire, firefighting and flames. The competencies developed will enable you to use flames from various sources appropriately and protect yourself, others and the environment from fire accidents both at school and out of school.

Think

The significance of fire in daily life activities and the potential risks of uncontrolled fire.

Fire

Fire is the state at which an ignited material combines with oxygen and gives off light, heat, flame and *combustion* products. When controlled, fire can be beneficial and used for cooking, heating or industrial processes.

Task 3.1

Search reliable online sources for the components required to start a fire. Then, write a summary describing the role of each component.

Components needed to start fire

For a fire to start, a combination of heat, oxygen and fuel is needed in the suitable proportion. The three components are usually referred to as the *fire triangle* as illustrated in Figure 3.1. If any of them is missing, no fire will start or continue burning.

Note: A fuel is a substance that releases energy when burned. Examples of fuels are wood, charcoal, natural gas, kerosene, petrol, diesel and coal.

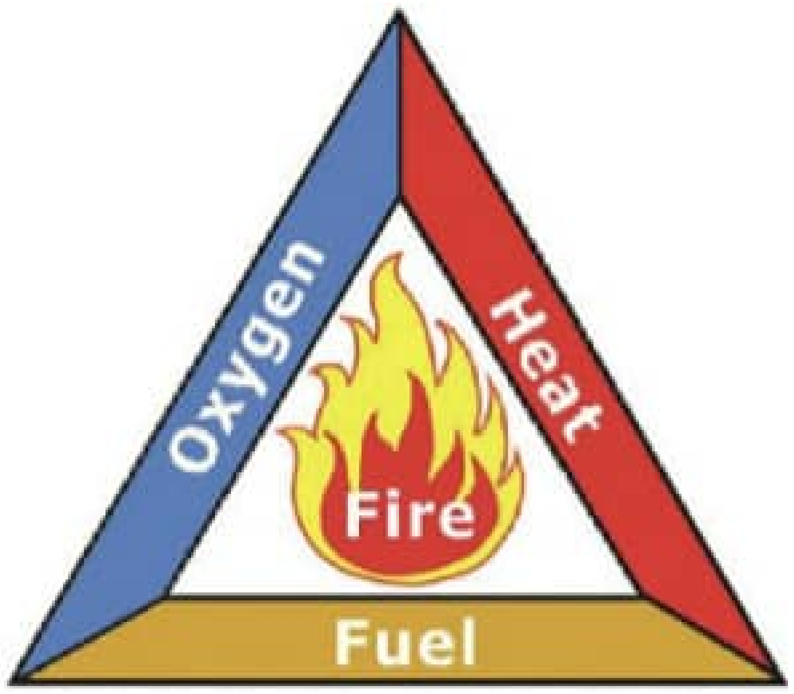


Figure 3.1: *Fire triangle*

Task 3.2

Use available resources in your environment to demonstrate the necessity of each component of a fire triangle. *Note*: The environment should be controlled to ensure safety.

Firefighting

Uncontrolled fires can be dangerous and destructive leading to injuries, property damage and environmental harm. *Firefighting* is the act of extinguishing fires and mitigating their effects. Materials which are used to put out fires are called *fire extinguishers*. Firefighting usually involves eliminating at least one of the three components in the fire triangle. Most firefighting equipment work by cutting off the oxygen supply to the fire to effectively extinguish it. In addition, each type of fire will need appropriate firefighting techniques and materials. The use of a wrong fire extinguisher for a particular fire can result in the fire spreading instead of being put out. Table 3.1 shows the classification of different fires according to the burning materials and the appropriate extinguishers.

Table 3.1: *Classification of fires and the appropriate extinguishers*

Class	Burning materials	Appropriate fire extinguishers
A	Ordinary solid combustible materials such as paper, wood and clothing	• Use water • Any type of portable fire extinguisher except carbon dioxide
B	Flammable liquids such as petrol, alcohol, kerosene and oil-based paints	• If the fire is small, use a fire blanket or sand • If the fire is large, use dry powder, foam or carbon dioxide extinguisher

Class	Burning materials	Appropriate fire extinguishers
C	Flammable gases such as butane and propane	• Dry powder extinguisher • Carbon dioxide extinguisher
D	Combustible metals such as magnesium, sodium and lithium, especially in powder form	• Dry powder extinguisher • Foam extinguisher
F	Cooking oils and fats	• Wet chemical extinguishers

Note: Fires caused by electricity are not given their own full classes as they can fall into any of the classifications. After all, it is not the electricity that burns but the surrounding material that has been set alight by the electric current. Electricity can be a source of any of the fire classes A, B, C, D and F. Normally, the fire caused by an electrical fault is extinguished based on the burning material. Thus, dry powder, carbon dioxide gas and dry sand can be used to extinguish the fire. Before extinguishing the fire caused by an electrical fault, the main electrical switch should be turned off.

Task 3.3

A Form One Student has been using a cooker for several weeks without cleaning. On one occasion while cooking a small fire broke out on the upper surface of the cooker and spread to the nearby curtains.

Questions

(a) Identify the sources of ignition and fuel.

(b) What type of fire extinguisher is most suitable for the fire and why?

Portable fire extinguishers

A portable fire extinguisher is an equipment used to put out fire and can be easily moved from one place to another. It is usually hung in an upright position in automobiles and on the walls of buildings such as schools and other public buildings. A portable fire extinguisher consists of a metal container that contains the extinguishing agent (substance) stored at high pressure. Different portable fire extinguishers are shown in Figure 3.2.

Figure 3.2: *Portable fire extinguishers*

Portable fire extinguishers should have ratings or codes on them. This means the extinguisher has a good firefighting capacity for specific classes of fires. Table 3.2 shows the different types of fire extinguishers, the chemical compositions of the extinguishing agents and the classes of fire for which they are either suitable or not suitable.

Table 3.2: *Chemical composition of the extinguishing agents of different portable fire extinguishers*

Type of fire extinguishers	Chemical composition of the extinguishing agents	Suitable for	Not suitable for
Air Pressurised Water (APW)	Ordinary tap water pressurised by air	Class A fire	Fire classes B, C and D (will spread the flame)
Dry chemical (DC)	Fine sodium bicarbonate powder pressurised by nitrogen	Classes A, B and C fires	Aircraft and electronics fires (Corrosive to metals such as aluminium) **Note:** Although it is safe to use indoors, it can obscure vision

Type of fire extinguishers	Chemical composition of the extinguishing agents	Suitable for	Not suitable for
Carbon dioxide	Carbon dioxide gas under extreme high pressure	Classes B and C fires and fire caused by electrical faults	Class A fire (material can re-ignite)
Halon	Bromochloro-difluoro-methane	Class A fire and fire caused by electrical faults	Classes B and C fires (least suitable)
Foam	Proteins and fluoro-proteins	Classes A and B fires	Fire caused by electrical faults
Wet chemical	Potassium acetate	Class F fire	Fires caused by electrical faults
ABC	Monoammonium phosphate with a nitrogen carrier	Classes A, B and C fires	Fire on electronic equipment

Note: For small Class A fire (of combustible solids), water can be used to easily put out the flames. Small fires that involve flammable liquids should be put out using sand or fire blanket. These cut off the oxygen supply. Water should never be used to put out Class B (flammable liquids) fires since it would spread the flame. Water is denser than flammable liquids, hence when poured, flammable liquids float above it.

How to use a portable fire extinguisher

Portable fire extinguishers should be used in the right ways to quickly put out fires. The extinguisher should aim at the base part of a *flame*. Figure 3.3 shows parts of a flame.

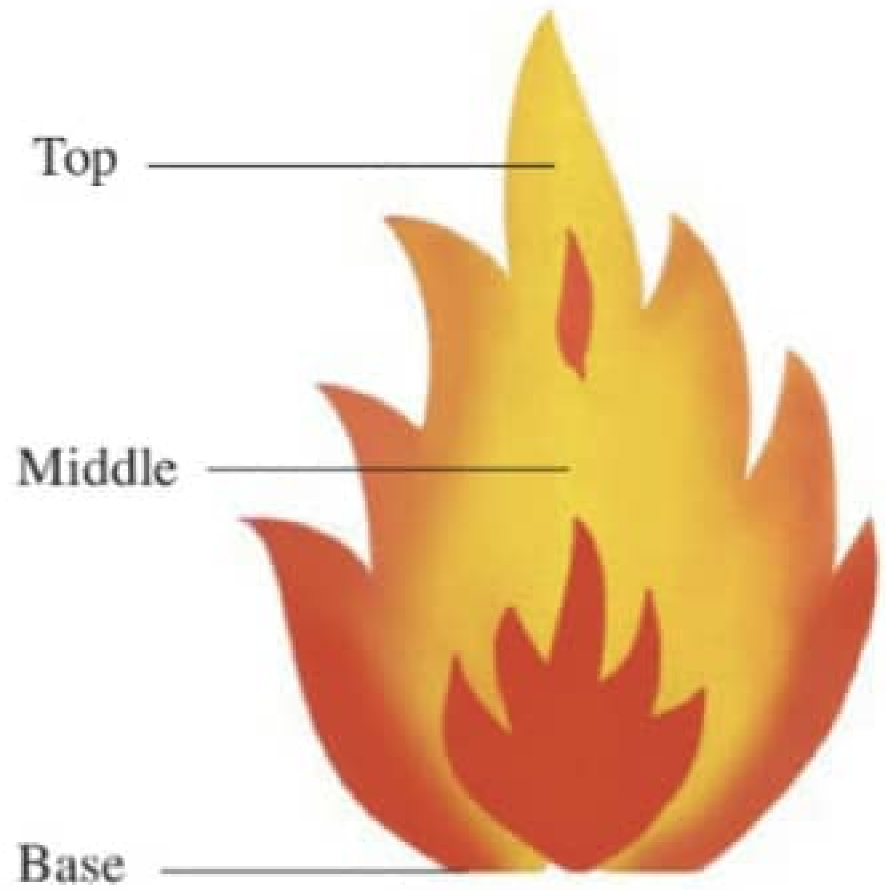


Figure 3.3: *Parts of a flame*

The following are the stages to be followed during putting out fire using a portable fire extinguisher (Figure 3.4).

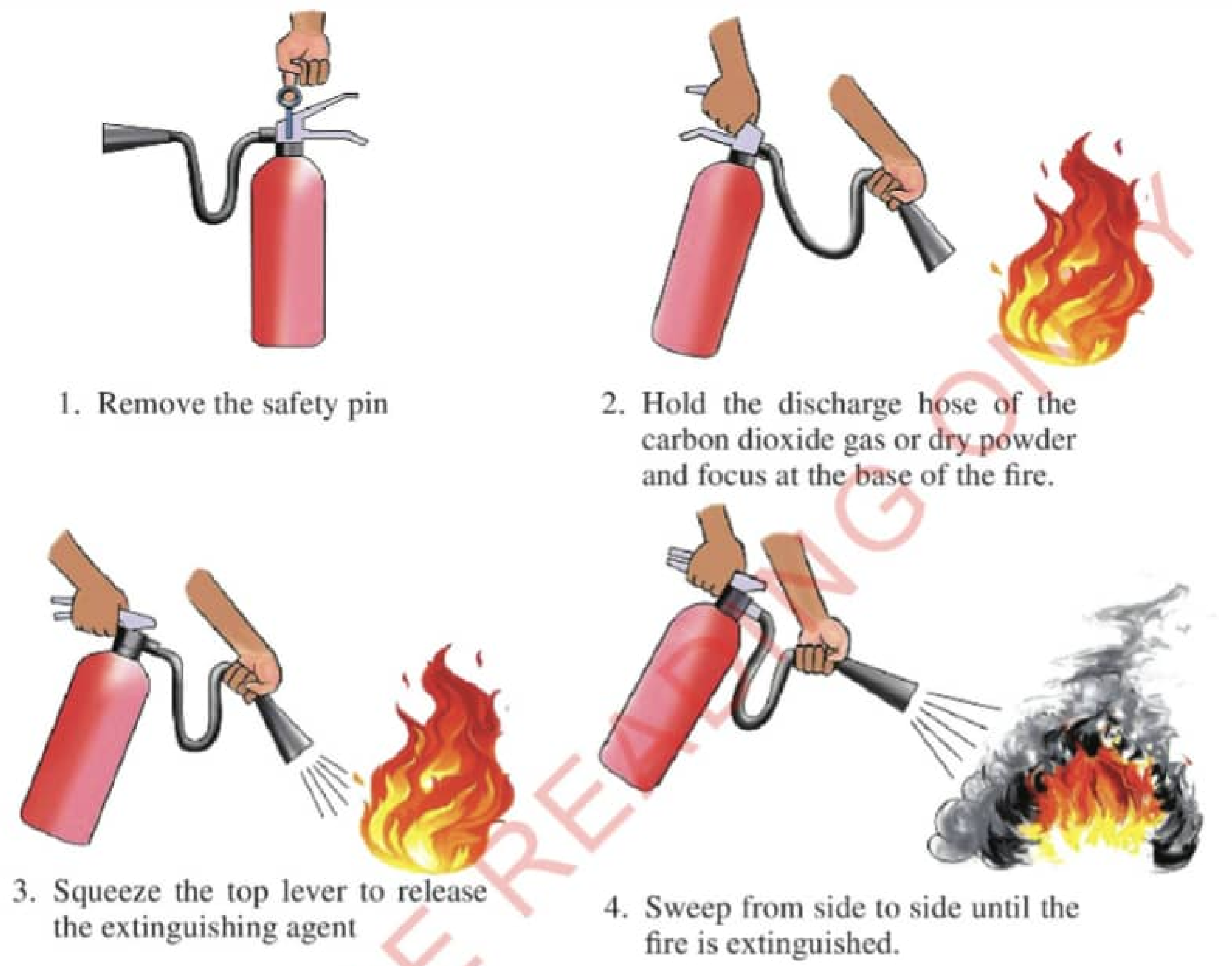


Figure 3.4: *How to use the portable fire extinguisher*

Precautions

The following precautions should be taken when using portable fire extinguishers:

1. Keep a reasonable distance (at least 1.5 metres) from the fire as it may suddenly change direction.
2. For a person on fire, use a fire blanket instead of portable fire extinguisher.
3. Do not test a portable fire extinguisher at your school without permission.
4. Do not return a used portable fire extinguisher to its place.
5. Do not hold the discharge horn when using carbon dioxide type fire extinguisher as it become extremely cold that can lead to severe frost burns.
6. When a fire gets out of control, abandon it and notify the nearest firefighting squad (fire brigade).

Activity 3.1

Aim: To extinguish small fires using the appropriate fire extinguisher.

Requirements: Sand bucket, portable fire extinguishers such as carbon dioxide and wet chemical fire extinguishers, combustible materials such as kerosene, papers and cooking oil

Caution: Fire can cause accidents; therefore, safety measures must be observed.

Procedure

1. Light carefully the combustible material. This should be done in outdoors under the supervision of your teacher or laboratory technician.
2. Use the right fire extinguisher to put out the fire.
3. Repeat the steps 1 and 2 using different combustible materials, one at a time.

Question

Why is it important to use the right type of fire extinguisher to put out a fire?

Exercise 3.1

1. A truck carrying kerosene was involved in an accident and unfortunately fire broke out. With explanation, suggest the fire extinguishers not suitable to use.
2. A Form One Student reads a statement from a certain book, "If the clothes worn by your friend catch fire, cover them with a fire blanket". Why a fire blanket is to be used in that case?
3. With a support of vivid examples, explain how you will educate the community to avoid the causes of fire accidents.

Flames

Task 3.4

Using books from the library and online sources, search for different heat sources that produce flames and write a brief report using the following hints:

1. Types of heat sources commonly used in the laboratory.
2. Types of flames produced by each heat source named in 1.

A flame is a zone or region of burning gases that produces heat and light. It is the visible glowing part of fire. The flame is formed as a result of burning a fuel. The colour and temperature of the flame depends on the type of the fuel used, source of the flame and the oxygen supply. Flames can be classified depending on the light they produce. Based on this, there are two main types of flames; *luminous* and *non-luminous* flames. The luminous flame is bright and yellow in colour (Figure 3.5). Although the luminous flame requires oxygen when burning, the oxygen supply is usually not enough to completely burn up the fuel. For this reason, it produces a black substance known as *soot*. The luminous flame is called a safety flame because it is easily seen, therefore, is less likely to cause accidents.

Figure 3.5: *Luminous flame*

The non-luminous flame is blue in colour (Figure 3.6). This is because there is a sufficient supply of oxygen and so the fuel burns completely. Such a flame does not produce any soot. It produces more heat than the luminous flame and it is noisy. The non-luminous flame can easily cause accidents in the laboratory since one may not be aware that the burner is on.

Figure 3.6: *Non-luminous flame*

Different heat sources produce different flames (Figure 3.7). For example, a candle and a spirit lamp produce luminous flames, whereas a gas stove produces a non-luminous flame. However, some of the heat sources can be regulated to produce both types of flames. For instance, a kerosene stove, gas stove and Bunsen burner **can be regulated to** produce either luminous or non-luminous flame.

Candle

Spirit burner

Gas stove

Figure 3.7: *Different flames produced by different heat sources*

Activity 3.2

Aim: To produce different types of flames from different heat sources.

Requirements: **Lighter, various heat sources such as charcoal burner, candle, hurricane lamp, kerosene stove, spirit burner, gas stove, tin lamp and the Bunsen burner**

Procedure

1. **Light up one heat source after another under the supervision of the teacher. Care should be taken when handling the heat sources.**
2. Carefully identify the types of flames produced by the different heat sources.
3. **Record your observations in the following table:**

Heat sources	Types of flame

Question

What are the characteristic features of each type of the observed flames?

Bunsen burner

A Bunsen burner is a laboratory heat source consisting of a vertical metal tube connected to a gas source. The burner is named after the German chemist, Robert Bunsen, who invented it in 1855. The Bunsen burner is able to produce a very hot flame of temperatures up to 1000 °C from a mixture of gas and air. The air is allowed into the burner through adjustable holes at the base of the vertical metal tube known as ***barrel or chimney***. The ***collar*** is used for the adjustment. Other parts of the Bunsen burner include the jet, gas inlet and the base. Figure 3.8 shows the parts of the Bunsen burner.

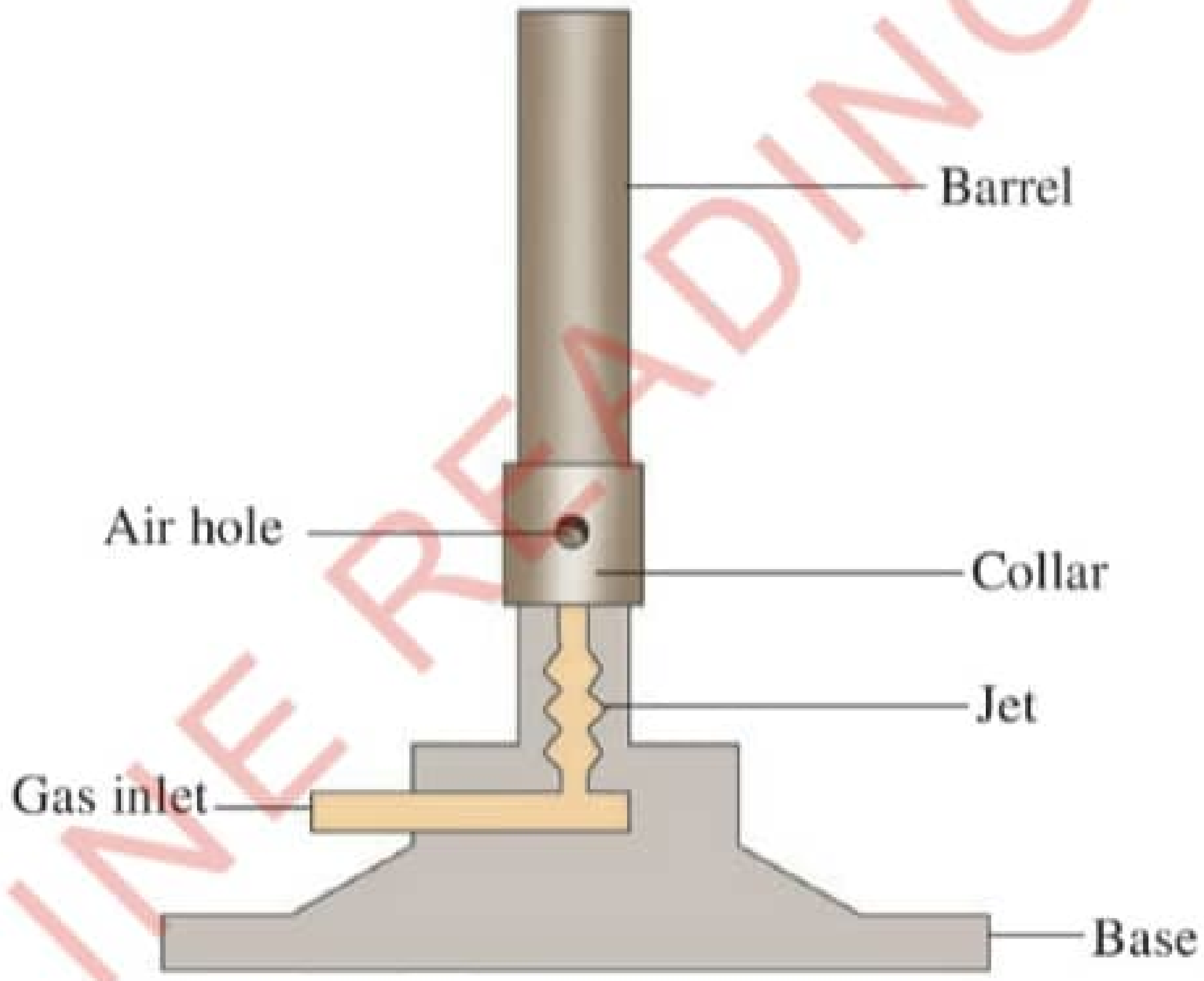


Figure 3.8: ***Parts of the Bunsen burner***

How the Bunsen burner works

Gas enters the burner through a tube connected to a jet inside the base. Air enters the burner through the adjustable air holes. The amount of air coming in can be varied by turning the collar. At the top of the barrel, the mixture of gas and air burns to produce a flame. The type of flame obtained depends on the amount of oxygen available for burning. When the air holes are closed, there is less oxygen supply and the flame is luminous. When they are partly open, there is more oxygen supply and the flame is medium. When the air holes are fully open, there is enough oxygen supply and the flame is non-luminous. Figure 3.9 shows the two types of flames obtained after adjustment of the air holes of the Bunsen burner.

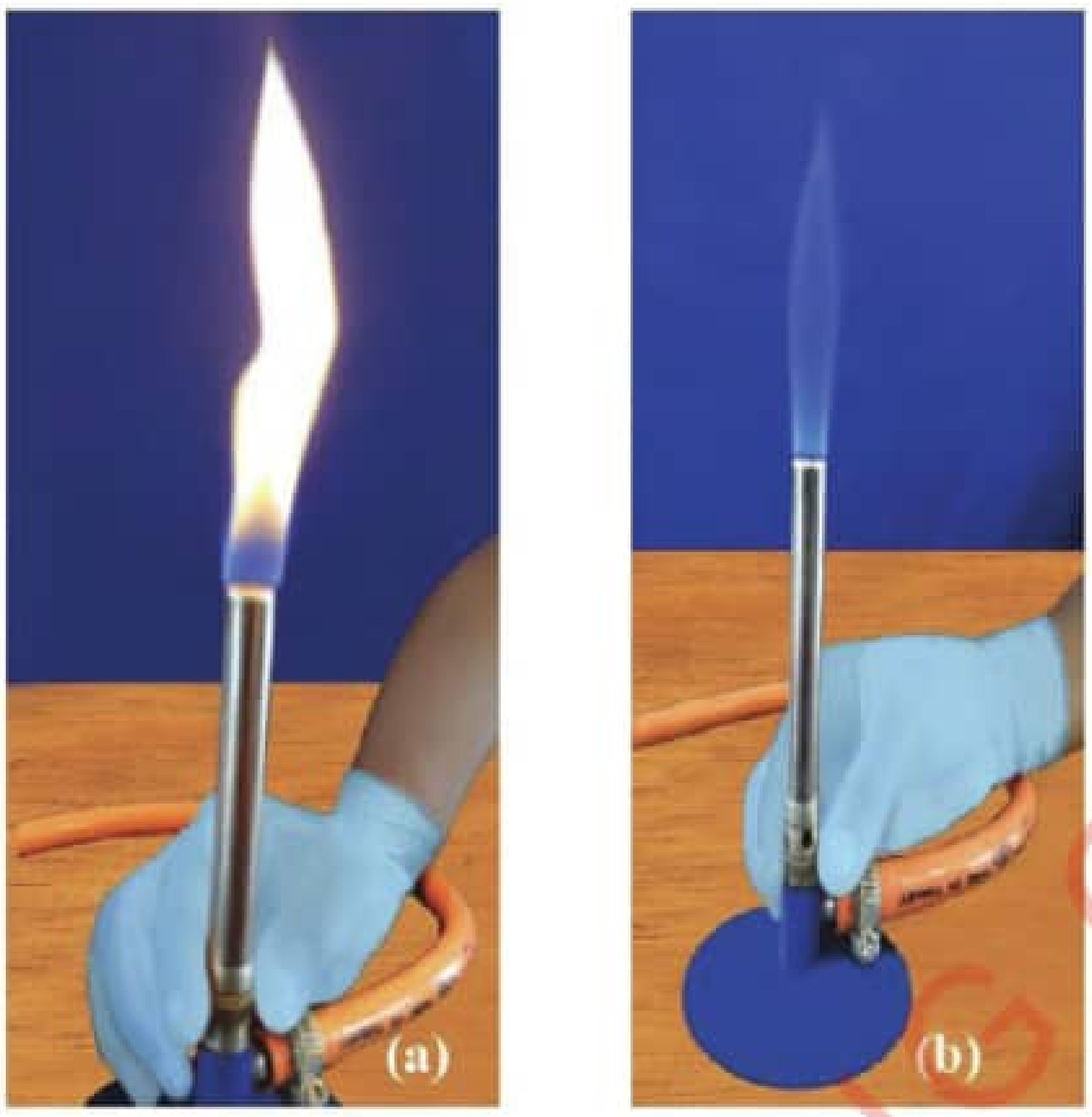


Figure 3.9: *Types of flames when the air holes are (a) closed (b) opened*

Activity 3.3

Aim: To light a Bunsen burner.

Requirements: Matchbox or lighter, wooden splint and Bunsen burner

Caution: Keep your face away from the tip of the barrel when you light the Bunsen burner. You should also not allow gas to come out for long time without lighting the burner as it can lead to serious fire accident.

Procedure

1. Close the air holes of the burner.
2. Connect the burner to the gas tubing.
3. Light a match or wooden splint and hold it on top of the barrel.
4. Slowly turn on the gas. The gas will light up if the tap is connected properly to the tubing. Ask for your teacher's assistance if it does not light up.
5. Open the air holes halfway and then fully. Record your observations.

Questions

1. What happened to the flame when you opened the air holes halfway?
2. What happened to the flame when the air holes were fully opened?

Note: Burning back or striking back is an event that occurs when a gas burns at the jet. It occurs when lighting the Bunsen burner while the air holes are open. This can be corrected by turning off the gas supply, then relight the burner.

Parts of flames

Both luminous and non-luminous flames have different parts. The luminous flame consists of four parts, namely thin outer zone, yellow middle zone, inner unburnt zone and blue outer zone as shown in Figure 3.10.

Figure 3.10: *Parts of a luminous flame*

The non-luminous flame has three parts; the colourless inner zone, blue-green middle zone and the pale purple-blue outer zone as shown in Figure 3.11. Inner zone is the region of unburnt gas. The blue-green middle zone is the region where part of the gas burns because there is no enough air to burn all the gas completely. The hottest part is at the tip of this region. Complete burning of the gas occurs in the pale purple-blue zone.

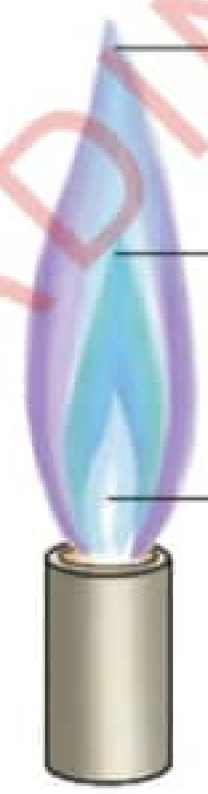


Figure 3.11: *Parts of a non-luminous flame*

Characteristics of flames

Heat sources produce either a luminous flame or non-luminous flame. These flames have different characteristics.

Activity 3.4

Aim: To identify the characteristics of luminous flame and non-luminous flame.

Requirements: Matches box or lighter, Bunsen burner, gas supply, candle, spirit burner and gas stove

Procedure

1. Light each heat source given one at a time using a matchbox or a lighter.

2. Observe the colour, size, shape and other characteristics of the flames produced by each heat source.

3. Fill Table 3.3 by putting a tick (√) to show where each characteristic belongs. The first one has been done for you as an example.

Table 3.3: *Characteristics of flames of a Bunsen burner*

S/N	Properties of the flames	Luminous flame	Non-luminous flame
1.	Bright yellow in colour	√	
2.	Light blue in colour		
3.	Produces soot		
4.	Steady		
5.	Noisy		
6.	Very hot (produces more heat)		
7.	Produces more light		
8.	Has unlimited supply to oxygen		
9.	Wavy and brightly visible		
10.	Formed when the air-hole is fully open		
11.	Has three visible zones		
12.	Has four visible zones		

Question

If you were to choose the type of flame to be used in your day to day cooking, which one will you choose? Give reasons.

Uses of different types of flames

Activity 3.5

Aim: To determine the type of flame that can be used for heating in the laboratory.

Requirements: Matches box or lighter, safety goggles, Bunsen burner, beakers, stopwatch, water, wire gauze and tripod stand

Caution: Before starting the activity, make sure you wear safety goggles and know where the nearby fire extinguishers are located in case of fire accident.

Procedure

1. Set two beakers of equal amounts of water on Bunsen burners whereby the first beaker is heated by a luminous flame and the second beaker is heated by a non-luminous flame.
2. Start a stop watch immediately after setting the two beakers on the heat sources.
3. Note the time when water starts boiling in each beaker.

Questions

1. Which beaker started boiling first?
2. Based on the activity performed, what type of flame is suitable for heating in the laboratory? Explain.

Uses of luminous flame

Luminous flame is mainly used for lighting because it is bright. It is also not very hot, therefore, it is safer for lighting than the non-luminous flame. Some heat sources such as tin lamp and hurricane lamp (Figure 3.12) produce luminous flame which is mainly used as a source of light and not for the purpose of heating or cooking.

Hurricane lamp Tin lamp

Figure 3.12: *Luminous flames from different heat sources*

Uses of non-luminous flame

The following are some of the ways in which the non-luminous flame is used:

1. The flame is used for heating purposes because it gives a lot of heat. Most heating in the laboratory is done using a non-luminous flame. A liquefied petroleum gas (LPG) burner may at times be used in the laboratory since it produces non-luminous flame.
2. The non-luminous flame is used in the flame test of certain chemical substances. The flame test involves introducing a sample of the desired substance to the non-luminous flame and observing the colour change. The flame colour can indicate what components the substance is made of. For example, sodium produces yellow flame and calcium produces brick red flame.

3. Non-luminous flame is suitable for welding (Figure 3.13) because it is very hot. Welding is the joining together of metal pieces or parts by heating their surfaces and pressing them together.

Figure 3.13: *Welding of a car bonnet*

4. A non-luminous flame is suitable for cooking (Figure 3.14) because it gives enough heat and it does not produce soot.

Figure 3.14: *A person cooking using a non-luminous flame*

Exercise 3.2

1. Why is the Bunsen burner commonly used in the laboratory as a source of heat?
2. Luminous flame is generally not useful in the laboratory. Give reasons.
3. Give a brief description of important safety measures that must be adhered to before lighting a Bunsen burner in a laboratory.

4. How does the choice of a flame type on a stove affect the cooking process and the lifespan of cooking pots?
5. If you encounter a problem with a Bunsen burner during an experiment such as difficult in lighting, what steps would you take to address the problem?

Chapter summary

1. The components needed to start fire are fuel, oxygen and heat.
2. Fire is the state of combustion in which ignited materials combine with oxygen to give off light, heat, flame and combustion products.
3. Firefighting is the extinguishing of hazardous/harmful fires.
4. A Bunsen burner is the most commonly used source of heat in the laboratory.
5. A flame is a zone or region of burning gases that produces heat and light.
6. There are two main types of flames, the luminous flame and non-luminous flame.
7. The luminous flame is yellow in colour, produces soot and is less hot than non-luminous flame.
8. The non-luminous flame is blue in colour, does not produce soot, and gives more heat.

Revision exercise 3

1. The following are the steps to follow in lighting the Bunsen burner. However, the steps are not in the correct order. Re-write them in the correct sequence.
   (a) To extinguish the flame, turn off the gas tap to stop the gas flow.
   (b) Light the gas at the top of the barrel with a lighted match stick.
   (c) Turn the collar to close the air hole completely.
   (d) Keep your face away from the top of the barrel.
   (e) Adjust the gas tap until the supply of gas is enough for a flame.
   (f) Turn on the gas fully to ensure that plenty of the gas enters the burner.
2. Write TRUE for a correct statement and FALSE for an incorrect statement. Give a reason for your answers.
   (a) A yellow flame is obtained when the air holes of the Bunsen burner are open.

(b) When the air holes are completely closed, a Bunsen burner produces a non-luminous flame.

(c) The safety flame of a Bunsen burner is the hottest part of the flame.

(d) A quiet flame is obtained when the air holes of the Bunsen burner are completely open.

(e) The yellow colour of a luminous flame is due to the presence of hot carbon particles.

(f) A non-luminous flame looks similar to a candle flame.

(g) Controlling the fuel gas flow allows one to change the flame colour of a gas burner.

(h) The dark zone of a luminous flame consists of unburnt gas.

(i) The hottest part of a non-luminous flame is just over the tip of the dark zone.

(j) Understanding fire triangle is fundamental in fire safety.

3. Students went to camp in an isolated area where they needed to make tea using a portable stove. Which type of flame would be selected, and what are the practical reasons behind their choice?

4. What determine the choice of an appropriate firefighting equipment in the event of fire accidents?

5. What are some common problems that could arise when using a Bunsen burner, and what solutions would you recommend?

6. Why is it not advised to use water for extinguishing Class B fire and the fire caused by electrical faults?

7. A student was preparing food for the family by a deep-frying method. Accidentally, the cooking pan tipped over and a huge fire spread on the kitchen floor.

   (a) Identify the fuel in this fire accident.

   (b) Which fire extinguishers would be suitable for putting out the fire? Explain.

8. If a gas cooker is involved in a fire accident, what steps can be taken to extinguish the fire safely?

Chapter Four

Matter

Introduction

The entire universe is composed of matter. Matter is the substance that makes up everything in our environment, including ourselves, water, air, metals, plants and animals. An understanding of matter creates the way for understanding its nature and how it can be used. In this chapter, you will learn about the states of matter, as well as physical and chemical changes of matter. The competencies developed will enable you to use the changes of states of matter to get solutions to daily challenges in life.

Think

Various processes in daily life that involve changes in states of matter.

States of matter

Task 4.1

1. Watch videos or simulations showing how particles are arranged in different states of matter.
2. Relate the arrangement of the particles observed with the properties of matter found in your environment.

Matter can exist in different states such as solid, liquid and gas. Plasma is often called the fourth state of matter distinct from the more commonly encountered solid, liquid and gas state. The mentioned states are referred to as fundamental states of matter because they are observable in everyday life and are defined by their volume, shape and the general properties of the substances. The state of matter depends largely on the changes in temperature and pressure. It is important to note that there are other states of matter that exist under extreme conditions, but not commonly observed in everyday life.

The solid, liquid and gas states are composed of particles namely *atoms* whereas a plasma state is composed of charged particles. The liquids, gases and plasma states of matter are classified as *fluids* because their particles have ability to flow due to their constant random motion.

Solids

A solid is a substance that has a definite shape and volume. It is usually hard and not easily deformed. Examples of solids include stones, firewood, cooking pots, pencils, chalks, sand and charcoal. Figure 4.1 shows examples of solids.

Figure 4.1: *Examples of solids*

Liquids

Liquid substances flow easily and have specific (fixed) volume. A liquid takes the shape of the container holding it. Examples of liquids include juice, milk, and water which are shown in Figure 4.2.

Figure 4.2: *Examples of liquids*

Gases

Gases have no specific shape or volume. They easily flow and expand indefinitely to occupy the space in which they are held. Examples of gases are oxygen, hydrogen, carbon dioxide and chlorine. Figure 4.3 shows balloons filled with gases.

Figure 4.3: *Balloons filled with gases (air)*

Plasma

Plasma is a state of matter which is not commonly encountered in our daily activities like other states of matter. This state occurs when a gas is extremely heated to form a mixture of positively *charged ions* and free *electrons*. Plasma is a charged gas that is less *dense* than solids or liquids and lacks the fixed shape and volume. It occurs naturally in a variety of phenomena, such as stars and lightning (Figure 4.4), in industrial processes and manufactured products like plasma televisions (TVs).

Figure 4.4: *Lightning*

Activity 4.1

Aim: To investigate the properties of matter.

Requirements: Beakers, analytical balance, stones, pieces of chalk, pieces of wood, metals, papers, balloons and liquids such as water, milk and ink

Procedure

1. Collect different items like stones, pieces of chalk, papers, water, milk, ink and balloons.

2. Carefully observe all the objects you have collected. Use your sense of sight to make your observations on the following:
   (a) Size
   (b) Shape
3. Take two beakers, label them as A and B, weigh each beaker and record its mass.
4. Fill Beaker A with water and re-weigh it. Record the mass and the volume of water.
5. Fill Beaker B with milk and re-weigh it. Record the mass and volume of milk.
6. Weigh the solids, one at a time and record the mass of each solid.
7. Take the solids, one at a time and dip them into the beaker filled with water.
8. Record the volume and mass of water displaced.
9. Weigh a balloon that is not inflated and record its mass.
10. Blow air into the balloon and record its mass.

Questions

1. Do all the substances have mass? Explain.
2. Give reasons why the solids displace some water in the beaker.

Exercise 4.1

1. What are the conditions for a substance to be called matter?
2. Why solids are not classified as fluid?
3. Why do gases expand more than solids for the same increase in temperature?

Changes in states of matter

Matter may change from one state to another. For example, when heated some solids melt to become liquids, while liquids change into gases (vapour) and vapour changes to plasma under high temperature. In some cases, the reverse changes take place. The plasma may undergo the process of cooling down to form recombined gas. The gas may be cooled or condensed to form liquid. The liquid may be cooled further or frozen to form a solid.

A substance that exists in the four states of matter is water. For example, surface water at a temperature below 0 °C exists as solid (ice). Ice changes into liquid (water) as the temperature rises, at 100 °C the water changes to vapour. As the temperature rises extremely, the vapour becomes ionised, forming plasma. Figure 4.5 shows the three states of water.

Figure 4.5: Three states of water

Melting and freezing

Melting is a process in which a substance changes from a solid to a liquid. When a solid is heated, the particles gain energy and vibrate fast. Eventually, they break free from their fixed positions and begin to move in clusters. The temperature at which a solid change to liquid at standard atmospheric pressure is called melting point. The temperature at the melting point remains constant until the entire solid has changed to a liquid. The melting point of a solid explains how strongly its particles are held together. Substances with high melting points have strong forces of attraction between their particles. Those with low melting points have weak forces of attraction between their particles.

Freezing is the process in which a substance changes from a liquid to a solid. The temperature at which a liquid change to solid is called freezing point. The freezing point of pure water is 0 °C, which is also the melting point of ice. Freezing point is slightly affected by changes in atmospheric pressure of a place.

Vapourisation and condensation

Vapourisation is the conversion of a substance from a liquid to a vapour. When the liquid is heated, the particles move faster as their average kinetic energy increases. Some of the particles at the surface of the liquid gain enough kinetic energy and escape into the air. As more of the liquid particles escape to form a gas, the liquid is said to evaporate. As the heating continue, all the particles will have enough kinetic energy to escape. The temperature at which the vapour pressure of the liquid is equal to the atmospheric pressure is called the boiling point. The boiling point describes how strongly the particles are held together in the liquid. Liquids with high boiling points have stronger forces of attraction between their particles than liquids with low boiling points. The boiling point of a liquid is affected by atmospheric pressure. For example, at low pressures, water may vapourise even at room temperature. When a vapour is cooled, the average kinetic energy of the particles decreases and the particles come closer. The forces of attraction between the particles become significant and cause the vapour to condense into a liquid. This process is called condensation.

Ionisation and deionisation

These are the processes involved in the formation and deformation of plasma. When a gaseous state of matter is subjected to a very high temperature, the atoms that form gases gain more kinetic energy, collide with each other and hence lose their electrons to become free ions. The distances between the ions are more far apart compared to that of gases and the resulting fluid is called plasma. On the other hand, deionisation refers to the recombination of free ions and electrons of the plasma to form atoms that eventually combine to form gas molecules. This occurs when temperature decreases in the system leading to the free ions losing their kinetic energies. This decreases their collisions, allowing them to gain electrons to form atoms, which eventually leads to the recombination of gas.

Sublimation and deposition

Sublimation occurs when a solid is heated and changes directly into a gas. In sublimation, a solid does not pass through the liquid state, instead it changes directly into a gas. The direct change of a gas to a solid is called deposition. In sublimation and deposition, the liquid state is bypassed. Figure 4.6 shows the processes involved in the inter-conversion of the states of matter.

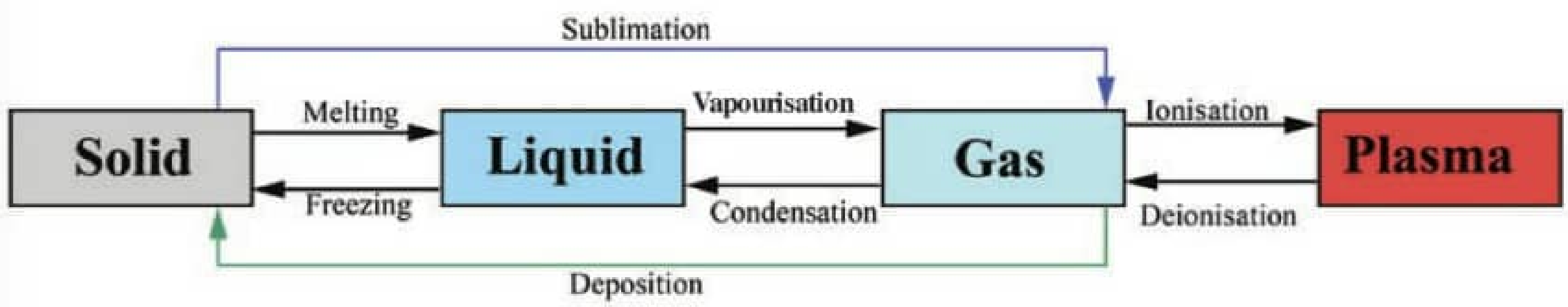


Figure 4.6: Inter-conversion of the states of matter

Activity 4.2

Aim: To investigate the changes of matter from one state to another.

Requirements: Watch glass, source of heat, beaker and ice cubes

Procedure

1. Put some ice cubes in a beaker and heat. Carefully observe the changes that take place.
2. Heat the substance further until it starts to change into vapour.
3. Cover the beaker with a watch glass.
4. Carefully remove the watch glass and record your observations on the surface of the watch glass.

Questions

1. What was the shape of the ice before heating?
2. What was the shape of the substance formed after heating the ice?
3. What happened when water was heated above its boiling point?
4. What happened after covering the heated substance in a beaker with the watch glass?

Importance of the changes in states of matter

Task 4.2

Watch videos illustrating the importance of the changes in states of matter and write a summary of what you have learnt.

The importance of changes of matter from one state to another can be explained in terms of the following:

Water cycle

Water cycle describes a continuous process that drives the movements of water on, above, and below the surface of the Earth. The ability of water to change from liquid to gas (vapour), from vapour to clouds and finally to rain is important to climatic conditions. Rain is formed through this process.

Refrigeration

Refrigeration involves the use of refrigerants. The refrigerants are chemicals found in refrigerators whose change in state involve the changes in energy. The changes of state from liquid to vapour absorb energy from the surroundings, and thus, cause the cooling effect. Air conditioners work in the same principle.

Refinery

Petroleum refinery and distillery work under the principle that liquids can change to vapour and then the vapour can be cooled to liquids. While simple distillation is employed in distilleries, fractional distillation is employed in petroleum refineries.

Metallurgy

Metallurgy involves the extraction and purification of metals from their ores and the manufacture of alloys. The changes of state from solid to liquid and back to solid make metallurgy possible.

Steam engines

A steam engine uses steam as the working fluid to perform work. Steam engines were used in the early trains and ships. The change of state from liquid water to steam makes steam engines operate.

Drying of materials

Drying of materials occurs when the liquid contained changes to vapour. An example is when wet clothes are exposed to the sun to dry.

Production of electricity

Coal is used to heat water in closed system of pipes into vapour. The produced vapour is led by pipes to drive the turbines that in turn produce electricity.

Exercise 4.2

1. Explain the roles of temperature in the changes of states of matter.
2. Why do cold foods not smell from a distance?
3. Provide real world examples of how the knowledge of changes in states of matter is applied in industries.

Particulate nature of matter

For a long time, matter was thought to be continuous and lack discrete particles. Later, it was discovered that matter is made up of particles. This was proved by a phenomenon known as the Brownian motion (Figure 4.7) which can be observed in liquids and gases.

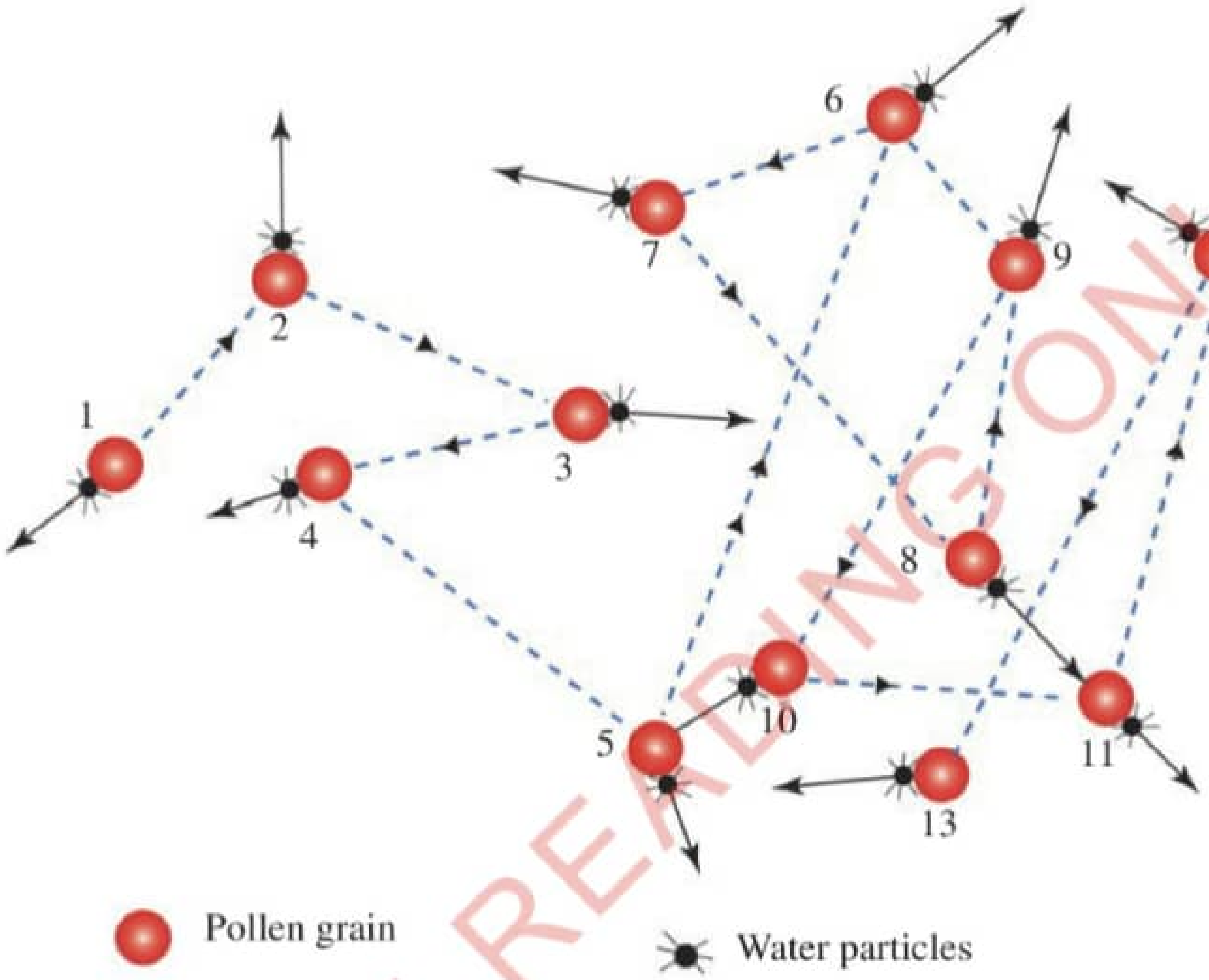


Figure 4.7: Brownian motion

The Brownian motion

In 1827, a botanist called Robert Brown observed through a microscope that pollen grains suspended in water move short distances in an irregular zigzag manner as shown in Figure 4.7. This is because they are constantly bombarded by water particles. This shows that matter is particulate in nature. Other examples which show that matter is made up of particles that are in constant motion include the spread of the smell of the food being cooked from the kitchen to a considerable distance, diffusion of potassium permanganate particles in water, and the spread of the perfume due to diffusion of the perfume vapours into the air. Diffusion is the movement of particles from an area of high concentration to that of low concentration.

Task 4.3

Watch the video clip illustrating the demonstration of Brownian motion and make observation on movement of particles. How does the motion observed relate to the motion of particles in a gas or liquid?

Activity 4.3

Aim: To demonstrate Brownian motion.

Requirements: Pollen grains, distilled water, light microscope, microscope slide and dropper

Procedure

1. Put a drop of distilled water on a microscope slide.
2. Put some pollen grains on top of the drop of water.
3. Place the slide under a microscope and observe any movement of the pollen grains.

Questions

1. Is the movement patterned or random?
2. Why do the pollen grains move in this manner?

Activity 4.4

Aim: To demonstrate that solids are made up of particles.

Requirements: Beakers, pipette, stirring rod, potassium permanganate crystals and water

Procedure

1. Dissolve about 0.5 g of potassium permanganate crystals in 100 cm^3 of water in a beaker. Stir to ensure that potassium permanganate has dissolved completely. Label it Beaker A.
2. Using a pipette, measure about 20 cm^3 of the solution made in Step 1 and pour it into another beaker. Label it Beaker B.
3. Add water to the solution in Step 2 until it gets to the 100 cm^3 mark.
4. Obtain 20 cm^3 of the solution made in Step 3 and put it in a beaker labelled C. Add water to it up to the 100 cm^3 mark.
5. Repeat this process two more times, each time taking 20 cm^3 of the previous made solution and adding water up to the 100 cm^3 mark. Label the beakers D and E, consecutively.
6. Arrange the beakers as shown in Figure 4.8 and observe the changes.

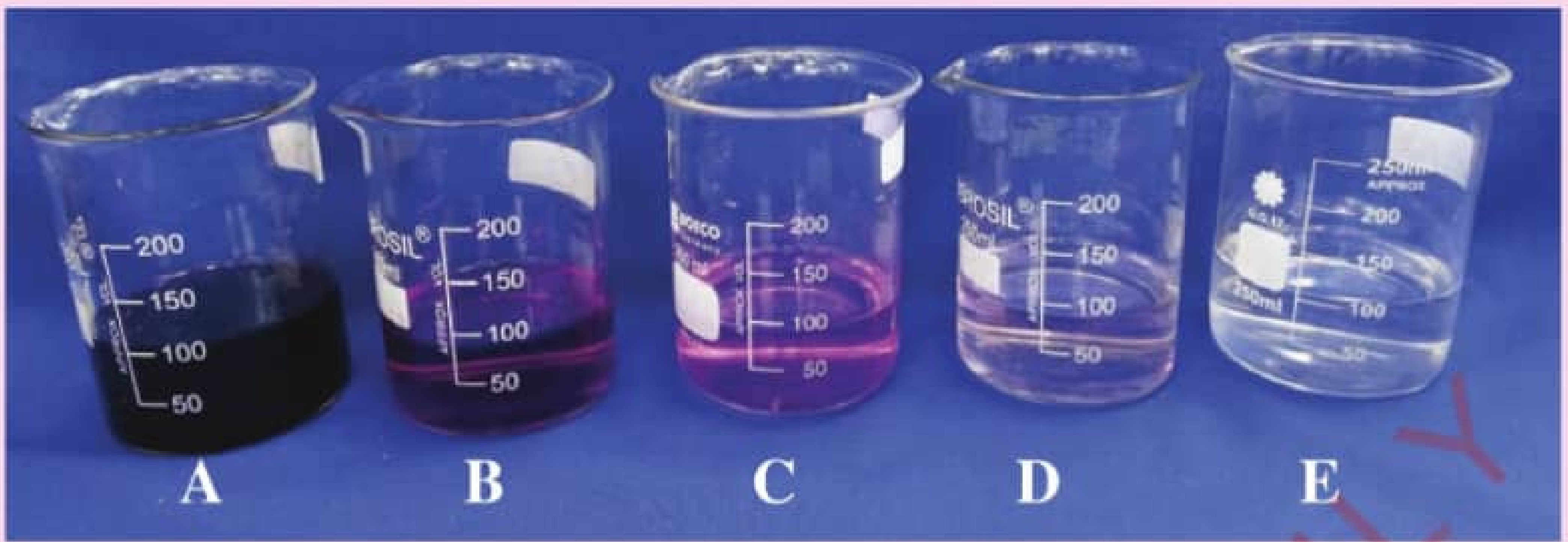


Figure 4.8: *Beakers containing different concentration of potassium permanganate*

Questions

1. What conclusion can you make from this experiment?
2. How does this experiment prove that solids are made up of particles?
3. How is the knowledge obtained from this activity applicable in real life?

Activity 4.5

Aim: To show that liquids are made of particles.

Requirements: Watch glasses, ether and perfume

Procedure

1. Place few drops of ether on a watch glass.
2. Move some distance away from the watch glass containing ether.
3. Use your senses to detect what happens after a short time.
4. Spray a perfume at one corner of the classroom and use your senses to detect what happens at another corner of the classroom.

Questions

1. Why was the smell of ether detected at different points in the classroom?
2. Can you smell the perfume from any corner of the classroom? Explain.

Task 4.4

Watch the simulation demonstrating the kinetic nature of matter. Observe how the particles move. Compare the motion of particles in solid, liquid and gas with their properties.

The particles of matter tend to display different characteristics depending on the physical state of the substance. Solids keep their shapes and are not easily squashed (compressed) or stretched, and so they tend to retain their volume. This is because the particles are packed so closely together that they can only vibrate but not move. The more energy they have, the more they vibrate. Heating can provide this energy, so that at a certain temperature, they vibrate so much that their regular arrangements break down. At this point (the melting point) the solid melts and changes to a liquid (Figure 4.9).

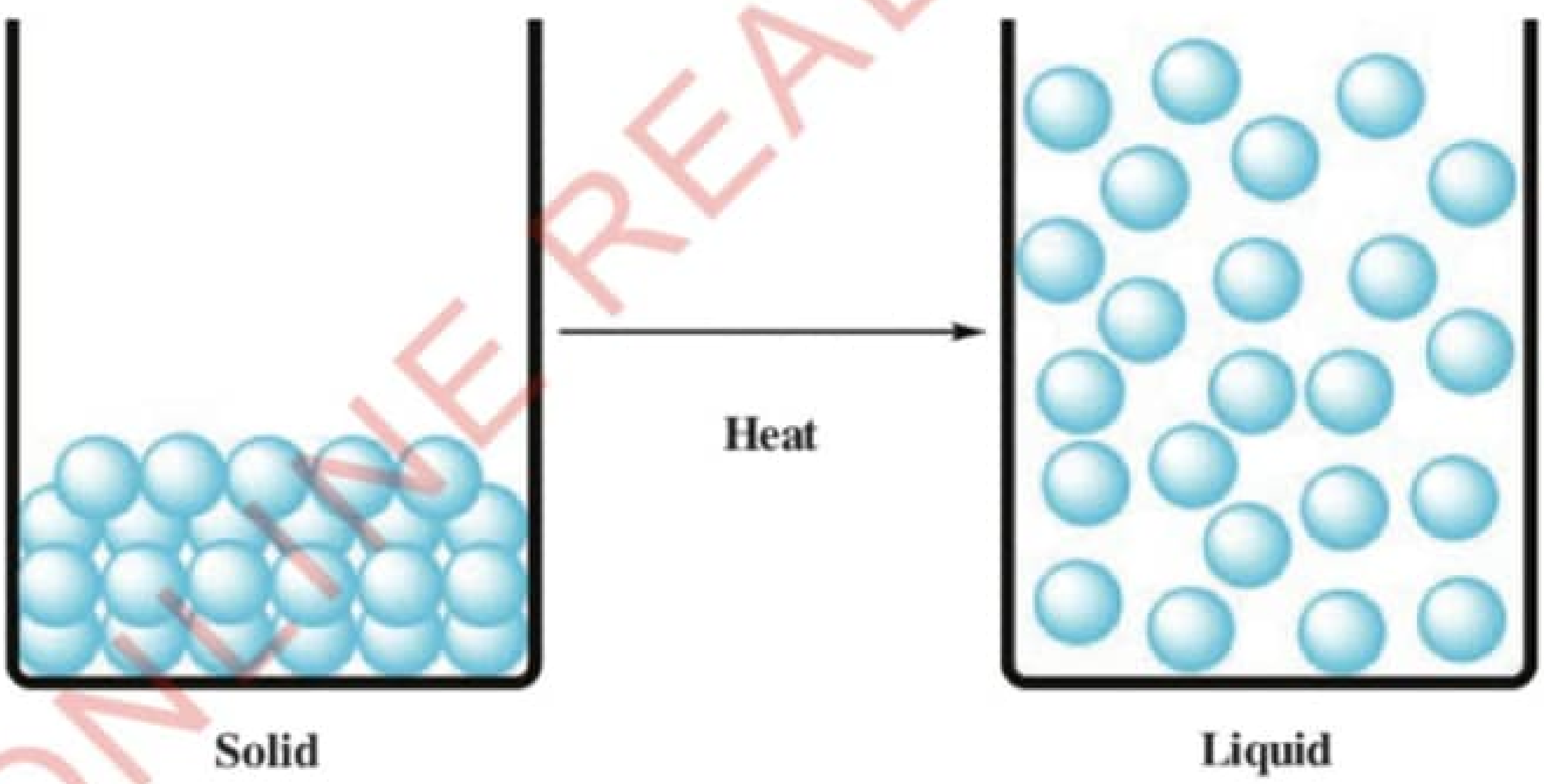


Figure 4.9: *Heating solid into liquid*

Liquids take the shapes of the containers holding it. They cannot be compressed since the particles are still close together but they have no definite order. However, the particles in liquids have some spaces between them and can move around or slide past each other. If energy (heat) is applied to a liquid, the particles move about fast until a temperature called the boiling point is reached. At the boiling point the particles have enough energy and the liquid changes to vapour (gas) (Figure 4.10).

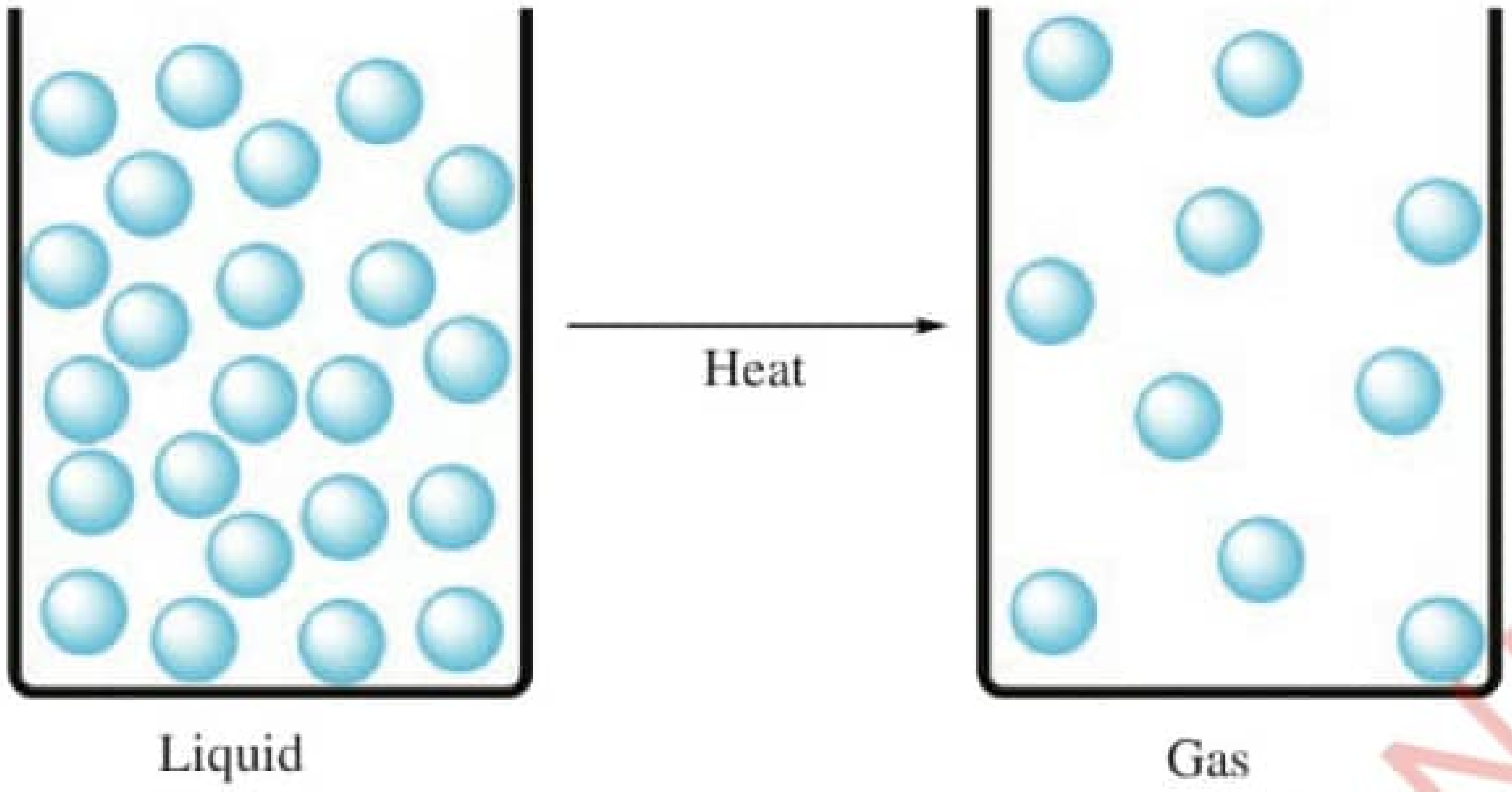


Figure 4.10: *Heating liquid into vapour (gas)*

The particles in the gaseous state are far apart from each other with no regular order. The particles in the plasma state are in charged forms (positive and negative) and scattered. The large spaces between the gas particles make gases compressible. The way particles behave in solids, liquids and gases is called the kinetic molecular behaviour.

Activity 4.6

Aim: To demonstrate the kinetic nature of matter.

Procedure

1. In groups, stay closely together so that your shoulders touch.
2. Can you individually move about freely?
3. Move into an empty classroom and walk slowly around the room while your hand touches on another's shoulder. How far are you able to move around?
4. Go out of your classroom and move around without touching each other. How far are you able to move?

Question

Compare your movements in the three scenarios with the kinetic nature of the three states of matter.

A summary of the different properties of gases, liquids and solids is given in Table 4.1.

Table 4.1: *Different properties of gases, liquids, solids, and plasma.*

Property	Solid	Liquid	Gas	Plasma
Shape	Retains a fixed shape	Takes the shape of its container	Has no definite shape, fills the shape of the container	Has no definite shape, fills the shape of the container
Volume	Has a fixed volume	Has a fixed volume	Takes the volume of its container	Takes the volume of its container
Movement of particles	Particles do not move except	Particles move or slide past	Particles move	Highly energetic,

Rusting

Rusting is a chemical change that occurs on the surface of iron materials and forms a reddish-brown coating. The reddish-brown coating is called *rust* (Figure 4.11)

Figure 4.11: *Rusted iron sheets*

There are three conditions that are necessary for rusting to take place, namely iron, oxygen and water. Therefore, rusting occurs only when iron is exposed to water and air containing oxygen.

Activity 4.8

Aim: To demonstrate conditions necessary for rusting of iron nails.

Requirements: Iron nails, oil, cotton wool, four test tubes, water and anhydrous calcium chloride

Procedure

1. Label four dry test tubes, A, B, C and D.
2. Put two new iron nails in each test tube.
3. Insert some cotton wool in Test tube A and add about 5 cm^3 of tap water to make the cotton wool wet.
4. Add about 10 cm^3 of boiled tap water to test tube B followed by 3 cm^3 of oil.
5. Insert some cotton wool in test tube C and place some crystals of anhydrous calcium chloride on the cotton wool.
6. Do not put anything in test tube D except nails.
7. Leave the set-up as shown in Figure 4.12, open to the air for three days.

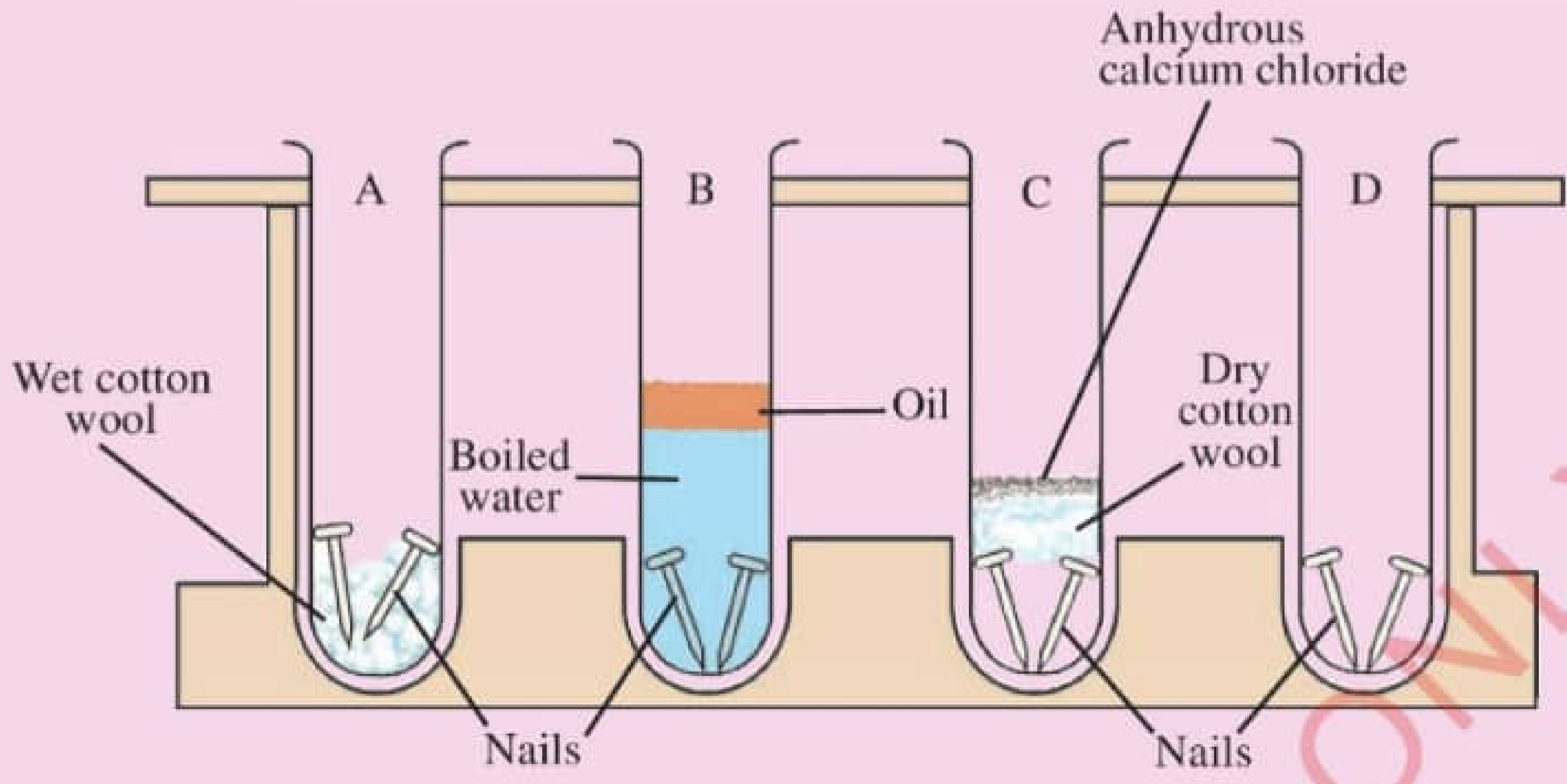


Figure 4.12: *Set up to demonstrate rusting of iron nails*

Questions

1. What are the observations in the test tubes after three days?
2. Why was boiled water and oil added in Test tube B?
3. What was the function of anhydrous calcium chloride in Test tube C?
4. From the results of the experiment, state the conditions necessary for rusting to occur.

Task 4.5

Categorise the following processes as either chemical or physical changes:

(a) Aluminium foil is cut into half
(b) Clay is moulded into a new shape
(c) Wood is burnt
(d) Food scraps are turned into compost
(e) Drying of wet clothes
(f) A match is lit
(g) Digestion of food in the body
(h) Grinding a piece of chalk
(i) Food is cooked
(j) Jewellery tarnishes (changes colour)

The differences between physical changes and chemical changes are described in Table 4.2

Table 4.2: *Differences between physical changes and chemical changes*

Physical changes	Chemical changes
Affect only physical properties of matter	Affect both physical and chemical properties of matter
Are temporary changes	Are permanent changes
Are easily reversible which means the original substance can be recovered	Are irreversible which means the original substance cannot be recovered
No new substance is formed	New substances are formed
The molecules are rearranged while their actual compositions remain the same	The molecular compositions of a substance completely change.
No energy is produced or absorbed	Energy is produced or absorbed

Chapter summary

1. Matter is anything that has mass and occupies space.
2. Matter exists as solid, liquid, gas and occasionally as plasma.
3. A solid is a substance that does not flow easily and retains its shape and size.
4. A liquid is a substance that flows, retains its volume but takes the shape of the container in which it is held.
5. A gas is a substance that has neither definite shape nor size.
6. Plasma is a charged gas that occurs when the gas is extremely heated.
7. Sublimation is the change of state of matter from a solid state directly to a gas or vapour without passing through a liquid state.
8. Matter is made up of particles that are able to display some forms of movements.
9. Diffusion is the movement of particles from an area of high concentration to an area of low concentration.
10. Brownian motion refers to the random movements of particles suspended in a liquid or gas.
11. The kinetic molecular behaviour is the way particles behave in solids, liquids and gases.
12. Physical changes are reversible. They only affect the physical properties of

substances, such as shape and state.

13. Chemical changes are irreversible. They affect the chemical properties of substances.
14. A chemical change results in the production of new substances while a physical change does not.

Revision exercise 4

1. What types of changes are these?
   (a) Cloud changing into rain
   (b) Magnetising of iron
   (c) Heating an iron rod
   (d) Rotting of mangoes
   (e) Decaying of teeth
2. A Form One Student has a balloon filled with a certain gas. If the student puts the balloon in the freezer overnight, what do you think will happen to the balloon? Explain.
3. When a container of coffee is opened in a room, people in different parts of the room may notice its smell. Explain.
4. Describe the role of changes in state of matter in water purification.
5. Explain how the changes in state of matter are linked to the formation of rain cloud in areas of low pressure and moist air.
6. How does temperature affect the state of matter?
7. Give four real-life examples that illustrate the kinetic nature of matter.
8. Explain how physical and chemical changes are involved in cooking process.
9. When a metal such as copper is heated, it expands. Explain what happens to the metal particles during expansion.
10. Explain how physical and chemical changes are involved in steel production and food processing industries.

Chapter Five

Elements, compounds and mixtures

Introduction

Everything in the universe is composed of one or more elements. Elements are the building units of compounds. When compounds combine physically, they form mixtures. In this chapter, you will learn about elements and their chemical symbols, binary compounds and mixtures, and separation of mixtures. The competencies developed will enable you to identify and classify substances which are useful in your daily life activities, regain useful pure substances out of impure substances, and make compounds and mixtures for day-to-day needs.

Think

The building blocks of all materials in the universe.

Elements and chemical symbols

Task 5.1

1. Watch animations showing different elements and write the meaning of element.
2. How are elements commonly written?

A limitless number of words can be written using the letters of the alphabet, which are the building blocks of all the words. Most of buildings are made by different bricks. The bricks are the building blocks of the buildings.

Task 5.2

1. Take a piece of chalk. Note its size.
2. Break the piece of chalk into two halves, note the size of each piece. Compare the size of the pieces to the original chalk.
3. Take the two halves of pieces of chalks and continue breaking them into halves until you can no longer break them.

Questions

1. How does this task help you to understand the concept of element?
2. Why is it difficult to break the pieces of chalk further?

Matter is made up of tiny particles called *atoms* which are the smallest particles of a chemical *elements* that can exist. An element is a pure chemical substance which cannot be split into simpler substances by a simple chemical process. There are 118 known chemical elements, most of which occur naturally, while few are man-made. There may be discoveries of new elements in the future. Common examples of elements are sodium, potassium, magnesium, sulfur, carbon, iron, silver, gold, copper, oxygen, nitrogen and hydrogen. Some common elements are shown in Figure 5.1.

Figure 5.1: *Some common elements*

Names and chemical symbols of elements

Elements have names that are usually represented by letters, commonly known as *chemical symbols*. The chemical symbols are abbreviations or short representations of element names. The use of symbols has made it easier for chemists, other scientists and students in studying chemistry and other sciences. Chemical symbols are often derived from the Latin or Greek names of the elements and may not have much similarity to the common English names. For example, the symbol for sodium is Na which is derived from the Latin name *Natrium* and the symbol for gold is Au which is derived from the Latin name *Aurum*. Chemical symbols are written according to the rules of the International Union of Pure and Applied Chemistry (IUPAC).

Task 5.3

Collect different fruits and identify their common names. Why do they have those names?

Criteria for assigning chemical symbols

The following are the criteria used for assigning chemical symbols:

1. An element may be represented by a chemical symbol that is derived from the first letter of its English name. See examples in Table 5.1.

Table 5.1: *Some elements with chemical symbols derived from the first letter of their English names*

Name	Chemical symbol
Boron	B
Carbon	C
Fluorine	F
Hydrogen	H
Iodine	I
Nitrogen	N
Oxygen	O
Phosphorus	P
Sulfur	S
Vanadium	V
Yttrium	Y

If different elements have the same first letter, for example cobalt, copper and calcium, it is necessary to differentiate the elements. In this case, another letter, usually the second or third from the name is used together with the first one (Table 5.2). The first letter will be capital, while the second will be a small letter.

Table 5.2: *Chemical symbols of some elements with two letters from their English names*

Name	Chemical symbol
Aluminium	Al
Argon	Ar

Name	Chemical symbol
Beryllium	Be
Calcium	Ca
Chlorine	Cl
Cobalt	Co
Helium	He
Lithium	Li
Magnesium	Mg
Manganese	Mn
Neon	Ne
Silicon	Si

In some cases, the chemical symbols are derived from Latin names instead of the common English names. Examples of elements with chemical symbols derived from Latin names are given in Table 5.3.

Table 5.3: Examples of elements with chemical symbols derived from their Latin names

Name	Latin name	Chemical symbol
Antimony	Stibium	Sb
Copper	Cuprum	Cu
Gold	Aurum	Au
Iron	Ferrum	Fe
Lead	Plumbum	Pb
Mercury	Hydrargyrum	Hg
Potassium	Kalium	K
Silver	Argentum	Ag
Sodium	Natrium	Na
Tin	Stannum	Sn
Tungsten	Wolfram	W

Significance of chemical symbols

The use of chemical symbols has made it easier for chemists and other people to:

1. quickly understand the elements being referred to, instead of memorising the full names.
2. write chemical equations in abbreviated forms instead of writing each element in its full name.
3. distinguish one element from the other.

Exercise 5.1

1. Why is it important to know chemical symbols at early stage of chemistry learning?
2. Why are chemical symbols used alongside the common names of elements?
3. What challenges might occur if different countries could use different names for the same element?

Metals and non-metals

Elements can be classified into metals and non-metals. The systematic method to represent and organise chemical elements in a table format is called *Periodic Table*. Metals are mostly found at the left hand side and at the middle of the Periodic Table. Most of the elements in these areas are distinguished by their brilliance, malleability, ductility, and strong thermal and electrical conductivity. Examples of metals are magnesium, aluminium, potassium, calcium, sodium, copper, zinc and iron. Non-metals are located at the upper right-hand portion of the Periodic Table. These elements do not display the properties of metals. Examples of non-metals are sulfur, carbon, chlorine, nitrogen and hydrogen. Most non-metals exist in gaseous state or liquid state with exception of a few such as carbon and sulfur which exist in solid state. Although hydrogen is located at the left hand side of the Periodic Table, it possess the characteristics of non-metals. Figure 5.2 shows the position of metals and non-metals in the Periodic Table.

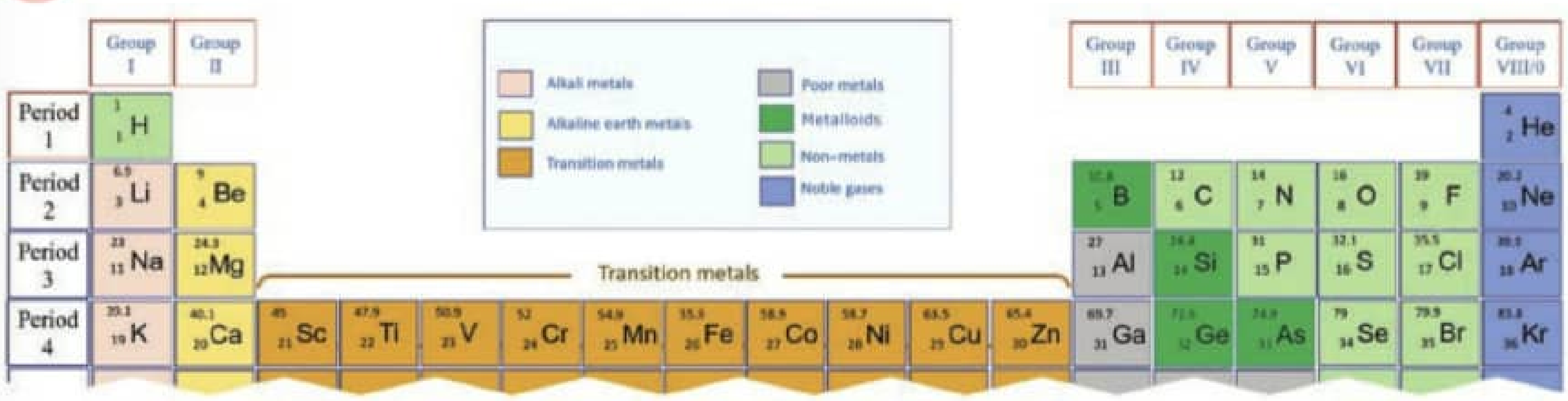


Figure 5.2: *Position of metals and non-metals in the Periodic Table*

Task 5.4

Collect the following materials: aluminium wires, copper wires, iron nails, charcoal, roll sulfur and zinc strips. Describe each material in terms of: (a) appearance (b) malleability (c) tensile strength and (d) sonority

In principle, metals are different from non-metals. The characteristics used to differentiate metals from non-metals are presented in Table 5.4.

Table 5.4: *Characteristics of metals and non-metals*

S/N	Metals	Non-metals
1.	Are good conductors of electricity	Are poor conductors of electricity. They are insulators
2.	Are good conductors of heat. This is why cooking utensils are made of metals like iron and aluminium	Are poor conductors of heat
3.	Have high melting points and boiling points	Have low melting points and boiling points
4.	Have high tensile strength	Have low tensile strength
5.	Are good conductors of sound. They are sonorous. That means they produce a typical metallic sound when hit with something	Are poor conductors of sound. They are not sonorous
6.	Are ductile, which means that they can be drawn into thin wires	Are not ductile. They cannot be drawn into thin wires
7.	Are malleable, which means that they can be hammered into thin sheets	Are not malleable
8.	Are lustrous. They produce shining surfaces when cut	Are non-lustrous except a few such as diamond

Note: There are exceptions to the above generalisations for some metals and non-metals. Sodium, potassium and mercury have different characteristics from those described in Table 5.4. These metals are light with very low melting points and boiling points. While most metals are solid at room temperature, mercury is usually in liquid form. Diamond as a non-metal has very high melting point and boiling point whereas graphite which is also a non-metal is a good conductor of heat and electricity.

Exercise 5.2

1. Give three examples of metals and non-metals, and explain their uses in daily life.
2. Why are cooking pots made of metals?
3. Describe properties of metals and explain how do they make metals useful in various applications?

Compounds and mixtures

A **compound** is a pure substance made up of two or more elements in a chemical combination. When a compound is made up of two components is referred to as a **binary compound**. Examples of binary compounds include carbon dioxide gas, common salt and water. When a binary compound is formed, two or more pure substances of the same or different types chemically combine. The combination of the components to make a compound are always in fixed ratios. For example, carbon dioxide is a compound which is made up of one atom of carbon and two atoms of oxygen. The number of the combining components will be fixed regardless the amount of carbon dioxide you have. More examples of the substances that make binary compounds are shown in Table 5.5.

Table 5.5: Examples of substances that combine to make binary compounds

S/N	First part	Second part	Compound
1.	Sulfur	Oxygen	Sulfur dioxide
2.	Sodium	Chlorine	Sodium chloride (common salt)
3.	Hydrogen	Oxygen	Water
4.	Nitrogen	Hydrogen	Ammonia
5.	Zinc	Sulfur	Zinc sulfide

Properties of binary compounds

1. The components in it have different properties from those of the resulting compound. For example, sodium chloride is a white solid compound made from a greenish yellow chlorine gas with a silvery soft sodium metal.
2. Whenever a compound is formed the substances in it combine in definite proportions.
3. A compound is formed only by a chemical reaction. Energy is taken in or given out in the form of heat and light when a compound is formed.

4. The components of a compound can only be separated by chemical means. Physical means can never separate a compound into its elements. Table 5.6 present some of the differences between elements and compounds.

Table 5.6: *Differences between elements and compounds*

S/N	Elements	Compounds
1.	Cannot be separated into simpler substances	The components can be separated by chemical means
2.	They are made up of similar atoms of the same type	Can be made up of two or more substances that are naturally different
3.	They maintain their characteristics when are in mixtures	Have different characteristics from their components when they are made
4.	Contain unique atomic numbers	Have varying total atomic numbers depending on their components
5.	Symbols represent the elements	Chemical formula represents a compound

Despite the fact that compounds differ with elements as shown in Table 5.6, there are some similarities among them. Compounds relate to elements in that they are both purely made from their constituents as homogenous substances and cannot be separated by physical means. Compounds also are similar to elements in that they are both made from atoms as their simplest building blocks and their components are mostly combined in fixed ratios.

A mixture is a physical combination of two or more substances in any ratio. Since mixtures are not chemically combined, they can be separated by physical means. The mixtures can be liquid-liquid (for example, oil and water), solid-liquid (for example, muddy water) or solid-solid (for example sand and salt). Mixtures can be homogeneous or heterogeneous. A homogeneous mixture has uniform composition, appearance and properties. An example is when a small amount of common salt is dissolved in a glass of water. After stirring, every section of the solution is identical in composition, appearance and physical properties. A heterogeneous mixture has different compositions, appearance and properties at various points in the mixture. For example, when sulfur powder and iron filings are mixed, they form a heterogeneous mixture. The mixture is physically combined and can be separated using a magnet bar as shown in Figure 5.3.

Figure 5.3: *Magnetic separation of a mixture of sulfur and iron filings*

Task 5.5

You are provided with the following substances: Common salt, mud, juices, milk, water and soft drinks. Classify the substances as either compounds or mixtures and give reasons for your responses.

Activity 5.1

Aim: To compare the properties of a mixture and a compound.

Requirements: Test tubes, stoppers, test tube rack, mortar and pestle, burner, retort stand and clamp, spatula, sheets of paper, powdered sulfur, magnet, coarse and fine powdered iron filings, dilute hydrochloric acid and an analytical balance

Caution: The gas being produced is poisonous in high concentrations, smell it by wafting it towards your nose.

Procedure

1. Take a spatulaful of powdered sulfur and mix it with a spatulaful of coarse powdered iron in a clean and dry test tube. Stopper the test tube and shake it well.
2. Take out half of the mixture and put it on a sheet of paper.
3. Place a magnet near the mixture and observe what happens. Record your observations.
4. Pour dilute hydrochloric acid in the test tube containing the remaining mixture. Record your observations.

5. Take 8 g of sulfur and 14 g of fine powdered iron filings and put them in a clean and dry test tube. Put a stopper onto it and shake the test tube to mix the two substances thoroughly.
6. Take out half of the mixture, put it in another test tube and add dilute hydrochloric acid. Record your observations.
7. Attach the test tube containing the remaining mixture to a clamp on a retort stand and put it in a fume chamber. Heat the mixture with the burner until it starts to glow. Record your observations. Remove the heat and allow the contents in the test tube to cool.
8. Crack the sample in the test tube and put the contents on a sheet of paper. Make sure not to break the test tube. Place a magnet near the compound and record your observations.
9. Put the compound in a mortar. Grind the contents to fine powder using a pestle. Divide the contents into two different test tubes.
10. Add some water to the first test tube and shake vigorously. Wait for two minutes and then record your observations.
11. Place the second test tube in the fume chamber. Add dilute hydrochloric acid until the test tube is half full.

Questions

1. Describe what takes place when a magnet is placed near a mixture of sulfur and iron.
2. What happens when dilute hydrochloric acid is added to a test tube containing a mixture of iron and sulfur?
3. What happens when a magnet is placed near the cooled product formed when the mixture of iron and sulfur is heated? Explain your answer.
4. What type of change takes place when dilute hydrochloric acid is added to the product formed when the mixture of iron and sulfur is heated?

Mixtures and compounds have different properties. Table 5.7 shows some of the differences between mixtures and compounds.

Table 5.7: *Differences between mixtures and compounds*

S/N	Mixture	Compounds
1.	The components can be separated from one another by physical methods	The constituent elements cannot be separated by physical methods but can be separated by chemical methods
2.	Mixtures may vary widely in compositions. The components are mixed in any proportions	Compounds have always definite/ fixed compositions by mass of the elements. The proportions are fixed
3.	No chemical change occurs when mixtures are formed	Chemical changes are involved when compounds are formed
4.	The properties of mixtures are those of the individual components	The properties of compounds are very different from those of the individual elements

Exercise 5.3

1. A chemist categorised the following items as elements, mixtures and compounds. What could be the reasons behind the assigning:
   (a) rock salt as a mixture and table salt as a compound.
   (b) muddy water as a mixture and pure water as a compound
   (c) charcoal as an element and sugar as a compound
2. Provide examples of how elements, compounds and mixtures are used in everyday life.

Solutions, suspensions and emulsions

Liquid mixtures can be described as solutions, suspensions or emulsion depending on their compositions.

Solutions

A *solution* is a homogeneous mixture of two or more substances which are *solvent* and *solute*. A solvent is the component of the solution that dissolves the solute. It is the component that is usually present in large amount in a solution. A solute is the component in solution that is dissolved in the solvent. It is the component that is usually present in small amount in a solution. For example, when water dissolves sugar, the solution of sugar and water is formed. In this case sugar is the solute and water is the solvent.

Activity 5.2

Aim: To make solutions.

Requirements: Beakers, spatula, water, common salt, sugar and glucose

Procedure

1. Put a small amount of each solid in a different beaker.
2. Add water to every beaker and stir until no more solid can be seen. What do you observe?

Question

Why was stirring important in the process?

Types of solutions

A solution can either be unsaturated, saturated or supersaturated. An *unsaturated* solution is the one in which the solvent can dissolve more solute at a given temperature and pressure. A *saturated* solution is the one in which the solvent can dissolve no more solute at a given temperature and pressure. A *supersaturated* solution is the one that holds more solute than the maximum amount it can dissolve at a given temperature and pressure.

Note: Saturation depends on temperature. As the temperature increases, the kinetic energy of solvent molecules increases and this creates more space for dissolving more solute particles.

Activity 5.3

Aim: To distinguish among the saturated, unsaturated and supersaturated solutions.

Requirements: Beaker, Bunsen burner, tripod stand, spatula, glass rod, trough, water and common salt

Procedure

1. Measure about 100 cm^3 of water and pour it in a beaker.
2. Add a spatulaful of the common salt in the beaker and stir. Does the salt dissolve?
3. Continue adding more salt to the solution and stirring until no more salt can dissolve.
4. Place the solution on the tripod stand and heat gently. Continue stirring while heating.
5. Stop heating when the salt dissolves.
6. Place the beaker in a trough that is half-filled with cold water and allow to cool for 5 minutes. Record your observations.

Questions

1. What type of solution is formed when a spatulaful of table salt dissolves in 100 cm^3 of water?
2. Which type of solution is obtained at room temperature when no more salt can dissolve?
3. What name is given to the final solution?

Applications of saturation

The concept of saturation can be applied when:

1. separating certain mixtures in laboratories.
2. extracting some minerals, such as extracting common salt from sea water.
3. cooking and salting food.
4. using detergent during laundry.
5. dissolving sugar in tea.

Classification of solutions into the three states of matter

Solutions can exist in the three states of matter namely, solid, liquid and gaseous states. The solutes and solvents can be in any state, that is, liquid, solid or gas. Examples of solutions in the three states of matter are listed in Table 5.8.

Table 5.8: *Examples of types of solutions in the three states of matter*

Solute	Solvent	Examples
Solid	Gas	Naphthalene slowly sublimes in air to form a solution
Solid	Liquid	Sugar in water and salt in water
Solid	Solid	Steel and other metal alloys
Liquid	Gas	Water in air
Liquid	Liquid	Ethanol (alcohol) in water and various hydrocarbons in each other (petroleum)
Liquid	Solid	Mercury in gold and hexane in paraffin wax
Gas	Gas	Oxygen and other gases in air
Gas	Liquid	Carbon dioxide in water (carbonated water)
Gas	Solid	Hydrogen in metals

Task 5.6

Identify various solutions available at your school and home and classify them into solid, liquid and gas.

Uses of solvents

Solvents have a wide range of uses, especially in homes, institutions such as schools and colleges, hotels and in manufacturing industries. They are usually used in cleaning. This is because they form a solution with the dirt (solute). Solvents are also used during varnish removal, stain removal, bleaching, thinning paints and degreasing. Moreover, solvents are used in chemical synthesis (manufacturing). Water is the most common solvent as it has the capacity to dissolve many solutes. Figure 5.4 shows examples of solvents in their containers.

Figure 5.4: *Solvents in containers*

Suspensions

A *suspension* is a heterogeneous mixture of a liquid (solvent) and fine particles of a solid. The solute particles do not dissolve, but get suspended in the solvent. When the suspension is not disturbed, the solute particles will settle down and become visible. For example, when you add flour into water during cooking, you have to keep on stirring, otherwise the flour will settle. Other examples of suspensions are; muddy water, blood, sand particles suspended in water, and chalk powder suspended in water. The suspensions composed of either liquid droplets or fine solid particles floating in a gas are called *aerosols*. Suspensions are used in many aspects in our lives. Most of them are stored in containers with the instructions *'Shake well before use'*. They include medicines (syrups), body sprays and insecticides. Examples of commonly used suspensions are shown in Figure 5.5.

Figure 5.5: *Containers with commonly used suspensions*

Table 5.9 presents the differences between solutions and suspensions.

Table 5.9: *Differences between solutions and suspensions*

S/N	Solutions	Suspensions
1.	Homogeneous mixtures	Heterogeneous mixtures
2.	Transparent/clear	Opaque/not clear
3.	Solute particles completely dissolved in a solvent	Solute particles settle if the suspension is undisturbed
4.	Components can be separated by evaporation	Components can be separated by filtration

Activity 5.4

Aim: To prepare suspensions.

Requirements: Mud, water, chalk powder, spatula, copper carbonate, beakers and glass rod

Procedure

1. Put a small amount mud, chalk powder and copper carbonate in separate beakers.
2. Add enough water to cover the solids and stir until the solid mixes with the water.
3. Leave the mixture to settle. What do you observe?

Questions

1. Which criteria can be used in selecting the materials for making a suspension?
2. Where in real-life can the experience of doing Activity 5.4 be applied?

Emulsions

An emulsion is a mixture of liquids that do not completely mix with each other. Liquids that do not mix are said to be *immiscible*, whereas those that mix are *miscible*. An emulsion is usually formed from two liquids, one water-based liquid and the other oil-based liquid. When shaken, the oily liquid forms droplets suspended in the water-based liquid. The harder the emulsion is shaken, the smaller the droplets, so that the emulsion may appear to be a homogeneous solution. Examples of emulsion include milk which contains drops of butterfat in water; and emulsion paint which contains drops of coloured oils in water.

Separation of mixtures

Task 5.7

Watch videos demonstrating different methods used for separation of mixtures. Identify the methods presented and relate them with those used in your daily life.

Many mixtures contain useful substances mixed with unwanted materials (impurities). In order to obtain the useful substances, chemists often separate the mixtures using different methods. The separation method depends on the components of the mixture and their properties. Some of the methods that are used in separating the mixtures are commonly applicable in our day-to-day living. The methods include handpicking, sieving, winnowing, filtration, decantation and evaporation. For example, handpicking and winnowing can be used to obtain rice that is free from sand particles. Likewise, clean and safe water can be obtained after filtering boiled or chemically treated water. These are just few examples of day-to-day useful ways of separating mixtures applied at home.

The principles for separating mixtures are also widely applied in various fields including mining, agriculture and other large-scale industries. For instance, decantation, filtration and sedimentation are some of the common methods that are widely used in purification of water for public consumption. Some of the methods of separating mixtures that can be demonstrated in a school laboratory include decantation, filtration, evaporation, distillation, layer separation, sublimation, chromatography and solvent extraction.

Decantation

The water in the oceans, lakes, rivers, ponds and dams are heterogeneous mixtures that contain sand and other insoluble matter that later settle at the bottom when the water is calm. This process of some of the components of a mixture settling at the bottom is called *sedimentation* and it is the first step in decantation. *Decantation* is the process of separating a heterogeneous mixture of a liquid and a solid by pouring out the liquid only and leaving the solid at the bottom of the container. This method works well when the solid component is made up of large particles. If the particles are small, the process may not give a good separation.

Activity 5.5

Aim: To separate substances by decantation.

Requirements: Muddy water and beakers

Procedure

1. Put some muddy water in a beaker and let it stand for some time.
2. Carefully pour out the clear water from the top into another beaker as shown in Figure 5.6.

Figure 5.6: *Decantation of muddy water*

Questions

1. What do you observe when the muddy water is left to stand for some time?
2. What precautions have to be taken into account to ensure effective separation?

Filtration

This is the method used to separate a heterogeneous mixture of a solid and a liquid. Some chemical reactions produce insoluble solids in liquid. The mixture of this kind can be separated by filtration. The solid is separated from the liquid using a porous filter, such as filter paper. The solid obtained is the *residue* and the liquid is the *filtrate*.

Activity 5.6

Aim: To separate substances by filtration.

Requirements: Sand, water, filter paper, filter funnel, conical flask, retort stand and clamp

Procedure

1. Fold the filter paper to form a cone shape as shown in Figure 5.7.

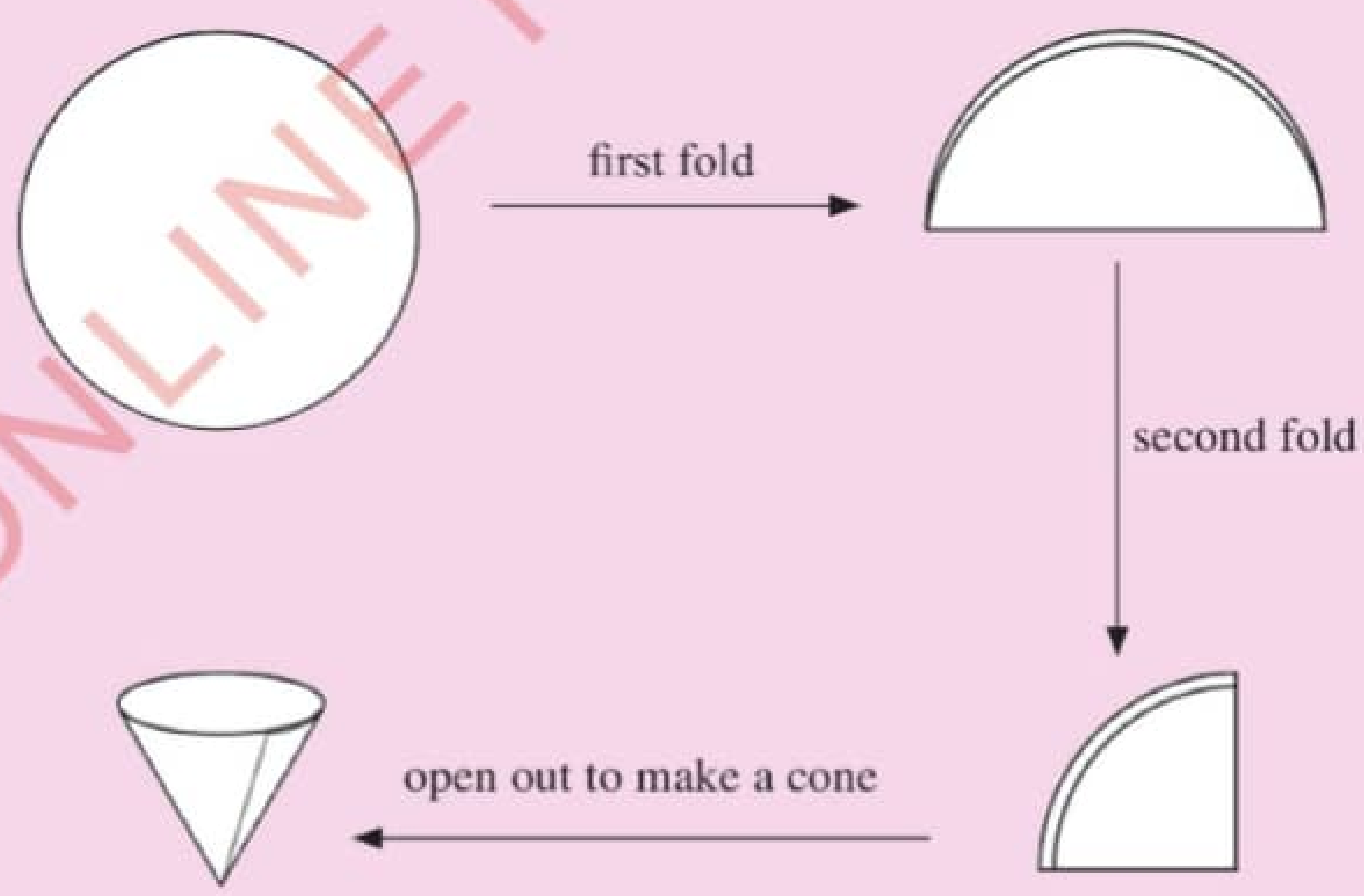


Figure 5.7: *Steps in folding a filter paper*

2. Place the cone shaped filter paper into the filter funnel clamped on a retort stand.
3. Put some sand and water in a beaker to make a mixture.
4. Pour the mixture of sand and water into the filter funnel connected to the mouth of a conical flask.

5. Leave the experiment to proceed as shown in Figure 5.8.

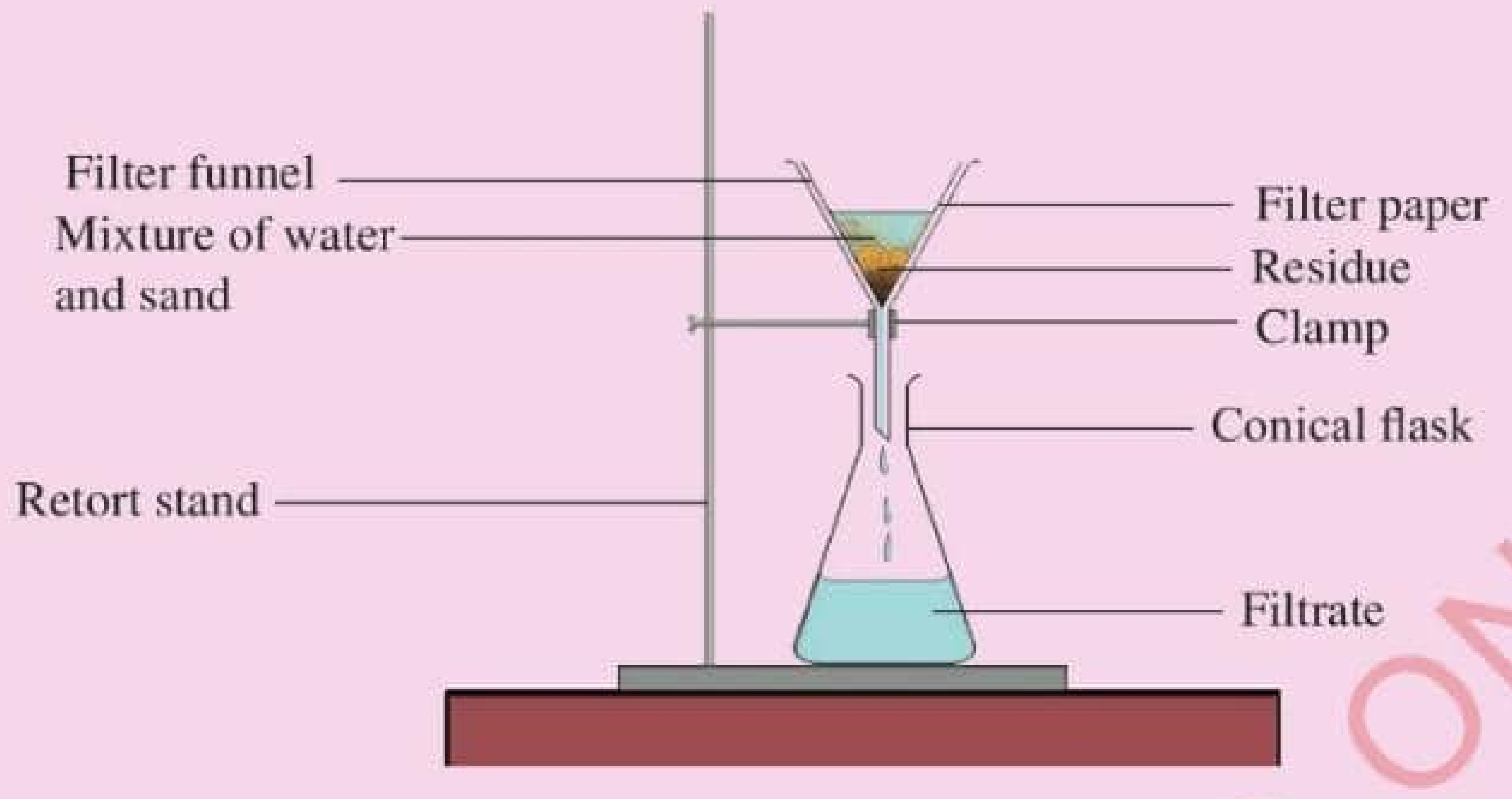


Figure 5.8: *Filtration of a mixture*

Questions

1. What is the nature of the filtrate obtained?
2. Which component of the mixture is the residue?
3. What will happen if the filter paper is not well fitted into the filter funnel?

Evaporation

Evaporation is the process of separating a solute from a liquid solution through heating. During evaporation, the solvent is converted from liquid to gas through heating as shown in Figure 5.9. The solute remains as a residue. For example, when a solution of common salt and water is heated, the water evaporates and the common salt remains as a residue.

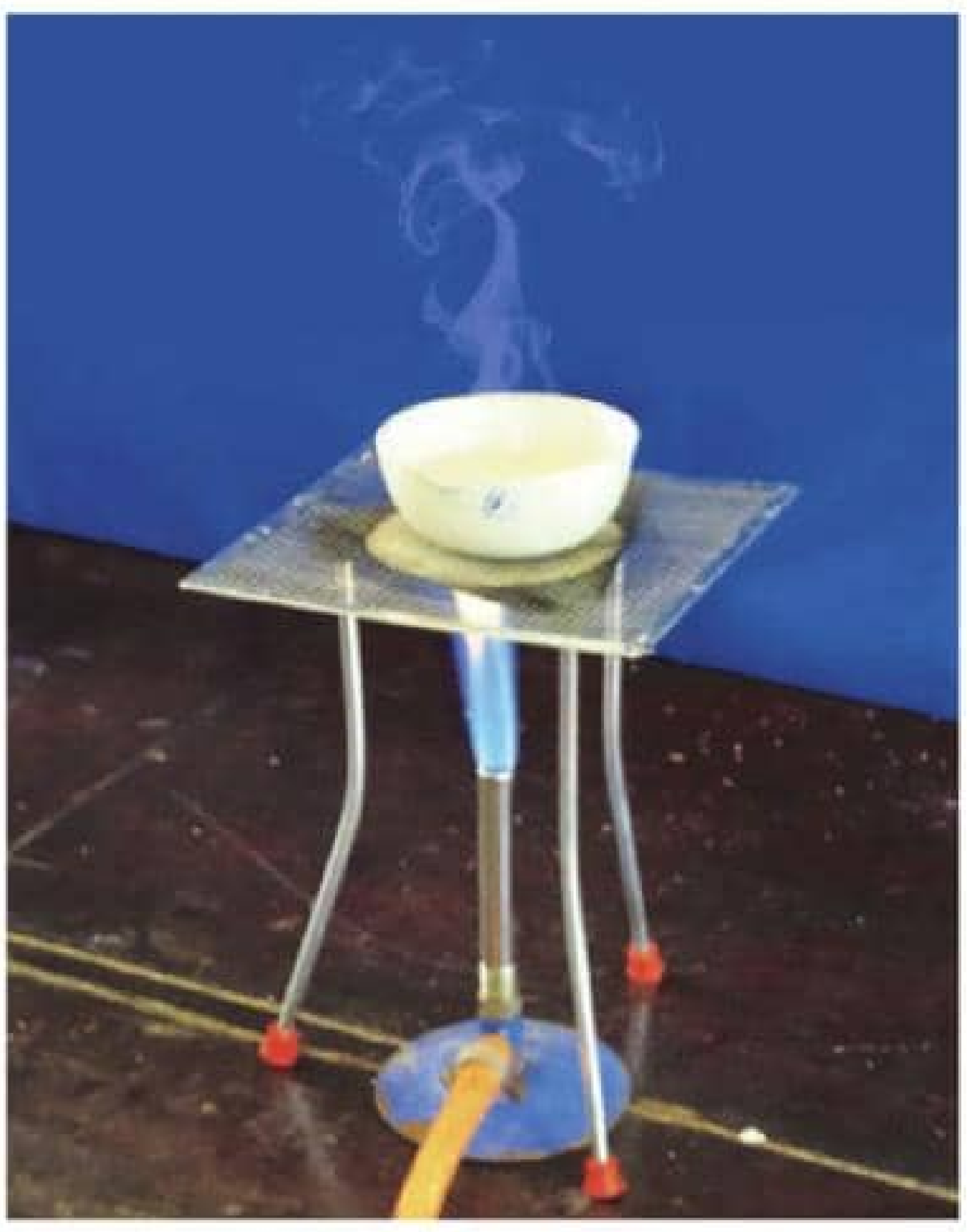


Figure 5.9: *Separation of a mixture by evaporation*

Activity 5.7

Aim: To separate substances by evaporation.

Requirements: Evaporating dish, beaker, tripod stand, Bunsen burner, wire gauze and salt solution

Procedure

1. Pour some of the salt solution into the evaporating dish and place it on a wire gauze on a tripod stand.
2. Heat the solution until all the water evaporates.
3. Let the resulting product completely cool.

Questions

1. What was your observation when the resulting product was allowed to cool?
2. Why was evaporation suitable for separating such a mixture?

Distillation

Distillation refers to the process of separating the components of a mixture by heating a liquid to suitable temperature until it forms vapour (a gas) and then cooling it back to liquid. The liquid from a cooled vapour is called the *distillate*. There are two types of distillation; simple distillation and fractional distillation.

Simple distillation

This is a process of obtaining a single liquid from a suspension or solution by boiling the mixture to obtain vapour and then condensing it back to a liquid. An example is distilling water from muddy water (Figure 5.10). The condenser helps to cool the vapour by means of the water that flows in and out of it. The distillate obtained is clean water and therefore, it is colourless.

Activity 5.8

Aim: To recover clean water from muddy water.

Requirements: Liebig condenser, distilling flask, beaker, thermometer, burner, retort stand and clamp, stopper, tripod stand, wire gauze, muddy water and source of running water

Procedure

1. Set up the apparatus as shown in Figure 5.10.
2. Pour the muddy water into the flat-bottomed flask, stopper it and insert a thermometer through the stopper.
3. Heat the mixture until it vaporises and a clear liquid collects in the beaker as you run water through the Liebig condenser.

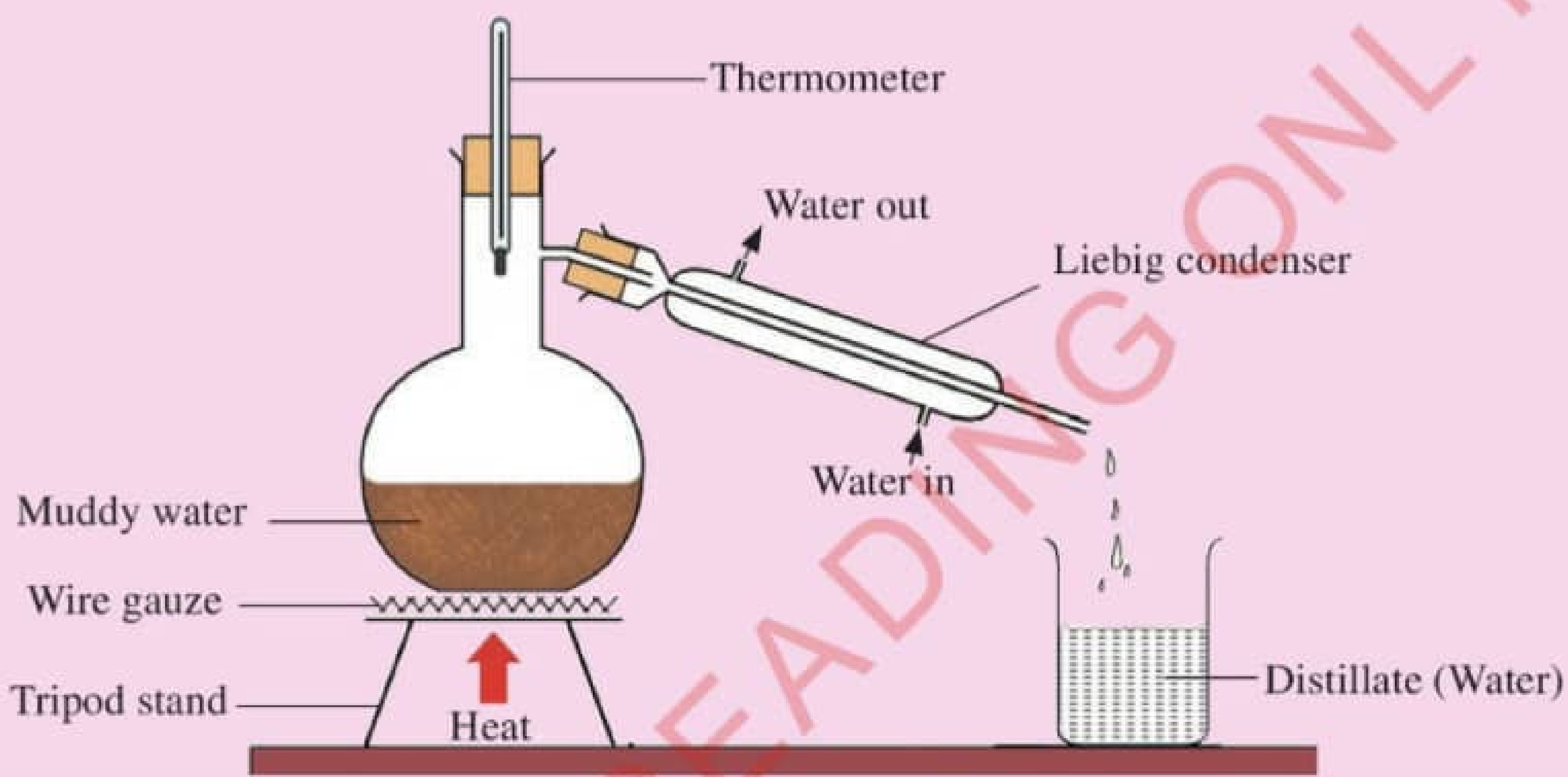


Figure 5.10: ***Simple distillation of muddy water***

Questions

1. What role does the Liebig condenser play?
2. If you do not have a Liebig condenser, what will you use to carry out this simple distillation?

Fractional distillation

This is a method of separating a homogeneous mixture of two or more liquids by means of a fractionating column. The liquids are said to be miscible. A mixture of ethanol and water is a good example. Since the liquids have a reasonable difference in boiling points, the solution can be separated by careful heating. The fractionating column separates the two liquids, the one with a lower boiling point moves to the upper part of the column and ***distils*** over. The liquid with the lower boiling point is collected first as a distillate. Each component collected is known as a ***fraction***. Figure 5.11 shows the set-up of fractional distillation.

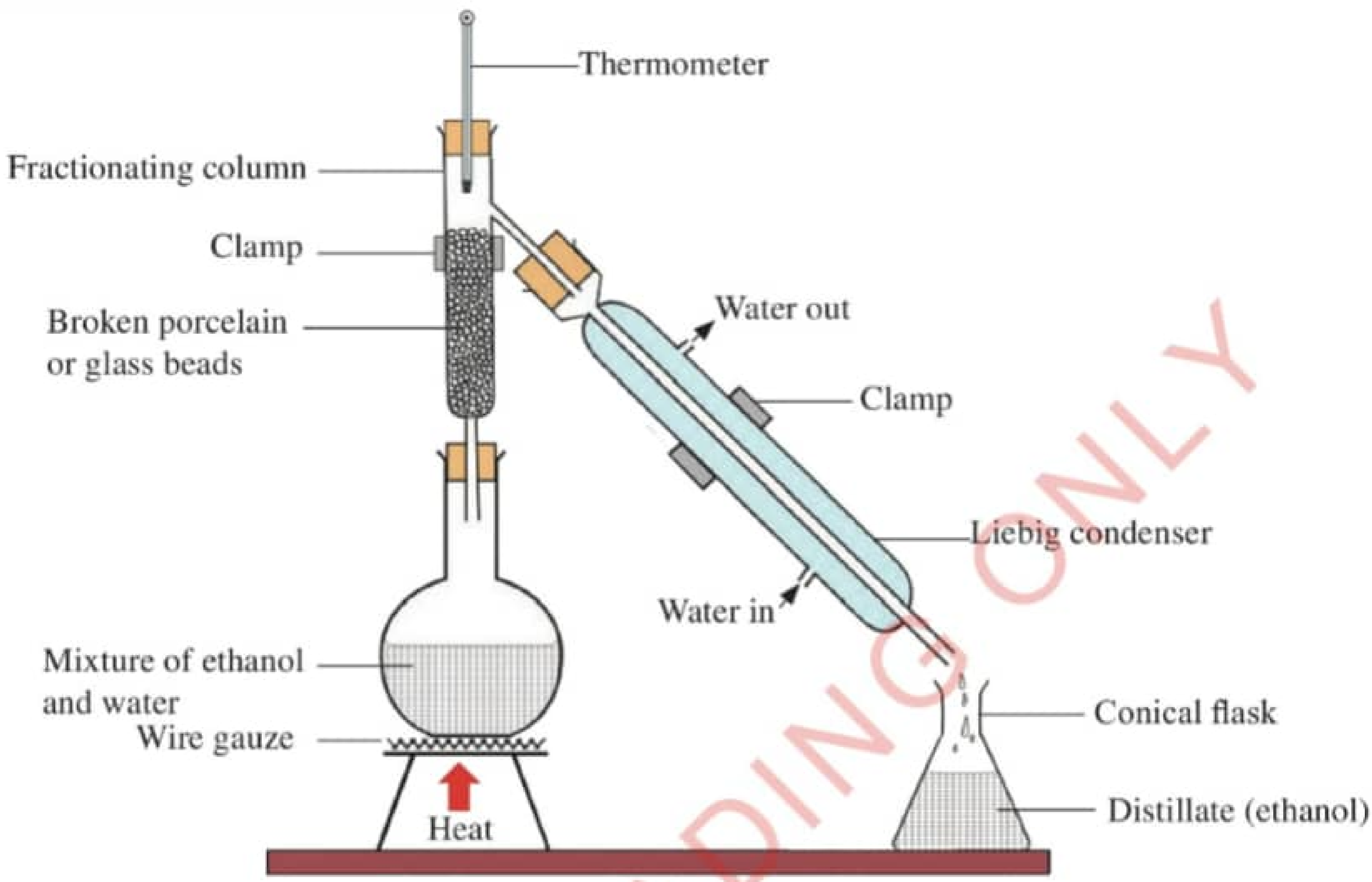


Figure 5.11: *Fractional distillation*

Activity 5.9

Aim: To separate two miscible liquids.

Requirements: Condenser, fractionating column, retort stands and clamps, thermometer, burner, tripod stand, wire gauze, conical flasks, ethanol and water

Procedure

1. Set the experiment as shown in Figure 5.11. Note that, the setting of this experiment can be done by a teacher or a technician.
2. Heat the mixture of ethanol and water. Note the temperature at which the first liquid is distilled. Record all your observations.

Questions

1. Which liquid distils first and at what temperature?
2. What role does the fractionating column play?
3. Why was it possible for the two liquids to be separated?

Layer separation

This method is used for separating immiscible liquids using a separating funnel. Immiscible liquids are those that do not mix, but form distinct layers when put together. The most dense liquid settles at the bottom, while the least dense remains at the top of the separating funnel. By opening the tap, the liquid at the bottom is released first into an empty container (Figure 5.12).

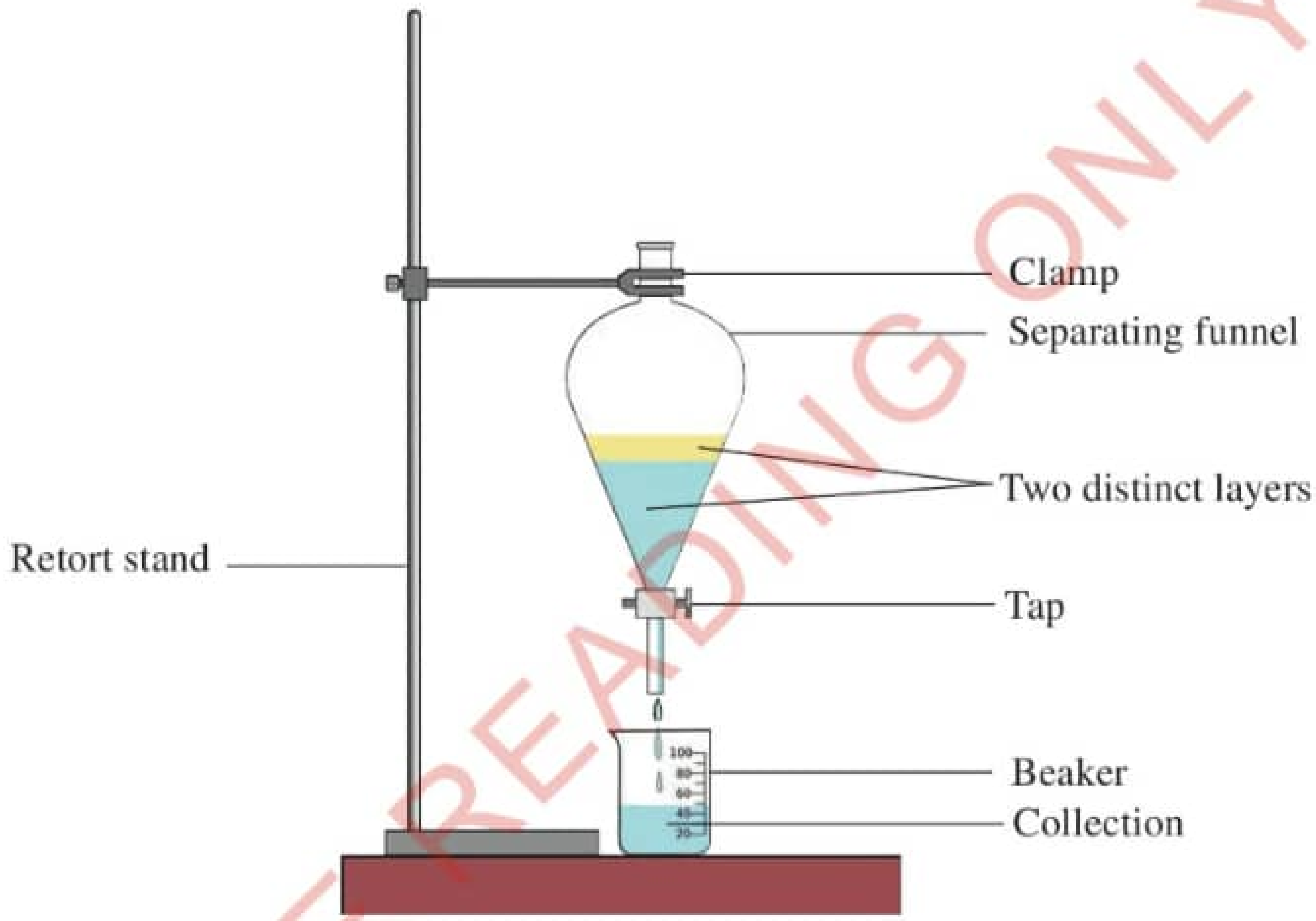


Figure 5.12: *Separating immiscible liquids*

Activity 5.10

Aim: To separate immiscible liquids.

Requirements: Separating funnel, kerosene, cooking oil, beakers, stirrer, water, stopper and retort stand with a clamp

Procedure

1. Pour water into a beaker.
2. Add kerosene and stir using a stirrer. What happened?
3. Pour the mixture into a separating funnel and wait for some few minutes. What do you observe?
4. Open the tap of the separating funnel slowly and drain away the liquid at the bottom first into an empty beaker.
5. Repeat steps 1 to 4 with water and cooking oil.

Questions

1. In each case, name the liquid that you drained first. Give reasons.
2. What factors led to the separation of the liquids?

Sublimation

Sublimation is the process whereby a solid changes its state directly to a gas, usually on heating. The solid that forms when the vapour cools is called a *sublimate*. This process can be used to separate mixtures in which one of the substances sublimes. Iodine and ammonium chloride are among the few compounds that change directly from solid to gas when heated. The reverse process of vapour changing to solid on cooling is called deposition.

Activity 5.11

Aim: To separate mixtures by sublimation.

Requirements: Beakers, wire gauze, tripod stand, burner, round-bottomed flask, iodine, ammonium chloride, sand, common salt and ice cubes

Note: These experiments should be carried out in a fume chamber.

Procedure

1. Place a mixture of iodine and sand in a beaker.
2. Put some ice cubes in a round-bottomed flask.
3. Set the apparatus as shown in Figure 5.13.
4. Heat the mixture and note what happens.
5. Repeat the experiment using a mixture of common salt and ammonium chloride.

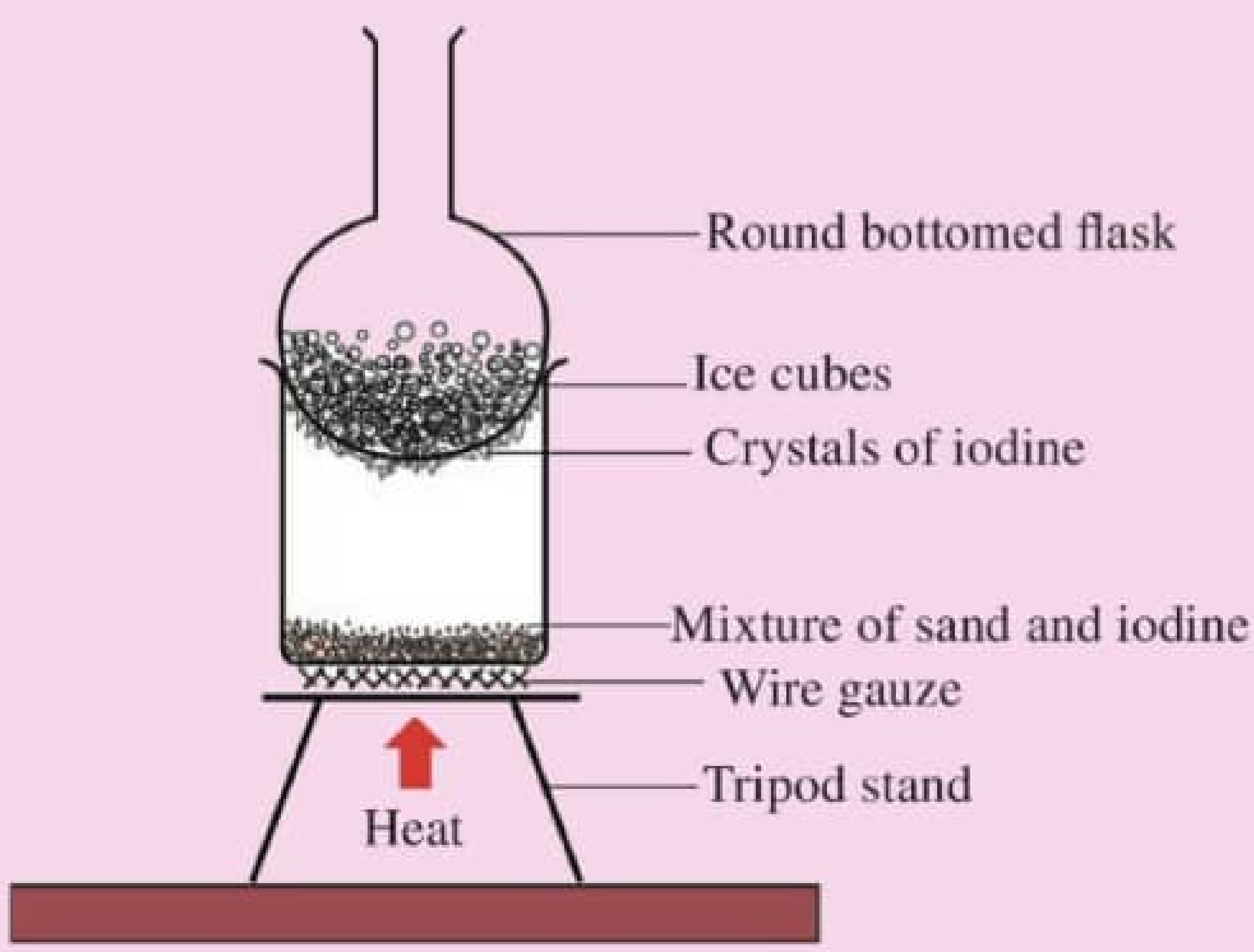


Figure 5.13: *Sublimation*

Questions

1. What happened when the mixture of iodine and sand was heated?
2. What happened when the mixture of common salt and ammonium chloride was heated?
3. Why were ice cubes put into a round bottomed flask?

Chromatography

Chromatography is the process of separating mixtures using a solvent and an immobile substance. The moving solvent is called a **mobile phase** and may be a liquid or gas. The immobile substance is called a **stationary phase**, which is either a solid or a liquid supported on a solid. The mobile phase flows through the stationary phase and carries the components of the mixture with it. Different components are separated at different rates. The substance that is separated during chromatography is called the **analyte**. There are different types of chromatography, such as paper chromatography, thin layer chromatography and gas chromatography. In paper chromatography, the stationary phase is a uniform **adsorbent** paper. The mobile phase is a suitable liquid solvent or mixture of solvents.

Activity 5.12

Aim: To carry out simple paper chromatography.

Requirements: Filter paper, beakers, extracts from plant parts such as leaves, roots or stem bark, acetone or ethanol, droppers and watch glass

Procedure

1. Take a rectangular piece of filter paper of a size that can fit in a beaker when placed upright.
2. Draw two straight lines one at the base and another at the upper part.
3. Mark three areas on the base line and on each mark put a drop of different plant extract. Label the spots A, B and C and make sure there is a space between the spots.
4. Place the paper, with baseline at the bottom, into a beaker containing acetone or ethanol. Ensure that the solvent level is below the baseline, and cover the beaker with a watch glass.
5. When the solvent has risen to the upper line, remove the paper from the beaker and place it in a dry beaker. Leave it to stand for about 30 minutes.
6. Mark the levels to which the pigments have risen for each spot. See Figure 5.14.

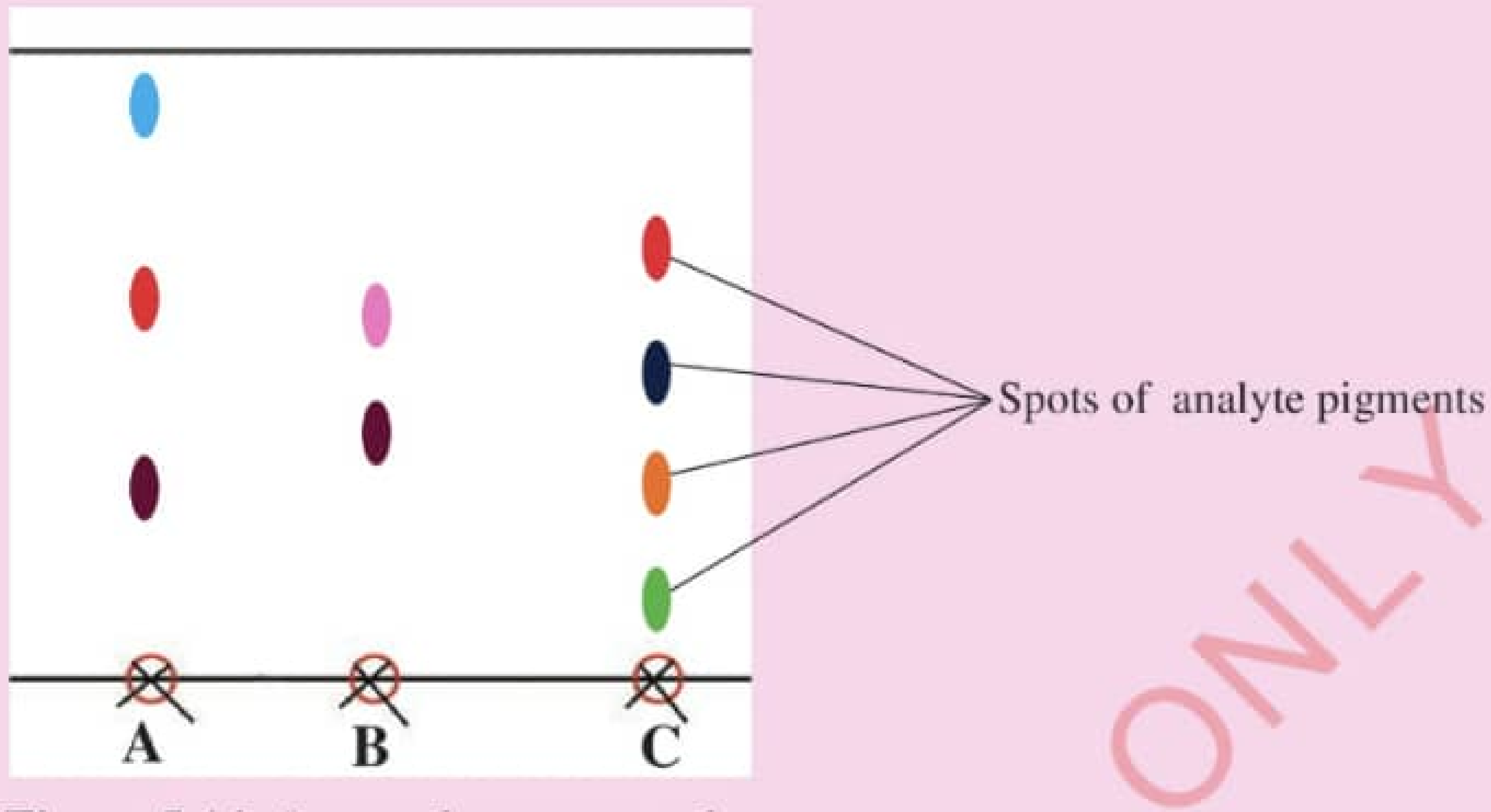


Figure 5.14: *Paper chromatography*

Questions

1. What is the role of the acetone or ethanol in this experiment?
2. Can the type and quality of paper influence the separation results? Explain.

Uses of chromatography

Chromatography is used in many different ways. It is used to separate components of a mixture, and for identification and quantification of different chemical substances. It can be used by security agents and medical personnel to analyse drugs in blood and urine samples. It is also used to detect causes of pollution in water, soil, air and organisms. Figure 5.15 summarises some of the uses of chromatography.

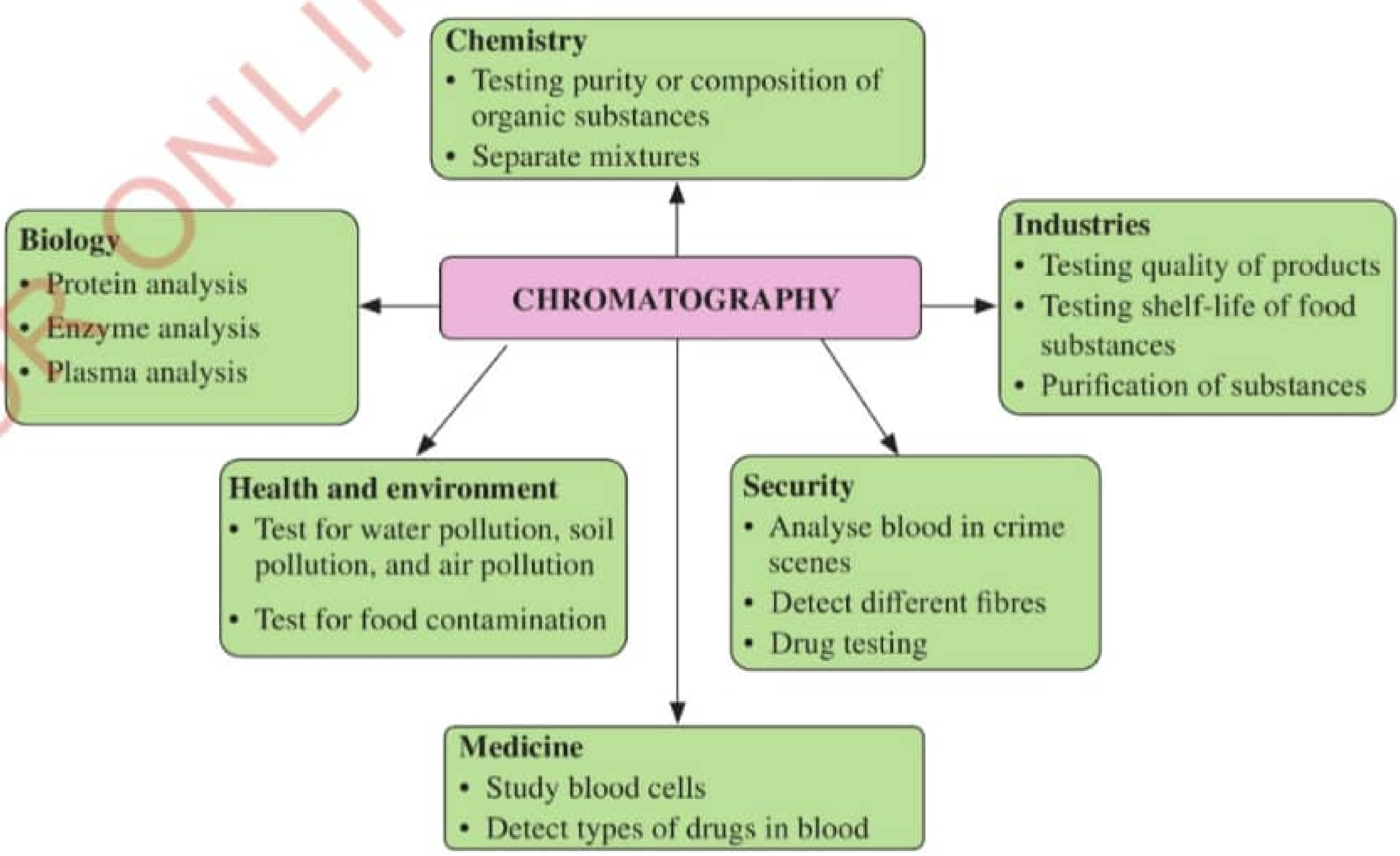


Figure 5.15: *Some of the uses of chromatography*

Solvent extraction

This is the process of separating components of a mixture using a solvent that dissolves selected components. For example, extracting essential oils from plant seeds using a liquid that dissolves the oils. The extracted oils are then distilled to let the solvent evaporate. Another example is preparation of tea which involves extracting tea components from tea leaves using hot water.

Note: In solvent extraction, the following conditions should be considered:

1. The solvent chosen should dissolve the component of interest.
2. The solvent must not react with the component of interest.
3. The solution formed between the component of interest and the chosen solvent must allow for easy separation of the desired component.

Activity 5.13

Aim: To extract oil from groundnuts.

Requirements: Ethanol, groundnuts, mortar and pestle, fractionating column and beaker

Procedure

1. Crush a handful of groundnuts in the mortar.
2. Pour in about 200 cm^3 of the ethanol and mix well.
3. Allow the mixture to settle for a while then decant the liquid into a beaker.
4. Use fractional distillation to separate the ethanol from the oil.

Questions

1. What was the reason for mixing ethanol with the groundnuts?
2. Why is fractional distillation used to separate the ethanol from the oil?

Activity 5.14

Aim: To demonstrate the applications of separation of mixtures.

Requirements: Fruit pulp (various fruits), milk, saucepan, clean water, filter funnel, filter paper, beakers and muddy water

Procedure

1. Mix the fruit pulp with some water and filter into a beaker.
2. Pour some of the muddy water in the filter and collect the clean water.
3. Boil milk in a saucepan, let it cool and extract the cream by filtering.

Questions

1. What was the product obtained in Step 1?
2. How can you apply the knowledge of separating mixtures in everyday life?

Project

Explore methods for the separation of mixtures in various industries and scientific fields.

Chapter summary

1. An element is a pure substance that cannot be broken into simpler substances by ordinary chemical means.
2. Chemical symbols are representations of the names of elements using letters.
3. The first letter of the chemical symbol must always be a capital letter.
4. Elements that have identical first letters are usually differentiated by a second or a third letter. The second or third letter is taken from the element's name, and is always a small letter.
5. Chemical symbols are derived from the Latin names or Greek names of the elements and some English names.
6. A compound is a pure substance made up of two or more elements in a chemical combination.
7. A mixture is a physical combination of two or more substances in any proportion.
8. A solution is a homogeneous mixture composed of a solute dissolved in a solvent.
9. A suspension is a heterogeneous mixture of liquid and fine solid particles.

10. The methods of separating mixtures are determined by the components of interest. One or more methods can be used to obtain the components of interest out of the mixtures.

11. Liquid-solid mixtures can be separated by evaporation, decantation, distillation or filtration.

12. Liquid-liquid mixtures can be separated using fractional distillation, separating funnel, or chromatography.

13. Solid-solid mixtures can be separated using a variety of methods such as using a magnet, sublimation, by using a suitable solvent and by winnowing.

Revision exercise 5

Choose the correct answer in items 1–8.

1. The chief cook added sugar to the cup of coffee. The resulting solution can be termed as:
   (a) Homogenous mixture
   (b) Binary compound
   (c) Heterogeneous mixture
   (d) Uniform suspension

2. Your young brother was given a syrup with a prescription that it should be well shaken before use. This is because the syrup is a:
   (a) clear solution.
   (b) suspension.
   (c) homogeneous mixture.
   (d) filtrate.

3. Table salt in water makes a homogenous mixture. What is the kind of the resulting mixture?
   (a) Solute
   (b) Solvent

(c) Suspension

(d) Solution

4. Which of the following is a compound?

(a) Helium (He)

(b) Hydrogen (H_2)

(c) Carbon monoxide (CO)

(d) Neon (Ne)

5. An element is defined as a:

(a) pure substance made up of two or more types of atoms.

(b) substance composed of multiple elements.

(c) substance composed of identical atoms.

(d) mixture of compounds.

6. Which of the following is a heterogeneous mixture?

(a) Saltwater

(b) Air

(c) A fruit salad

(d) Vinegar

7. The components of a mixture of iron and sand can be separated by:

(a) Boiling

(b) Filtration

(c) Chemical reaction

(d) Magnetic attraction

8. Which process among the following is involved in decantation?

(a) Sedimentation

(b) Melting

(c) Saturation

(d) Distillation

9. Write TRUE for a correct statement and FALSE for an incorrect statement.
   (a) A magnet can be used to separate metals from non-metals.
   (b) An unsaturated solution is the one in which the solvent has less than the maximum amount of solute that can dissolve at a given temperature.
   (c) Solutions can only be in liquid form.
   (d) You can obtain sugar crystals from a solution by decantation.
   (e) Ethanol and water are miscible.
   (f) Sublimation is the process by which a substance changes from a solid to a liquid form.
   (g) Emulsion is a clear solution.
   (h) Distillation is a process of separating a mixture of substances with different boiling points.
   (i) A compound is a pure substance made up of two or more elements in a chemical form.
   (j) The first letter of the chemical symbol must be a capital letter.
10. Name the process that can be used to separate each of the following substances:
   (a) Iodine and sand
   (b) Kerosene and water
   (c) A mixture of petrol and diesel
   (d) Components of the stem bark extract
   (e) Oil from seeds
11. Briefly explain how you can differentiate a solution and a suspension by their appearance.
12. What role does the knowledge on boiling point play in distillation process?
13. What factors determine the choice of a specific separation method for a given mixture?
14. Suppose you are given a mixture and you are asked to determine whether it is homogeneous or heterogeneous.

(a) What could be your criteria for the analysis?

(b) Make a description of how you will separate the;

(i) homogeneous mixtures.

(ii) heterogeneous mixtures.

15. How are suspensions utilised in the food and beverage industry?

16. Provide examples of how chromatography is used in the analysis of pharmaceuticals

17. Describe the applications of solutions, suspensions, and emulsions in daily life.

18. The time taken for sugar to dissolve in cold tea differs from that in hot tea. Explain

Glossary

Adsorbent	a substance or material that is used to adsorb other substances.
Aerosol	is a suspension of fine solid particles or liquid droplets in air or another gas.
Agriculture	is the practice of growing crops and keeping animals.
Analytical chemistry	is concerned with studies of instruments and methods used to separate, identify, and quantify matter.
Apparatus	a special tool or equipment used in a laboratory.
Artificial	man-made (not natural).
Amospheric pressure	the pressure exerted by the atmosphere.
Atom	smallest particles of a chemical element that can exist.
Barrel	is a cylindrical tube, forming part of an object such as a Bunsen burner.
Brownian motion	the random movement of particles in a liquid or gas.
Bunsen burner	a small adjustable gas burner used in laboratories.
Chemical change	a change that affects the chemical properties of substances in which new products are formed.
Chemical process	a method or means of changing one or more chemical compounds. Such a process involves chemical reactions.
Chemical symbol	representation of the name of element using letters.
Chemical safety signs	safety symbols found on chemical containers or places where hazardous chemicals are found.
Chromatography	the process of separating the components of a mixture by passing it in solution or suspension through a medium in which the components move at different rates.
Collar	a band near the base of a Bunsen burner that can be rotated to control air supply to the burner.

Compound	a substance that consists of two or more elements that are chemically combined.
Compression	the action of being condensed or reduction in volume.
Condensation	the cooling of vapour or gas to form liquid.
Contaminant	any potentially harmful substance in the environment.
Cosmetics	products used for cleansing, beautifying, promoting attractiveness or altering one's appearance.
Decantation	process of carefully pouring a liquid from a container, leaving the sediments at the bottom of the container.
Decomposition	the breaking down of a chemical compound into elements or smaller compounds.
Detergent	a liquid or solid compound or mixture of compounds used for cleaning.
Disinfect	to make clean and free from infections, especially by the use of chemical substances.
Distillation	the process of separating a liquid mixture into its components by boiling it to evaporation, then condensing the resulting vapour to obtain one of the components.
Element	a pure substance that cannot be split into simpler substances by a simple chemical process.
Emulsion	a mixture of liquids that do not dissolve each other or mix well.
Evaporation	the process of turning from a liquid to a vapour or gas.
Experiment	a scientific procedure undertaken to make a discovery, test a hypothesis or demonstrate a known fact.
Fertilisers	chemicals or natural substances added to soil to increase its fertility.
Fire extinguisher	an apparatus used to put out fire.

Fire triangle	the three components that must all be present for fire to start.
First Aid	help given to a sick or injured person until professional medical help is available.
First Aid Kit	a box that contains items used to give help to sick or injured person.
Flame	a region of burning gases that produces heat and light.
Fractionating column	apparatus used in the distillation of liquid mixtures to separate them into their components or fractions.
Fume chamber	a ventilated enclosure in a laboratory in which potentially harmful experiments are done to limit exposure to dangerous fumes.
Gas	a fluid with no definite shape or volume, but that is able to expand indefinitely.
Geology	field which apply chemistry principles and theories in exploring the chemical composition and processes of earth and other planets.
Hazard	something that could be dangerous or could cause damage.
Heimlich manoeuvre	a first aid procedure for dislodging an obstruction from a person's throat. Repeated thrusts are applied to the abdomen, near the top of the stomach.
Heterogeneous	non-uniform in composition, appearance and properties.
Homogeneous	uniform in composition, appearance and properties.
Immiscible	not mixing well or not forming a homogeneous mixture when mixed.
Inorganic compound	compound that does not contain carbon-hydrogen bond.
Kinetic molecular behaviour	the manner in which particles in matter behave.
Laboratory	a special room or building used for scientific experiments.

Liquefy	to make or to become liquid.
Liquid	a fluid (not a gas) with a definite volume, but that takes the shape of the container holding it.
Luminous flame	the yellow flame of a burner that has a limited supply of oxygen and which produces soot.
Mass	a measure of the quantity of matter in an object.
Matter	anything that has mass and occupies space.
Melting point	the temperature at which a solid change to a liquid at standard atmospheric pressure.
Metal	a hard, usually shiny, material that is a good conductor of heat and electricity.
Miscible	capable of being mixed very well.
Mixture	a physical combination of two or more substances in any ratio.
Molecule	a group of two or more atoms held together by chemical bonds.
Non-luminous flame	the blue flame of a burner with an adequate supply of oxygen and which produces more heat than the luminous flame.
Non-metal	chemical element that does not have the typical characteristics of metals.
Organic chemistry	is the area of chemistry that studies carbon compounds.
Organic compound	compound that contains carbon-hydrogen bond.
Particulate matter	relating to or in the form of very tiny particles.
Periodic Table	the systematic method to represent and organise chemical elements in a table format.
Pharmacy	a facility or location where drugs and other medical services are sold or dispensed, or the science or practice of the preparation and dispensing of medical drugs.

Physical change a change that affects only the physical properties of a substance.

Physical chemistry a branch of chemistry concerned with the way in which the physical properties of substances depend on and influence their chemical structures, properties and reactions.

Plasma an ionized gas consisting of positively charged ions and free electrons.

Properties a set of attributes or settings that are used to describe an object.

Reagent a substance that takes part in a chemical reaction.

Recovery position a position used to prevent an unconscious person from choking. The body is placed such that the face is downwards and slightly to the side supported by bent limbs.

Resuscitate to revive an unconscious person.

Rust a reddish-brown substance that forms on the surface of iron materials due to a chemical reaction with water and air.

Saturated a solution that contains as much solute as it can dissolve at a particular temperature.

Shock position a position used when a person experiences shock. The person lies on the back with the legs elevated about 8-12 inches above the head to help blood flow to the heart.

Solid a substance that has a definite shape and volume.

Solute the component of a solution that is dissolved in the solvent.

Solubility the degree to which a substance dissolves in a solvent to make a solution.

Solution a homogeneous mixture of two or more substances in which one of the substances is a solvent and the other is a solute.

Solvent	the component of a solution that dissolves the other substance(s).
Solvent extraction	separation of compounds from materials using liquids in which they are soluble.
Soot	a black substance produced by the incomplete burning of matter.
Spirit lamp	a lamp that burns volatile spirits used in some laboratories for heating purposes.
Sublimate	a solid deposit of a substance which has sublimed.
Sublimation	the change of state of matter from solid directly to gas.
Suspension	a heterogeneous mixture of a liquid and finely divided solids in which the solids do not dissolve in the liquid but are suspended in it.
Temperature	the degree of hotness or coldness of a body or an environment which can be measured using a thermometer.
Thermometer	an instrument used for measuring temperature.
Vaccine	a substance used to provide immunity against one or several diseases.
Vapour pressure	a pressure exerted by the vapour in equilibrium with its liquid or solid phase at a given temperature.

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Appendix

Below are the safety signs that can be found in the laboratory. These safety signs must be obeyed to ensure all activities are carried out safely in the laboratory.

 No entry	 Explosive	 High voltage	 Unsafe water
 No smoking	 Corrosive	 Health hazard	 Exit
 Caution	 Flammable/ inflammable	 Strong radiation	 Eye protection required
 Fragile	 Toxic	 Biohazard	 Eye and face protection required

Index